$\Rightarrow f'(x) = \frac{1}{2} - \cos x$

$\therefore f'(x) = 0 \Rightarrow \cos x = \frac{1}{2} \Rightarrow x = \frac{\pi}{3}, \frac{5\pi}{3}$ are critical number.

i) $f'(x) < 0$ when $0 < x < \frac{\pi}{3}$ and $\frac{5\pi}{3} < x < 2\pi \Rightarrow f$ is decreasing on $\left[0, \frac{\pi}{3}\right]$ and $\left[\frac{5\pi}{3}, 2\pi\right]$ $f'(x) > 0$ when $\frac{\pi}{3} < x < \frac{5\pi}{3}$

$\therefore f$ is increasing on $\left[\frac{\pi}{3}, \frac{5\pi}{3}\right]$.

ii) The sign of f' change from negative to positive at $x = \frac{\pi}{3}$ (decreasing then increasing) $\Rightarrow f\left(\frac{\pi}{3}\right) = \frac{\pi}{6} - \frac{\sqrt{3}}{2}$ is local minimum value. The sign of f' change from positive to negative at $x = \frac{5\pi}{3}$ (increasing then decreasing)

$\Rightarrow f\left(\frac{5\pi}{3}\right) = \frac{5\pi}{6} - \frac{\sqrt{3}}{2} \leftarrow$ local maximum value.

k. $f(x) = \frac{\ln x}{x} \Rightarrow f'(x) = \frac{1 - \ln x}{x^2}$, $f'(x) = 0 \Rightarrow \frac{1 - \ln x}{x^2} = 0$

$\Rightarrow \ln x = 1 \Rightarrow x = e \Rightarrow x = e$ is critical number

From the table we concluded that:

i) $f'(x) > 0$ when $0 < x < e \Rightarrow f$ is increasing on $(0, e]$.
$f'(x) < 0$ when $x > e \Rightarrow f$ is decreasing on $[e, \infty)$.

	0	e	∞
1 - lnx	+ + +	0	− − −
x^2	+ + +		+ + +
f'(x)	+ + +	0	− − −

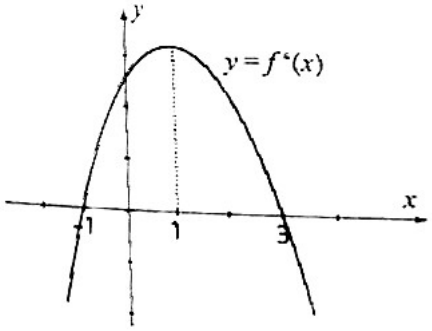
ii) The sign of f' change from positive to negative at $x = e$ (increasing then decreasing) $\Rightarrow f(e) = \frac{1}{e}$ is local max.

l. $f(x) = \frac{x}{x^2 + 2x + 1}$ is exercise left for you.

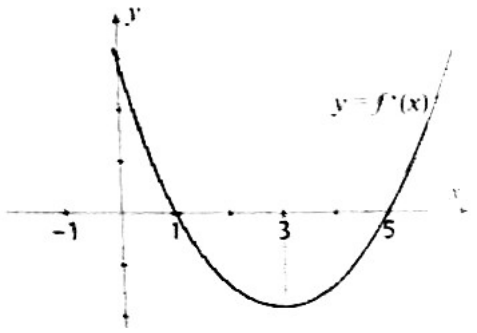
Determining monotonic and extreme value of function using graphs of derivative of function

Illustrative Example

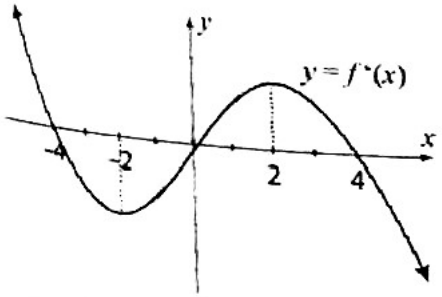
13. Suppose $f(x)$ is differentiable at all $x \in (-\infty, \infty)$. Then from the graph of $y = f'(x)$ given below, find:
   i) The intervals on which the function f is increasing.
   ii) The intervals on which the function f is decreasing.
   iii) The extreme value.



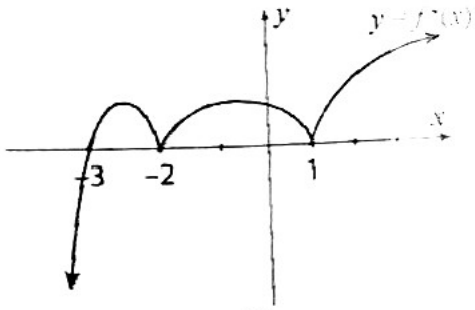
a)



b)



c)



d)

Solution: If the graph lies above the x-axis $f' > 0$. If the graph lies below the x-axis $f' < 0$.

a) i) since $x \in (-1, 3) \Rightarrow f'(x) > 0 \Rightarrow f$ is increasing on $[-1, 3]$.

ii) since $x \in (-\infty, -1)$ and $x \in (3, \infty) \Rightarrow f'(x) < 0$.

$\therefore f$ is decreasing on $(-\infty, -1]$ and $[3, \infty)$.

iii) Since $f'(-1) = 0$ and $f'(3) = 0$, $x = -1$, and $x = 3$ are critical numbers.

The sign of $f'(x)$ change from negative to positive at $x = -1$

$\Rightarrow (-1, f(-1))$ is local minimum value.

Sign of $f'(x)$ change from positive to negative at $x = 3$
$\Rightarrow (3, f(3))$ is local maximum value.

Note: you can see that $f''(1) = 0$.

b) Using sign chart of $f'(x)$

	$-\infty$		1		5		∞
$y = f'(x)$		+ + +		− − −		+ + +	
$y = f(x)$		Increase		Decrease		Increase	

i) $f(x)$ is increasing on $(-\infty, 1]$ and $[5, \infty]$.
ii) $f(x)$ is decreasing on $[1, 5]$.
iii) $f'(1) = 0$ and $f'(5) = 0$
The sign of $f'(x)$ change from positive to negative at $x = 1$
$\Rightarrow f(x)$ attains is local maximum value at $x = 1$.
Sign of $f'(x)$ change from negative to positive at $x = 5$
$\Rightarrow f(x)$ attains is local minimum value at $x = 5$.

Note: We can observe that $f''(3) = 0$.

c) Using sign chart of $f'(x)$

	$-\infty$		-4		0		4		∞
$y = f'(x)$		− − −		− − −		+ + +		− − −	
$y = f(x)$		Decrease		Decrease		Increase		Decrease	

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Digital Library for Ethiopian students
Link: https://telegram.me//digitallibrary4ethiopianstudents or digitallibrary4ethiopianstudents
YouTube Digital[0,1] Ethiopia

i) $f'(x) > 0$ on $(-\infty, -4)$ and $(0, 4)$
$\Rightarrow f(x)$ is increasing on $(-\infty, -4]$
ii) $f'(x) < 0$ on $(-4, 0)$ and $(4, \infty)$
$\Rightarrow f(x)$ is decreasing on $[-4, 0]$ and $[4, \infty)$.
iii) $x = -4$, 0 and 4 are local extreme value
$\Rightarrow (-4, f(-4))$ and $(4, f(4))$ are local maximum and $(0, f(0))$ is local minimum value.

d) Using sign chart:

	$-\infty$		-3		-2		1		∞
$y = f'(x)$		− − −		+ + +		+ + +		+ + +	
$y = f(x)$		Decrease		Increase		Increase		Increase	

i) $f'(x) > 0$ on $(-3, -2)$, $(-2, 1)$ and $(1, \infty) \Rightarrow f(x)$ is increasing on $[-3, \infty)$.
ii) $f'(x) < 0$ on $(-\infty, -3) \Rightarrow f(x)$ is decreasing on $(-\infty, -3]$.

iii) $f'(-3) = 0$, $f'(-2) = 0$, and $f'(1) = 0 \Rightarrow x = -3, -2$ and 1 are critical numbers. But only at $x = -3$ is local extreme value because no sign change at $x = -2$ and $x = 1$.

$\therefore f(-3) = -20 \leftarrow$ local minimum value.

14. Which of the following is local minimum point of the function $f(x) = -3x^4 + 4x^3 + 12x^2 + 1$ . . . (UEE)

A. $(-2, -33)$ B. $(-1, 4)$ C. $(0, 1)$ D. $(2, 33)$

Solution: $f'(x) = -12x^3 + 12x^2 + 24x = -12x(x^2 - x - 2)$

$f'(x) = 0 \Rightarrow -12x(x - 2)(x + 1) = 0$.

$\therefore x = 0, 2$ and -1 are critical numbers $\Rightarrow f''(x) = -36x^2 + 24x + 24$.

i) $f'(0) = 0$ and $f''(0) = 24 > 0$ $(0, 1)$ is local minimum value.

ii) $f'(-1) = 0$ and $f''(-1) = -36 < 0 \Rightarrow (-1, 6)$ is local maximum.

iii) $f'(2) = 0$ and $f''(2) = -72 < 0 \Rightarrow (2, 33)$ is local maximum

Answer: C

The second derivative test for local extreme

i) If $f'(c) = 0$ and $f''(c) > 0$, then f *has a local minimum* at c.

ii) If $f'(c) = 0$ and $f''(c) < 0$, then f *has a local maximum* at c.

Illustrative Example

15. Using the second derivative test, find the extreme value of:

a. $f(x) = 4x^3 + 21x^2 + 36x + 7$ b. $f(x) = -2x^3 - 6x^2 - 5$

Solution:

a. $f'(x) = 12x^2 + 42x + 36 = 6(2x + 3)(x + 2)$ and $f''(x) = 24x + 42$

$f'(x) = 0 \Rightarrow x = \dfrac{-3}{2}$ and $x = -2$.

$f'\left(\dfrac{-3}{2}\right) = 0$ and $f''\left(\dfrac{-3}{2}\right) = 78 > 0 \Rightarrow \left(\dfrac{-3}{2}, f\left(\dfrac{-3}{2}\right)\right)$ is local minimum and $f'(-2) = 0$ and $f''(-2) = -6 < 0 \Rightarrow (-2, f(-2))$ is local maximum.

b. Exercise left for you.

16. Let $f(x) = x^4 - 6x^2 + 1$ on $[-2, 3]$ be given, then which of the following a local maximum value of f . . . (UEE)

A. $f(0)$ B. $f\left(\sqrt{3}\right)$ C. $f(3)$ D. $f\left(-\sqrt{3}\right)$

Solution: $f'(x) = 4x^3 - 12x = 4x(x^2 - 3)$

$f'(x) = 0 \Rightarrow 4x(x^2 - 3) = 0 \Rightarrow x = 0, \sqrt{3}$ and $-\sqrt{3}$.

Thus, $x = -2, -\sqrt{3}, 0, \sqrt{3}$ and 3 are candidate for maximum.
$f(-2) = -7, f(3) = 28$ is local maximum,
$f(0) = 1, f\left(\sqrt{3}\right) = f\left(-\sqrt{3}\right) = -8$ is local minimum.

Answer: C

Note: The range of f on $[-2, 3]$ is $[-8, 28]$
The range of f on $(-\infty, \infty)$ is $[-8, \infty)$.

17. The maximum value of $f(x) = x^2 - 6x$ on the interval $[-4, 4]$.
A. $f(-4)$ B. $f(3)$ C. $f(-4)$ D. $f(1)$

Solution: $f'(x) = 2x - 6$
$f'(x) = 0 \Rightarrow 2x - 6 = 0 \Rightarrow x = 3$. Thus, $x = 3$ is critical number, $x = 3, -4, 4$ are candidates for maximum
$\Rightarrow f(-4) = 16 + 24 = 40, f(3) = -9$ and $f(4) = 16 - 24 = -8$.

Answer: A

18. Find the value of k, for which the function will have extreme value at the indicated point.

a. $f(x) = \dfrac{x+3}{x^2+k}$, at $x = 2$ b. $f(x) = \dfrac{x^2 - 2x + 6k}{x-2}$, at $x = -1$

Solution: Recall relative extreme value occur at critical number.

a. $f(x) = \dfrac{x+3}{x^2+k} \Rightarrow f'(x) = \dfrac{x^2 + k - (2x)(x+3)}{(x^2+k)^2} = \dfrac{-x^2 - 6x + k}{(x^2+k)^2}$,

since at $x = 2$ is extreme value of function, so that $f'(2) = 0 \Rightarrow f'(2) = \dfrac{-4 - 12 + k}{(4+k)^2} = 0 \Rightarrow -16 + k = 0 \Rightarrow k = 16$

b. $f(x) = \dfrac{x^2 - 2x + 6k}{x-2} \Rightarrow f'(x) = \dfrac{(2x-2)(x-2) - (x^2 - 2x + 6k)}{(x-2)^2}$

$\Rightarrow f'(x) = \dfrac{2x^2 - 6x + 4 - x^2 + 2x - 6k}{(x-2)^2} = \dfrac{x^2 - 4x + 4 - 6k}{(x-2)^2}$

Since at $x = -1$ is extreme value of f, so that $f'(-1) = 0$.

$\Rightarrow f'(-1) = \dfrac{1 + 4 + 4 - 6k}{(-3)^2} = 0 \Rightarrow 9 - 6k = 0 \Rightarrow k = \dfrac{3}{2}$

For what values of k so that $f(x)=\dfrac{x}{x^2+k}$ will have extreme values at $x=2$ . . . (UEE)

A. 0 B. 2 C. 4 D. 27

Solution: $f'(x)=\dfrac{x^2+k-(2x)x}{(x^2+k)^2}=\dfrac{k-x^2}{(x^2+k)^2}$ since at $x=2$ is an extreme value of f so that $f'(2)=0$

$\Rightarrow f'(2)=\dfrac{k-4}{4+k}=0\Rightarrow k-4=0\Rightarrow k=4,$ **Answer: C**

Concavity and inflection points

In the previous section we used to first derivative test help us to determine monotonicity intervals, local maximum value and local minimum value. Now, here that second derivative test is also important to determine on what interval the function is bending up (concave up) or bending down (concave down ward). Function is:

- **Concave up** then the graph of f is bending upward
- **Concave down** when the graph of f is bending down ward

Second derivative test for concavity

Let f be a function whose second derivative exists on an open interval I, then:

i) If $f''(x)>0$ for all x in I, then the graph of f is concave upward in I.

ii) If $f''(x)<0$ for all x in I, then the graph of f is concave down ward in I.

Point of inflection

The point where the graph of f changes from concave up to concave down or concave down to concave up is called point of inflection.

If $(c, f(c))$ is point of inflection of the graph of f, then either $f''(c)=0$ or $f''(c)$ does not exist.

Illustrative Example

20. Find:
    i) concavity interval ii) inflection point
    iii) local extreme value for each of the following.
    a) $f(x)=x^4-4x^3+5$ d) $f(x)=(\ln x)^2$
    b) $f(x)=4x^3-12x^2+9$ e) $f(x)=5+\sin x$, on $[0, 2\pi]$

c) $f(x) = \dfrac{4}{x^2+3}$

Solution: Use second derivative test

a) $f(x) = x^4 - 4x^3 \Rightarrow f'(x) = 4x^3 - 12x^2 \Rightarrow f''(x) = 12x^2 - 24x$

$f'(x) = 0 \Rightarrow 4x^2(x-3) = 0 \Rightarrow x = 0$ and 3 are critical numbers

$f''(x) = 0 \Rightarrow 12x(x-2) = 0 \Rightarrow x = 0$ and 2 is inflection point

Using sign chart

	$-\infty$		0		2		∞
$12x$	− − −		0	+ + +		+ + +	
$x-2$	− − −			− − −	0	+ + +	
$y = f''(x)$	+ + +		0	− − −	0	+ + +	
Graph of f	Concave up			Concave down		Concave up	

From the table (2nd derivative test)

i) The graph is concave up on $(-\infty, 0)$ and $(2, \infty)$. The graph is concave down on $(0, 2)$.

ii) Since $f''(0) = 0$ and $f''(2) = 0$, therefore $(0, f(0))$ and $(2, f(2))$ are inflection point.

iii) Since $x = 0$ and $x = 3$ are critical number so that from 2nd derivative for local extreme value

$f'(0) = 0$ and $f''(0) = 0$. Therefore, not extreme value

$f'(3) = 0$ and $f''(3) = 36 > 0$. Therefore, has local minimum.

∴ The only local extreme value is $(3, f(3)) = (3, -22)$

b. $f(x) = 4x^3 - 12x^2 + 9 \Rightarrow f'(x) = 12x^2 - 24x \Rightarrow f''(x) = 24x - 24$

$f'(x) = 0 \Rightarrow 12x(x-2) = 0 \Rightarrow x = 0$ and 2 are critical numbers

$f''(x) = 0 \Rightarrow 24x - 24 = 0 \Rightarrow x = 1$ is inflection point

The graph of f has:

i) **Concave up** when $f''(x) > 0 \Rightarrow 24x - 24 > 0 \Rightarrow x > 1$

ii) **Concave down** when $f''(x) < 0 \Rightarrow 24x - 24 < 0 \Rightarrow x < 1$

iii) **local extreme if**

1) $f'(0) = 0$ and $f''(0) = -24 < 0 \leftarrow$ local max at 0.

∴ $(0, f(0)) = (0, 9) \leftarrow$ local maximum value

2) $f'(2) = 0$ and $f''(2) = 48 - 24 = 24 > 0 \rightarrow$ local min

∴ $(2, f(2)) = (2, -7) \leftarrow$ local minimum value

c. $f(x)=\dfrac{4}{x^2+3}\Rightarrow f'(x)=\dfrac{-8x}{(x^2+3)^2}$

$$\Rightarrow f''(x)=\frac{-8(x^2+3)^2-(2(x^2+3)(2x)(8x))}{(x^2+3)^4}$$

$$\Rightarrow f''(x)=\frac{-8(x^2+3)+32x^2}{(x^2+3)^3}=\frac{24x^2-24}{(x^2+3)^3}=\frac{24(x^2-1)}{(x^2+3)^3}$$

$f'(x)=0\Rightarrow -8x=0\Rightarrow x=0\leftarrow$ critical number

$f''(x)=0\Rightarrow x^2-1=0\Rightarrow x=1$ and $x=-1\leftarrow$ inflection

Now, using sign chart table for $f''(x)$

	$-\infty$		-1		1		∞
$24(x^2+3)^3$	+ + +		+ + +		+ + +		
$x-1$	− − −		− − −		0	+ + +	
$x+1$	− − −	0	+ + +		+ + +		
$y=f''(x)$	+ + +	0	− − −	0	+ + +		
						Concave up	

i) f is concave up on $(-\infty,-1)$ and $(1,\infty)$.
f is concave down on $(-1,1)$

ii) Point of inflection at $x=-1$ and $x=1$ $(-1,f(-1))=(-1,1)$ and $(1,1)$.

iii) $f'(0)=0$ and $f''(0)=\dfrac{-8}{9}<0$ Therefore, local max at $x=0$.

d. $f(x)=(\ln x)^2\Rightarrow f'(x)=\dfrac{2\ln x}{x}\Rightarrow f''(x)=\dfrac{2-2\ln x}{x^2}$

$$\Rightarrow f'(x)=0\Rightarrow\frac{2\ln x}{x}=0\Rightarrow x=1\leftarrow\text{critical number}$$

$$\Rightarrow f''(x)=0\Rightarrow\frac{2-2\ln x}{x^2}=0\Rightarrow 2-2\ln x=0\Rightarrow\ln x=1\Rightarrow x=e$$

Using sign chart
Then, from the table we concluded that the graph of f is:

	0		e	
$2(1-\ln x)$	+ + +		0	− − −
x^2	+ + +			+ + +
$y=f''(x)$	+ + +		0	− − −
Graph of f (x)	***Concave up***			*Concave down*

i) Concave up on (0, e) and concave down on (e, ∞)
ii) At $x=e$ is inflection point $(e, 1)$
iii) Since $f'(1)=0$ and $f''(1)=2>0\leftarrow$ local. min at $x=1$.
$\therefore (1,f(1))=(1,0)$ is local minimum value.

e. $f(x)=5+\sin x\Rightarrow f'(x)=\cos x\Rightarrow f''(x)=-\sin x$

i) $f'(x) = 0 \Rightarrow \cos x = 0 \Rightarrow x = \frac{\pi}{2}, \frac{3\pi}{2}$ ← critical number

ii) $f''(x) = 0 \Rightarrow -\sin x = 0 \Rightarrow x = 0, \pi, 2\pi$ ← inflection point

From the table we conclude that the graph of f is:

	0		π		2π
$-\sin x$	− − −	0	+ + +		
$f''(x)$	− − −	0	+ + +		

i) Concave up on $(\pi, 2\pi)$ and concave down on $(0, \pi)$

ii) At $x = \pi$ is inflection point $(\pi, 5)$

iii) $f'\left(\frac{\pi}{2}\right) = 0$ and $f''\left(\frac{\pi}{2}\right) = -\sin\left(\frac{\pi}{2}\right) = -1$

$\therefore$ local maximum at $x = \frac{\pi}{2}$.

$$f'\left(\frac{3\pi}{2}\right) = 0 \text{ and } f''\left(\frac{3\pi}{2}\right) = -\sin\left(\frac{3\pi}{2}\right) = 1$$

$\therefore$ local minimum at $x = \frac{3\pi}{2}$

That is $\left(\frac{\pi}{2}, f\left(\frac{\pi}{2}\right)\right) = \left(\frac{\pi}{2}, 6\right)$ is local maximum value

$\left(\frac{3\pi}{2}, f\left(\frac{3\pi}{2}\right)\right) = \left(\frac{3\pi}{2}, 4\right)$ is local minimum value

f. $f(x) = x^4 e^{2x} \Rightarrow f'(x) = 4x^3 e^{2x} + x^4 e^{2x}(2) = e^{2x} x^3 (4 + 2x)$

$\Rightarrow f''(x) = 2e^{2x} x^3 (4 + 2x) + 3x^2 e^{2x} (4 + 2x) + e^{2x} x^3 (2)$

$= 4x^2 e^{2x} \quad (x^2 + 4x + 3)$

$f'(x) = 0$ when $e^{2x} x^3 (4 + 2x) = 0 \Rightarrow x = 0, -2$ are critical numbers

$\Rightarrow f''(x) = 0$ when $4x^2 e^{2x} (x^2 + 4x + 3) = 0$

$\Rightarrow 4x^2 e^{2x} (x + 3)(x + 1) = 0$

$\Rightarrow x = 0, -3, -1$ are inflection points

	−∞		−3		−1		0		∞
$e^{2x}4x^2$	+ + +		+ + +		+ + +	0	+ + +		
$x + 3$	− − −	0	+ + +		+ + +		+ + +		
$x + 1$	− − −		− − −	0	+ + +		+ + +		

Using sign chart from 2nd derivative:

From the table we see that:

i) $f''(x) > 0$ on $(-\infty, -3)$, $(-1, 0)$ and $(0, \infty)$

∴ The graph of f (x) is concave up on $(-\infty, -3)$, $(-1, 0)$ and $(0, \infty)$ similarly $f''(x) < 0$ on $(-3, -1) \Rightarrow$ the graph of $f(x)$ is concave down on $(-3, -1)$.

ii) At $x = -3$ and $x = -1$ are inflection point of the graph of function. But at $x = 0$ the graph has no inflection because no sign change at 0.

iii) $f'(-2) = 0$ and $f''(-2) = -64e^{-4} < 0 \Rightarrow (-2, 16e^{-4})$ is local maximum value

Note: For $a, b \in \mathbb{R}$

i) $f(x) = ax + b \Rightarrow f'(x) = a \Rightarrow f''(x) = 0$ (*concavity doesn't exist*)

ii) $f(x) = x^4 + a \Rightarrow f'(x) = 4x^3 \Rightarrow f''(x) = 12x^2 > 0$ for all $x \neq 0$

∴ It is concave up on $(-\infty, \infty)$.

∴ No inflection point even if $f''(0) = 0$ because no sign change.

17. The graph of $f(x) = x^4 - 6x^2 + 1$ has an inflection point at . . . (UEE)

A. $(-1, 1)$ B. $(1, -4)$ C. $\left(\sqrt{3}, 1\right)$ D. $(0, 1)$

Solution:

$f(x) = x^4 - 6x^2 + 1 \Rightarrow f'(x) = 4x^3 - 12x \Rightarrow f''(x) = 12x^2 - 12$.

Then $f''(x) = 0 \Rightarrow 12(x^2 - 1) = 0 \Rightarrow x = -1, 1$ **Answer: B**

$\Rightarrow (-1, f(1)) = (-1, -4)$ and $(1, f(1)) = (1, -4)$ are inflection.

18. Which of the following is true about $f(x) = x^3 - 3x^2 + 3x + 1$.

A. f is strictly decreasing on $(-\infty, \infty)$

B. f is strictly increasing on $(-\infty, \infty)$

C. f has local extreme value at $x = 1$

D. $(1, 0)$ is the inflection point of the graph

Solution:

$f(x) = x^3 - 3x^2 + 3x + 1 \Rightarrow f'(x) = 3x^2 - 6x + 3 = 3(x-1)^2 > 0$

$\Rightarrow f'(x) > 0$ for all x, except at $x = 1$.

∴ f is strictly increasing on $(-\infty, \infty)$. Even if $f'(1) = 0$, is critical number but $x = 1$ is not local extreme value because f' has no sign change at 1 similarly $\Rightarrow f''(x) = 6x - 6 \Rightarrow f''(x) = 0$ when $6x - 6 = 0 \Rightarrow x = 1$. Now, $f''(x) > 0$ when $x > 1$ and $f''(x) < 0$ when $x < 1$.

∴ $(1, f(1)) = (1, 2)$ is inflection point. **Answer: B**

Minimization and Maximization problem

One of the most common application of calculus involves the determination of minimum and maximum value. Consider the following terms you hear frequently. Some of them are: maximize, profit, minimize cost, least time, least size, maximize strength, maximize distance, . . . etc.

I. Maximizing and minimizing of product and sum of number

Illustrative Example

Finding two non-negative numbers

19. a. Find the two positive number whose the sum is 15 and the product is maximum.

Solution: $x \geq 0, y \geq 0$, and $x + y = 15$, with product (x, y) max.

Sum: $x + y = 15 \Rightarrow y = 15 - x$, product (P) $= xy \leftarrow$ to be max.

Product: $P(x) = x(15 - x) = 15x - x^2$

$\Rightarrow P'(x) = 15 - 2x, 0 \leq x \leq 15 \Rightarrow P'(x) = 0 \Rightarrow 15 - 2x = 0$

$$\Rightarrow x = \frac{15}{2}, y = 15 - \frac{15}{2} = \frac{15}{2}$$

$\therefore$ The two numbers are $x = \frac{15}{2}$ and $y = \frac{15}{2}$ whose product is maximum.

b. Find two positive number, whose sum is 18 and the product is maximum.

Solution: Exercise left for you. (Answer, 9 and 9)

c. Find two positive number whose product is 192 and the sum is minimum.

Solution:

$x \geq 0, x \geq 0, xy = 192$, and $x + y$ is to be minimum. $xy = 192 \Rightarrow y = \frac{192}{x}$ put in y for $x + y \Rightarrow$ sum $(S(x)) = x + \frac{192}{x}$

$$\Rightarrow S'(x) = 1 - \frac{192}{x^2}, S'(x) = 0 \Rightarrow x^2 = 192$$

$$\Rightarrow x = 8\sqrt{3}, \text{ so that } y = \frac{192}{x} = \frac{192}{8\sqrt{3}} = \frac{8\sqrt{3}}{3}$$

d. Find two number whose product is 192 the sum of the second number and three times the first is minimum.

Solution: Let x be the first and y be the second number.

Then $xy = 192$ and $y + 3x$ is to be minimized $\Rightarrow y = \frac{192}{x}$ put in y

for $y + 3x \Rightarrow f(x) = \frac{192}{x} + 3x \Rightarrow f'(x) = 3 - \frac{192}{x^2}$

$\Rightarrow f'(x) = 0$ if $x = 8$, then $y = 24$.

e. What positive number x minimize the sum of x and twice its reciprocal?

Solution:

$$f(x) = x + \frac{2}{x} \Rightarrow f'(x) = 1 - \frac{2}{x^2} \Rightarrow f'(x) = 0 \Rightarrow x = \sqrt{2}$$

$\therefore$ The minimum values is $x + \frac{2}{x} = \sqrt{2} + \frac{2}{\sqrt{2}} = \frac{4}{\sqrt{2}} = 2\sqrt{2}$.

20. **Maximizing area of rectangle and minimize perimeter**

a. Find the length (x) and width (y) of a rectangle with perimeter 100 m that maximize area.

Solution:

Perimeter (P) $= 2x + 2y = 100 \Rightarrow x = y = 50 \Rightarrow y = 50 - x$

Area (A) $= xy \Rightarrow A(x) = x(50 - x) = 50x - x^2 \Rightarrow A'(x) = 50 - 2x$

$\Rightarrow A'(x) = 0$ when $x = 25$, so that $y = 25$.

$\therefore$ length $x = 25$ and width w $= 25$ that gives the maximum area.

b. Find the length (x) and width (y) of rectangle with area 64 m^2 that minimize the perimeter.

Solution:

Area of rectangle $= xy = 64 \Rightarrow y = \frac{64}{x}$ Perimeter: $2x + 2y$ to be minimized $P(x) = 2x + 2\left(\frac{64}{x}\right) = 2x + \frac{128}{x}$

$\Rightarrow P'(x) = 2 - \frac{128}{x^2} \Rightarrow P'(x) = 0$ when $2 - \frac{128}{x^2} = 0 \Rightarrow 2x^2 = 128$

$\Rightarrow x^2 = 64 \Rightarrow x = 8$, then $y = \frac{64}{x} = \frac{64}{8} = 8$.

$\therefore$ Length $x = 8$ and width $y = 8$ that gives the minimum perimeter.

21. **Finding the point closest to the graph of *f*.**

Find the point on the graph of the function that are closest to the given point.

a. $f(x) = x^2 + 1, (0, 4)$ d. $f(x) = x^2 + 2x, (-1, 0)$

b. $f(x) = 4 - x^2, (0, 2)$ e. $f(x) = 2x - 4, (1, 3)$

c. $f(x) = x^2, \left(2, \frac{1}{2}\right)$ f. $f(x) = (x + 1)^2, (5, 3)$

Solution: Let d be the distance between the point (0, 4) and a point (x, y) on the graph $y = x^2 + 1$ is given by

$d = \sqrt{(x-0)^2 + (y-4)^2}$ quantity to be minimized.

$\Rightarrow d(x) = \sqrt{x^2 + (x^2 + 1 - 4)^2} = \sqrt{x^4 - 5x^2 + 9}$ put $x^2 + 1$ in y.

Let g $(x) = x^4 - 5x^2 + 9 \Rightarrow g'(x) = 4x^3 - 10x \Rightarrow g'(x) = 0$ when $2x$ $(2x^2 - 5) = 0 \Rightarrow x = 0, \sqrt{\frac{5}{2}}$ and $-\sqrt{\frac{5}{2}}$ are critical number. To identify, use second derivative test

$\Rightarrow g''(x) = 12x^2 - 10 \Rightarrow g''(0) = -10 < 0$ (maximum)

but $g''\left(\sqrt{\frac{5}{2}}\right) > 0$ and $g''\left(-\sqrt{\frac{5}{2}}\right) > 0$ minimum.

$\therefore$ The point $\left(-\sqrt{\frac{5}{2}}, \frac{7}{2}\right)$ and $\left(\sqrt{\frac{5}{2}}, \frac{7}{2}\right)$ are closes to (0, 4).

b. Let d be the distance between the point (0, 2) and a point (x, y) on $y = 4 - x^2$. $d = \sqrt{(x-0)^2 + (y-2)^2}$

$\Rightarrow d(x) = \sqrt{x^2 + (4 - x^2 - 2)^2} = \sqrt{x^4 - 3x^2 + 4}$

Let g $(x) = x^4 - 3x^2 + 4$ to be minimized

$\Rightarrow g'(x) = 4x^3 - 6x = 2x\,(2x^2 - 3) = 0 \Rightarrow x = 0, -\sqrt{\frac{3}{2}}$ and $\sqrt{\frac{3}{2}}$

$\Rightarrow g''(x) = 12x^2 - 6 \Rightarrow g''(0) = 6$, so, $x = 0$ yield relative max.

$\Rightarrow g''\left(-\sqrt{\frac{3}{2}}\right)=12$, so that $x=-\sqrt{\frac{3}{2}}$ and $\sqrt{\frac{3}{2}}$ yield minimum.

$\therefore$ The point $\left(-\sqrt{\frac{3}{2}},\frac{5}{2}\right)$ and $\left(\sqrt{\frac{3}{2}},\frac{5}{2}\right)$ are point on the graph closest to (0, 2).

c. Let d be the distance between the point $\left(2,\frac{1}{2}\right)$ and point (x, y) on $y=x^2$.

$$\Rightarrow d=\sqrt{(x-2)^2+\left(y-\frac{1}{2}\right)^2}$$

$$\Rightarrow d(x)=\sqrt{(x-2)^2+\left(x^2-\frac{1}{2}\right)^2}=\sqrt{x^4-4x+\frac{17}{4}}$$

Let $g(x)=x^4-4x+\frac{17}{4}\Rightarrow g'(x)=4x^3-4\Rightarrow g'(x)=0\Rightarrow x=1$

$\Rightarrow g''(x)=12x^3\Rightarrow g''(1)=12>0$, thus $x=1$ gives min. dist.

$\therefore$The point (1, 1) is closest to $\left(2,\frac{1}{2}\right)$.

d. Exercise left for you.

e. Let d is distance b/n the point (1, 3) and a point (x, y) on $y=2x-4$. $\Rightarrow d=\sqrt{(x-1)^2+(y-3)^2}$

f. $\Rightarrow d(x)=\sqrt{(x-1)^2+(2x-4-3)^2}=\sqrt{5x^2-16x+50}$

Let $g(x)=5x^2-16x+50\Rightarrow g'(x)=10x-16\Rightarrow g'(x)=0\Rightarrow x=\frac{8}{5}$.

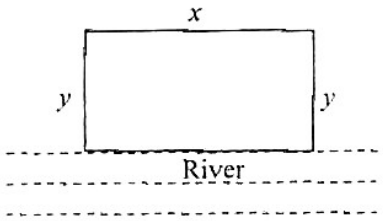
$\therefore$ The point $\left(\frac{8}{5},\frac{-4}{5}\right)$ is the point on the graph closest to (1, 3).

f. Exercise left for you.

22. **The shortest fence:** A farmer plans to fence a rectangular pasture adjacent to a river. The pasture must contain 180,000 m^2 in order to provide enough grass for the herd. What dimensions would requires the smallest amount of fencing if no fencing is needed along river.

Solution: i) Area $= xy \Rightarrow 180{,}000 \Rightarrow y - \dfrac{180{,}000}{x}$

ii) $2y + x \leftarrow$ quantity to be minimized (perimeter)



$$\Rightarrow A(x) = 2\left(\frac{180{,}000}{x}\right) + x = \frac{360{,}000}{x} + x$$

$$\Rightarrow A'(x) = 1 - \frac{360{,}000}{x^2} \Rightarrow A'(x) = 0$$

when $x = 600$

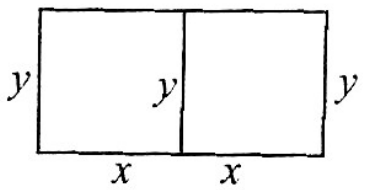
Then $y = \dfrac{180{,}000}{x} = \dfrac{180{,}000}{600} = 300\text{m}$

$\therefore$ The smallest amount of fencing when $y = 300$ m and $x = 600$m.

23. **Maximizing area:** A rancher (farmer) has 200 m of fencing with which to enclose two adjacent rectangular corrals (field). What dimensions should be used so that the enclosed area will be a maximum.

Solution: Perimeter $= 4x + 3y = 200 \Rightarrow y = \dfrac{200}{3} - \dfrac{4}{3}x$,

Area $= 2xy \leftarrow$ quantity to be maximize



$$\Rightarrow A(x) = (2x)\left(\frac{200}{3} - \frac{4}{3}x\right) = \frac{400x}{3} - \frac{8x^2}{3}$$

$$\Rightarrow A'(x) = \frac{400}{3} - \frac{16x}{3} \Rightarrow A'(x) = 0 \quad \text{when}$$

$x = \dfrac{400}{16} = 25$, then $y = \dfrac{200}{3} - \dfrac{4}{3}x = \dfrac{200}{3} - \dfrac{4}{3}(25) = \dfrac{100}{3}$

$\therefore x = 25$ m and $y = \dfrac{100}{3}$ m yield maximum area.

24. **The maximum distance:** The mass connected to a spring moves along the x-axis so that its x-coordinate at time t is given by

$x(t) = \sin 2t + \sqrt{3}\cos 2t$. What is the maximum distance of the mass from the origin. (Gulick).

Solution: $x(t) = \sin 2t + \sqrt{3}\cos 2t$

$\Rightarrow x'(t) = 2\cos(2t) + -2\sqrt{3}\sin(2t) \Rightarrow x'(t) = 0$ if $\cos 2t$

$= \sqrt{3}\sin 2t \Rightarrow \tan(2t) = \dfrac{1}{\sqrt{3}} \Rightarrow 2t = 30°, 210°$

$\Rightarrow t = 15°$ and $t = 105°$ ←critical number.

Now using 2nd derivative test

$x''(t) = -4\sin(2t) - 4\sqrt{3}(2t)$, then $x'(150°) = 0$ and $x''(105°)$

$$= (-4)\left(-\frac{1}{2}\right) - 4\sqrt{3}\left(\frac{-\sqrt{3}}{2}\right) = 2 + 6 = 8 > 0.$$

Hence t = 15° gives maximum distance which is

$$x(15°) = \sin(30°) + \sqrt{3}\cos(30°) = \frac{1}{2} + \frac{3}{2} = 2.$$

Thus the maximum distance of the mass from the origin is 2 unit.

25. **Minimum Area:** The combined perimeter of an equilateral triangle and a square is 10 m. Find the dimension of the triangle and the square that produce minimum total area.

Solution: Let x be the length of the side of triangle and y be the length side of square. Then $3x + 4y = 10 \Rightarrow y = \dfrac{10}{4} - \dfrac{3}{4}x$

$\Rightarrow$ Area of triangle $= \dfrac{1}{2}(x)x\sin 60° = \dfrac{\sqrt{3}x^2}{4}$,

Area of square $= y^2$

Area sum $= \dfrac{\sqrt{3}x^2}{2} + y^2 \Rightarrow A(x) = \dfrac{\sqrt{3}}{4}x^2 + \left(\dfrac{10}{4} - \dfrac{3}{4}x\right)^2$

$\Rightarrow A'(x) = \dfrac{\sqrt{3}}{2}x + 2\left(\dfrac{10}{4} - \dfrac{3}{4}x\right)\left(-\dfrac{3}{4}\right)$

$\Rightarrow A'(x) = 0 \Rightarrow x = \dfrac{30}{4\sqrt{3} + 0}$ m

$\therefore x = \dfrac{10}{4\sqrt{3}+9}$ m: yield the minimum area

26. A twelve meter wire is to be used to form circle and square. How much wire should be used for the circle and how much wire for the square if the total area enclose is:
a. minimum b. maximum

Solution:

$\underbrace{\qquad + \qquad}_{12\text{ m}}$ $2\pi r$ $4x$

The total area is given by
A = (Area of square) + (Area of circle)
Let x be the side length and r is radius
$\Rightarrow A = x^2 + \pi r^2$ . . . (1).
Total length of wire is 12m (perimeter) given by: P = (perimeter of square) + (circumference of circle)

$$\Rightarrow 12 = 4x + 2\pi r \Rightarrow 2\pi r = 12 - 4x \Rightarrow r = \frac{6-2x}{\pi}$$

$$\Rightarrow A(x) = x^2 + \pi\left[\frac{6-2x}{\pi}\right]^2 \text{ substituting } \frac{6-2x}{\pi}$$

$$\Rightarrow A'(x) = 2x + \frac{2\pi}{\pi^2}[6-2x](-2) = 2x + \frac{8x-24}{\pi},\ 0 \le x \le 3$$

$$\Rightarrow A'(x) = 0$$

$$\frac{2x\pi + 8x - 24}{\pi} = 0 \Rightarrow x = \frac{24}{2\pi+8} = \frac{12}{\pi+4} \text{ and}$$

$r = \dfrac{6-2x}{\pi} = \dfrac{6}{\pi+4} \leftarrow$ radius of circle. Hence we conclude the sum of the area of circle and square minimum when the side of square is twice radius of circle.
Note, the maximum area occurs when $x = 0$. That is all the wire is used for the circle.

27. You have been asked to design to hold 1 L oil can shaped like a right circular cylinder can. Find the dimensions that will minimize the cost of the metal to manufacture the can.

Solution: Volume of cylinder $(v) = \pi r^2 h \Rightarrow v = \pi r^2 h = 1000\ cm^3$

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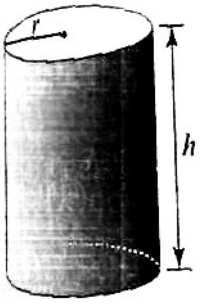
$1L = 1000\ cm^3 \Rightarrow h = \dfrac{1000cm^3}{\pi r^2}$.

Surface area of can: $A = 2\pi r^2 + 2\pi rh \leftarrow$ quantity that minimize.

$A(r) = 2\pi r^2 + 2\pi r\left(\dfrac{1000}{\pi r^2}\right) = 2\pi r^2 + \dfrac{2000}{r} \Rightarrow A'(r) = 4\pi r -$



Cylinder

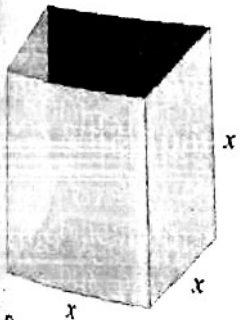
$\dfrac{2000}{r^2} \Rightarrow A'(r) = 0$ when $r = \sqrt[3]{\dfrac{500}{\pi}}$. Applying 2nd derivative test: $A''(r) = 4\pi + \dfrac{400}{r^3} > 0$. Hence, $r = \sqrt[3]{\dfrac{500}{\pi}}$ gives the minimum value. Thus,

$$h = \frac{1000cm^3}{\pi r^2} = 2\left(\sqrt[4]{\frac{500}{\pi}}\right) \leftarrow h = 2r$$

∴ To minimize the cost of the can, should be

$h = 2\left(\sqrt[3]{\dfrac{500}{\pi}}\right)$ and $r = \sqrt[3]{\dfrac{500}{\pi}}$.

28. Finding maximum value: A manufacture wants to design an open box having a square base and a surface area of $108m^2$. What dimensions will produce a box with maximum volume?



Open box with square base

Solution: Since the box has a square base its volume is $v = x^2h$...*quantity to be optimized.*

$\Rightarrow s$ = (Area of base) + (Area of 4 faces)

$= x^2 + 4xh = 108 \Rightarrow h = \dfrac{108 - x^2}{4x}$ solving for h in terms of x.

Now, $v = x^2h \Rightarrow v(x) = x^2\left(\dfrac{108 - x^2}{4x}\right) = 27x - \dfrac{x^3}{4}$

$\Rightarrow v'(x) = 27 - \dfrac{3x^2}{4} \Rightarrow v'(x) = 0.$

Therefore, $27-\frac{3x^2}{4}=0 \Rightarrow 3x^2 = 108 \Rightarrow x = \pm 6$ is critical number. Accept $x = 6$, reject $x = -6$ (length is non-negative).

Using 2nd derivative $v''(x) = -\frac{6x}{4} \Rightarrow v''(6) = -\frac{36}{4} < 0$.

$\therefore v$ is maximum when $x = 6$, and $h = \frac{108-x^2}{4x} = \frac{108-36}{24} = 3$

Examples from economics

Calculus makes important contribution to economic theory. First for maximizing profit and for minimize profit and for minimizing cost. Suppose that:

- R (x) = the revenue from selling x item.
- C (x) = the cost producing the x item.
- P (x) = R (x) – C (x) = the profit from selling x item.

The marginal revenue, marginal and marginal profit at this production level (x – items) are:

- $\frac{dr}{dx}$ = marginal revenue
- $\frac{dc}{dx}$ = marginal cost
- $\frac{dp}{dx}$ = marginal profit

29. A company has examined its revenue structure given by $R(x) = -\frac{1}{3}x^3 + 80x^2 + 72,000x$ where x is the number of units produced:

a. What production level will yield a maximum revenue.

b. What is the marginal revenue and when maximum.

Solution:

a. $R'(x) = -x^2 + 160x + 72,000$

$\Rightarrow R'(x) = -x^2 + 360x - 200x + 72,000$

$\Rightarrow R'(x) = -x(x - 360) - 200(x - 360) = (-x - 200)(x - 360)$

$\Rightarrow R''(x) = -2x + 160 \Rightarrow R''(360) = -720 + 160 < 0$

Therefore, $x = 360$ yield the maximum revenue.

29. Merginal revenue is $\frac{dR}{dx} = -x^2 + 160x + 72,000$ marginal revenue is maximum when $-2x + 160 = 0 \Rightarrow x = 80$

30. A steel plant produces x tone of steel per week at total cost in birr of $c(x) = \frac{1}{3}x^3 - 7x^2 + 111x + 8$. Find the marginal cost and its minimum.

Solution: $C'(x) = x^2 - 14x + 111$...*marginal cost*

$C''(x) = 2x - 14 = 0 \Rightarrow x = 7$.

∴ The marginal cost is minimum when $x = 7$.

31. **Harmonic motion:** The height of an object attached to a spring is given by equation $x(t) = \frac{1}{3}\cos(12t) + \frac{1}{3}\sin(12t)$. Find the maximum displacement.

Solution: Here: $x'(t) = -4 \sin(12t) + 4 \cos 12t$

$x'(t) = 0 \Rightarrow -4 \sin(2t) + 4 \cos(12t) = 0 \Rightarrow -\sin(12t) = -\cos(12t)$

$\Rightarrow \frac{\sin 12t}{\cos 12t} = \tan(12t) = 1 \Rightarrow 12t = \frac{\pi}{4} \Rightarrow t = \frac{\pi}{48}$

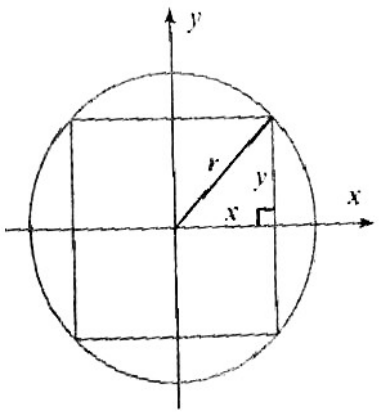
∴ The maximum displacement is

$$x\left(\frac{\pi}{48}\right) = \frac{1}{3}\cos\left(\frac{\pi}{4}\right) + \frac{1}{3}\sin\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{3}$$

32. Find the area of the largest rectangle that can be inscribed in a circle of:

a. $r = 10$ b. $r = 9$ c. $r = 2$

Solution: Let the dimension of rectangle length be $2x$ and height be $2y$. Consider the circle with center origin and radius r. Area of rectangle A = xy...*quantity optimized*, $y = \sqrt{r^2 - x^2}$



$\Rightarrow A(x) = x\sqrt{r^2 - x^2}$

a. $r = 10$

$\Rightarrow A(x) = x\sqrt{100 - x^2} = \sqrt{x^2}\sqrt{100 - x^2}$

$= \sqrt{100x^2 - x^4}$

Let $g(x) = 100x^2 - x^4 \Rightarrow g'(x) = 200x - 4x^3$, $g'(x) = 0$

$\Rightarrow 200x - 4x^3 = 4x(50 - x^2) = 0$ $x = 0$ and $x = 5\sqrt{2}$. Applying 2nd derivative: $g''(x) = 200 - 12x^2$

$\therefore$ when $x = 5\sqrt{2} \leftarrow$ maximum.

$\therefore$ The dimension of rectangle, length $= 2x = 10\sqrt{2}$ m and height $y = 2\sqrt{100-50} = 10\sqrt{2}$ m.

$\therefore$ Area of rectangle $= 4xy = 800$ m$^2 \leftarrow$maximum area.

b. $r = 9 \Rightarrow A(x) = \sqrt{81x^2 - x^4} \Rightarrow g(x) = 81x^2 - x^4$

$\Rightarrow g'(x) = 162x - 4x^3 \Rightarrow g'(x) = 0$ when $162x - 4x^3 = 0$

$\Rightarrow 2x(81 - 2x^2) \Rightarrow x = 0$ and $x = \dfrac{9}{\sqrt{2}}, y = \dfrac{9}{2}$

$\therefore$ length $2x = \dfrac{18}{\sqrt{2}}$ and height $2y = \dfrac{18}{\sqrt{2}}$

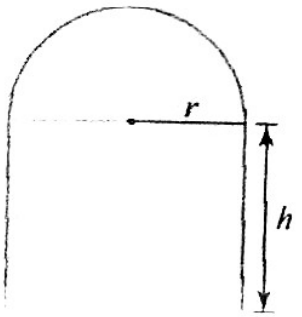
$\therefore$ Area of rectangle: $4xy = 4\left(\dfrac{18}{\sqrt{2}}\right)\left(\dfrac{18}{\sqrt{2}}\right) = 648\text{m}^2$ is maximum area.

c. Exercise left for you.

33. A window is constructed with top a semi circle and bottom is rectangle with total of perimeter of framing material is 16m. Find the dimension of the window to maximize the area.

Solution: Perimeter $= 2r + 2h + \pi r = 16\text{m} \Rightarrow h = 8 - r - \dfrac{1}{2}\pi r$

Area: $A = \dfrac{1}{2}\pi r^2 + 2rh \Rightarrow A(r) = \dfrac{1}{2}\pi r^2 + 2r\left(8 - r - \dfrac{1}{2}\pi r\right)$



$\Rightarrow A(r) = \dfrac{1}{2}\pi r^2 + 16r - 2r^2 - \pi r^2 = 16r - 2r^2 - \dfrac{1}{2}\pi r^2$

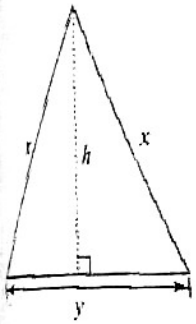
$\Rightarrow A'(r) = 16 - 4r - \pi r \Rightarrow A'(r) = 0 \Rightarrow 4r + \pi r = 16$

$\Rightarrow r = \dfrac{16}{4 + \pi}$ critical number and

$$h = 8 - r - \frac{1}{2}\pi r = 8 - r\left(1 + \frac{1}{2}\pi\right) = 8 - \frac{16}{4+\pi}\left(\frac{2+\pi}{2}\right) = \frac{16}{4+\pi}$$

34. An isosceles triangle has a perimeter of 16m. Find the length of the sides of a triangle that yield the maximum area.

Solution: Let the side length of x, x and y. Then,



Perimeter: $2x + y = 16$ and Area $= \frac{1}{2}hy$

$$h^2 + \frac{y^2}{4} = x^2$$

$$\Rightarrow h^2 = x^2 - \frac{y^2}{4} = \frac{4x^2 - y^2}{4} = \frac{(2x-y)(2x+y)}{4}$$

$$\Rightarrow h^2 = \frac{(16)(2x-y)}{4} \text{ put 16 in } 2x + y$$

$\Rightarrow h = 2\sqrt{2x-y} = 2\sqrt{(16-y)-y}$. Put $2x = 16 - y$

$\text{Area} = \frac{1}{2}hy \Rightarrow A(y) = \sqrt{16-2y}.y = \sqrt{16y^2 - 2y^3}$ ... $\sqrt{y^2} = y$

To optimize: let g $(y) = 16y^2 - 2y^3 \Rightarrow g'(y) = 32y - 6y^2$

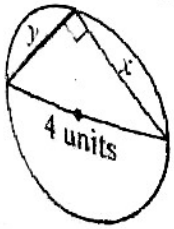
$\Rightarrow g'(y) = 0 \Rightarrow 2y(16 - 3y) = 0 \Rightarrow y = 0$ and $y = \frac{16}{3}$ *critical number*

$\Rightarrow x = \frac{16}{3}$ and $h = \frac{8\sqrt{3}}{3}$.

∴ The length of triangle should be $\frac{16}{3}, \frac{16}{3}$ and $\frac{16}{3}$ that yield maximum area which is $\frac{64}{9}\sqrt{3}\text{m}^2$.

35. What is the maximum area of a triangle which is inscribed in a circle of radius 2 units and one of its passing through the center of the circle (UEE)

A. 2 sq. unit B. 4π sq. unit C. 8 sq. unit D. 4 sq. unit



Solution: We have: $x^2 + y^2 = 16 \leftarrow$since one of side lies on center, $y = \sqrt{16 - x^2}$

Area of triangle $(A) = \frac{1}{2}xy$ … *to be optimized.*

$$A(x) = \frac{1}{2}x\sqrt{16-x^2} = \frac{1}{2}\sqrt{16x^2-x^4}$$

Note: $x = \sqrt{x^2}, x \geq 0$

To optimize: let g $(x) = 16x^2 - x^4$

$\Rightarrow g'(x) = 32x - 4x^3$

$g'(x)=0 \Rightarrow 4x(8-x^2) = 0 \Rightarrow x = 0$ and $x = \sqrt{8},\ y = \sqrt{8}$

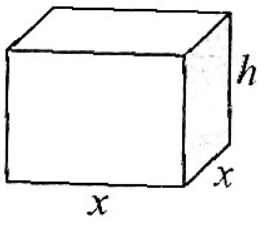
$\therefore$ Area (A) $= \frac{1}{2}\left(\sqrt{8}\right)\left(\sqrt{8}\right) = 4$ sq. unit.

Answer: D

36. An open top square base rectangular box container is to have a volume of $32cm^3$. What should be the height (h) of the box to obtain the minimum possible surface area (UEE)

A. $h = 2$ cm B. $h = 8$ cm C. $h = 4$ cm D. $h = 16$ cm

Solution: volume (v) = (base area) (height) $= x^2h = 32$, surface area $= x^2 + 4xh \leftarrow$ optimizing



$$A(x) = x^2 + 4x\left(\frac{32}{x^2}\right) = x^2 + \frac{128}{x}$$

$$\Rightarrow A'(x) = 2x - \frac{128}{x^2}$$

$\Rightarrow A'(x) = 0 \Rightarrow 2x^3 = 128 \Rightarrow x^3 = 64 \Rightarrow x = 4.$

$\therefore$ Therefore, $h = \frac{32}{x^2} = \frac{32}{16} = 2$ cm

$\therefore$ **Answer: A**

37. The total cost for ordering and storing x unit is given by C $(x) = 3x + \frac{30,000}{x}$ birr. How many units must be ordered to produce a minimum cost.

A. 50 B. 100 C. 300 D. 1000

Solution: $C'(x) = 3 - \frac{30,000}{x} \Rightarrow C'(x) = 0 \Rightarrow 3x^2 = 30,000$

$\Rightarrow x^2 = 10,000 \Rightarrow x = 100.$

Answer: B

38. A closed cylindrical can is to hold 1 ℓ (1000 cm^3) of liquid. How should we choose the height and radius to minimize the amount of material needed to manufacture the can.

Solution: Volume of cylinder $(v) = \pi r^2 h = 1000\ cm^3 \Rightarrow h = \dfrac{1000}{\pi r^2}$.

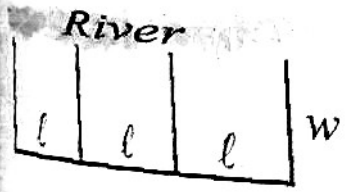
Surface: $2\pi r^2 + 2\pi rh$...*quantity to optimize*

$$\Rightarrow A(r) = 2\pi r^2 + 2\pi r\left(\frac{1000}{\pi r^2}\right) = 2\pi r^2 + \frac{2000}{r}$$

$$\Rightarrow A'(r) = 4\pi r - \frac{2000}{r^2} \Rightarrow A'(r) = 4\pi r - \frac{2000}{r^2}$$

$$A'(r) = 0 \Rightarrow r^3 = \frac{1000}{2\pi} \Rightarrow r = \frac{10}{\sqrt[3]{2\pi}}\ cm \text{ and } h = \frac{1000}{\pi r^2} = \frac{20}{\sqrt[3]{2\pi}}$$

39. A farmer wants to make three identical rectangular enclosure (same width and length) along a straight river as in the diagram show below. If he has a fencing material of 1,200 meters and the sides along the river need no fence what should be the dimension (length ℓ and width w) of each enclosure if the total area of the enclosure is to be maximized . . . (UEE)



A. $\ell = 100$ m, w = 225
B. $\ell = 300$ m, w = 75
C. $\ell = 200$ m, w = 200 m
D. $\ell = 200$ m, w = 150 m

Solution: Perimeter: $P = 3\ell + 4w = 1200 \Rightarrow w = 300 - \dfrac{3}{4}\ell$.

$$\text{Area (A)} = (3\ell)(w) \Rightarrow A(\ell) = 3\ell\left(300 - \frac{3}{4}\ell\right) = 900\ell - \frac{9}{4}\ell^2$$

$$\Rightarrow A'(\ell) = 900 - \frac{9}{2}\ell$$

$$A'(\ell) = 0 \Rightarrow 900 - \frac{9}{2}\ell = 0 \Rightarrow \ell = 200\text{m}$$

$$w = 300 - \frac{3}{4}(200) = 300 - 150 = 150 \text{ m}$$

∴ **Answer: D**

40. What are the value of a and b if $f(x) = \dfrac{ax^2 + b}{x^3}$ has extreme value at (3, 2).

A. $a = 6, b = 3$ C. $a = -9, b = 27$

B. $a = 9, b = -27$ D. $a = -6, b = 4$

Solution: We have $f(3) = 2 \Rightarrow \dfrac{9a + b}{27} = 2 \Rightarrow 9a + b = 54$ and

$$f'(x) = \frac{2ax(x^3) - 3x^2(ax^2 + b)}{x^6} = \frac{-ax^2 - 3b}{x^4}$$

$f'(3) = 0 \Rightarrow -9a - 3b = 0 \Rightarrow b = -3a$, so that, $\begin{cases} 9a + b = 54 \\ b = -3a \end{cases}$

$\Rightarrow 6a = 54 \Rightarrow a = 9, b = -27$

∴**Answer: B**

41. What are the value of a and b if $f(x) = ax^3 + bx^4$ has minimum value at (1, –1).

A. $a = 3, b = 4$ C. $a = -4, b = 3$

B. $a = 4, b = -3$ d D. $a = 3, b = -4$

Solution: Exercise left for you (**Answer: C**).

42. A rectangular page contained 24 square inches of print. The margins at the top and bottom of the page are to be $1\frac{1}{2}$ inch and the margins on the left and the right to be 1 inches. What dimensions of the page will produce the smallest area in inches.

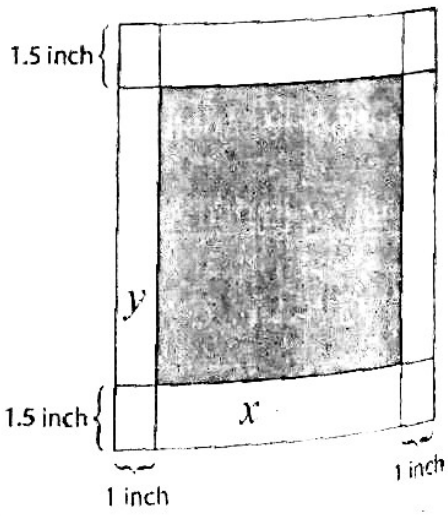
A. 6 by 4 B. 8 by 6 C. 9 by 6 D. 10 by 8

Solution: Let A be area minimized:

$A = (x + 2)(y + 3)$

$xy = 24$...*printed area inside margins*

$\Rightarrow y = \dfrac{24}{x}$



$\Rightarrow A(x) = (x+2)\left(\frac{24}{x}+3\right)$

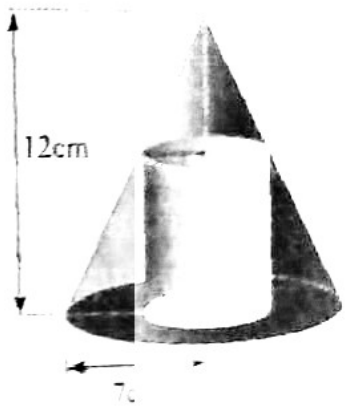
$= 30 - 3x + \frac{48}{x}$

$\Rightarrow A'(x) = 3 - \frac{48}{x^2} \Rightarrow A'(x) = 0 \Rightarrow 3x^2 - 48 = 0 \Rightarrow x^2 = 16 \Rightarrow x = 4$ and

$y = \frac{24}{x} = \frac{24}{4} = 6$.

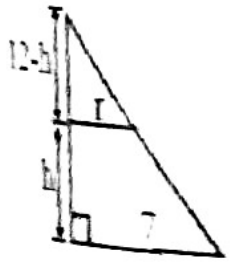
∴ Answer: 6

8. Find the radius and height of the cylinder of maximum volume that can be inscribed in a right circular cone of height 12 cm and radius of base 7 cm.



Solution: Let r = radius of cylinder, h = height of cylinder.

Then the volume of inscribed cylinder is given by



$V = \pi r^2 h$ *...to be optimized*

$\Rightarrow \frac{12-h}{r} = \frac{12}{7}$ *...using similarity pro*

triangle

$\Rightarrow h = 12 - \frac{12}{7}r$ *...solving for h*

$\Rightarrow V(r) = \pi r^2\left(12 - \frac{12}{7}r\right) = 12\pi r^2 - \frac{12}{7}\pi r^3, 0 \le r \le$

$\Rightarrow 12\pi r = 0$ and $2 - \frac{3}{7}r = 0 \Rightarrow r = \frac{14}{3}$ and $r = 0$.

Thus, $h = 12 - \frac{12}{7}\left(\frac{14}{3}\right) = 12 - 8 = 4$.

The inscribed cylinder has max. volume when

$h = 4$ cm.

Related Rates

$$\Rightarrow A'(x) = (x+2)\left(\frac{24}{x}+3\right)$$

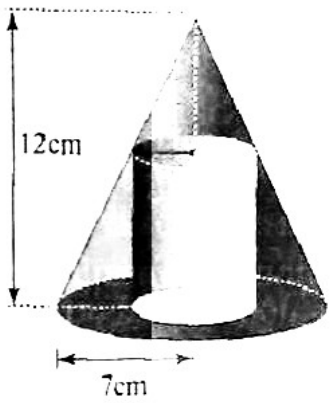
$$= 30 + 3x + \frac{48}{x}$$

$$\Rightarrow A'(x) = 3 - \frac{48}{x^2} \Rightarrow A'(x) = 0 \Rightarrow 3x^2 - 48 = 0 \Rightarrow x^2 = 16 \Rightarrow x = 4 \text{ and}$$

$$y = \frac{24}{x} = \frac{24}{4} = 6.$$

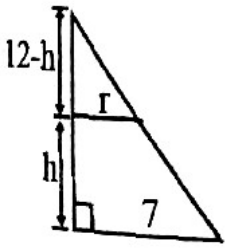
∴ Answer: C

43. Find the radius and height of the cylinder of maximum volume that can be inscribed in a right circular cone of height 12 cm and radius of base 7 cm.



Solution: Let r = radius of cylinder, h = height of cylinder.

Then the volume of inscribed cylinder is given by



$$v = \pi r^2 h \ldots \textit{to be optimized}$$

$$\Rightarrow \frac{12-h}{r} = \frac{12}{7} \ldots \textit{using similarity property of triangle}$$

$$\Rightarrow h = 12 - \frac{12}{7}r \ldots \textit{solving for } h$$

$$\Rightarrow v(r) = \pi r^2\left(12 - \frac{12}{7}r\right) = 12\pi r^2 - \frac{12}{7}\pi r^3, 0 \le r \le 7$$

$$\Rightarrow 12\pi r = 0 \text{ and } 2 - \frac{3}{7}r = 0 \Rightarrow r = \frac{14}{3} \text{ and } r = 0.$$

Thus, $h = 12 - \frac{12}{7}\left(\frac{14}{3}\right) = 12 - 8 = 4.$

∴ The inscribed cylinder has max. volume when $r = \frac{14}{3}$ cm and $h = 4$ cm.

Related Rates

Here we going to see the rates of change of two or more related variables that are changing with respect to time. **For example**: $y = x^3$, *differentiation both side with respect to time.*

$\Rightarrow \frac{dy}{dt} = 3x^2 \frac{dx}{dt}$ *...both rate of change with respect to time is related.*

Two Rates that are related

Illustrative Example

44. The radius of a circle is increasing uniformly at the rate of 3 cm/sec. Find the rate of change of the area when:

a. $r = 2\text{cm}$ b. $r = 6$ cm c. $r = 8$ cm

Solution: $A = \pi r^2$ differentiating with respect to time.

$$\Rightarrow \frac{dA}{dt} = 2\pi r \frac{dr}{dt}$$

a. $\frac{dA}{dt} = 2\pi(2\text{ cm})3\text{ cm/sec} = 12\pi\text{cm}^2/\text{sec}$

b. When $r = 6 \Rightarrow \frac{dA}{dt} = 2\pi(6)3\text{ cm/sec} = 36\ \pi\text{cm}^2/\text{sec}$

c. When $r = 8cm \Rightarrow \frac{dA}{dt} = 2\pi(8\text{ cm})(3\text{ cm/sec}) = 48\pi\text{ cm}^2/\text{sec}.$

45. The side of square sheet of metal is increasing at rate of 5 cm/min. Find the rate of change of the area when the is length x:

a. $x = 4$ cm b. $x = 9$ cm c. $x = 6$ cm

Solution: Area of square $A = x^2 \Rightarrow \frac{dA}{dt} = 2x\frac{dx}{dt}$:

a. $x = 4\text{ cm} \Rightarrow \frac{dA}{dt} = 2(4\text{ cm})\ 5\text{ cm/min} = 40\text{ cm}^2/\text{min}$

b. $x = 9\text{ cm} \Rightarrow \frac{dA}{dt} = 2(9)(5)\text{cm}^2/\text{min} = 90\text{ cm}^2/\text{min}$

c. $x = 6\text{ cm} \Rightarrow \frac{dA}{dt} = 2(6)(5)\text{cm}^2/\text{min} = 60\text{ cm}^2/\text{min}$

46. The side of an equilateral triangle is increasing at the rate of 4 cm/sec. Find the rate of change of the area when.

a. $x = \sqrt{3}$ cm b. $x = 8$ cm c. $x = 12$ cm

Solution: Let x be the side length of triangle:

Area (A) $= \frac{1}{2}x^2 \sin 60^\circ \Rightarrow A \Rightarrow \frac{\sqrt{3}}{4}x^2 \Rightarrow \frac{dA}{dt} = \frac{\sqrt{3}}{2}x\frac{dx}{dt}$, and

given $\frac{dx}{dt} = 4$ cm/sec

a. When $x = \sqrt{3} \Rightarrow \frac{dA}{dt} = \frac{\sqrt{3}}{2}\left(\sqrt{3}\right)(4) = 6 \text{ cm}^2/\text{sec}$

b. When $x = 8$ cm $\Rightarrow \frac{dA}{dt} = \frac{\sqrt{3}}{2}(8)(4) = 16 \text{ cm}^2/\text{sec}$

c. Exercise left for you.

47. A radius of spherical balloon is being inflated at the rate of 0.2 cm/sec. How fast is the volume increasing when the radius is

a. $r = 4$ cm b. $r = 3$ cm c. $r = 5$ cm

Solution: $v = \frac{4}{3}\pi r^3$ the volume of sphere

$$\Rightarrow \frac{dv}{dt} = 4\pi r^2 \frac{dr}{dt} \text{ and } \frac{dr}{dt} = 0.2 \text{ cm/sec}$$

a. When $r = 4$ cm $\Rightarrow \frac{dv}{dt} = 4\pi(16)(0.2) = 12.8\pi \text{ cm}^3/\text{sec}$

b. When $r = 3$ cm $\Rightarrow \frac{dv}{dt} = 4\pi(3)^2(0.2) = 7.2\pi \text{ cm}^3/\text{sec}$

c. Exercise left for you.

48. A spherical balloon is inflated at the rate of 4 cm^3/min. How fast is the radius of the balloon increasing when the radius:

a. $r = 3$cm b. $r = 6$ cm c. $r = 12$ cm

Solution: Volume of sphere, $v = \frac{4}{3}\pi r^3$

$$\Rightarrow \frac{dv}{dt} = 4\pi r^2 \frac{dr}{dt} \Rightarrow \frac{dr}{dt} = \frac{1}{4\pi r^2}\left(\frac{dv}{dt}\right) \text{ and } \frac{dv}{dt} = 4 \text{ cm}^3/\text{min}$$

a. When $r = 3$ cm $\Rightarrow \frac{dr}{dt} = \frac{1}{4\pi(9)}(4) = \frac{1}{9\pi}$ cm/min

b. When $r = 6$ cm $\Rightarrow \dfrac{dr}{dt} = \dfrac{1}{(4\pi)(36)}(4) = \dfrac{1}{36\pi}$ cm/min

c. Exercise left for you.

49. The edge of cube is increasing at the rate of 0.5 cm/sec. At what rate is the surface area increasing when the edge of the cube is 10 cm.

Solution: Let x be the edge of cube. Then surface area: $A = 6x^2$

$$\Rightarrow \frac{dA}{dt} = 12x\frac{dx}{dt} = 12(10)0.5 = 60 \text{ cm}^2/\text{sec}$$

50. The edge of cube is 5 cm long and volume is increasing at the rate of 2 cm^3/min. How fast the surface of cube is increasing. (UEE)

a. $\dfrac{2}{5}\text{cm}^2/\text{min}$ b. $\dfrac{8}{5}\text{cm}^2/\text{min}$ c. $\dfrac{25}{5}\text{cm}^2/\text{min}$ d. 8 cm^2/min

Solution: volume, $v = x^3$

$$\Rightarrow \frac{dv}{dt} = 3x^2\frac{dx}{dt} = 3x^2\frac{dx}{dt} \Rightarrow \frac{dx}{dt} \Rightarrow \frac{1}{3x^2}\left(\frac{dv}{dt}\right)$$

$\Rightarrow \dfrac{dx}{dt} = \dfrac{1}{3(5)^2}2 = \dfrac{2}{75}$ cm/min . And surface area, $A = 6x^2$

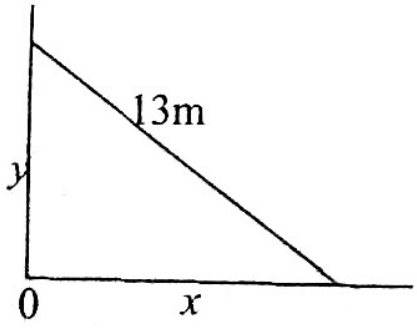
$$\Rightarrow \frac{dA}{di} = 12x\frac{dx}{dt} = 12(5)\left(\frac{2}{75}\right) = \frac{8}{5}\text{cm}^2/\text{min}$$

∴ **Answer: B**

51. A 13 m ladder is leaning against a wall of housel If the top of the ladder slices down the wall at a rate of 3 m/sec, how fast will the base (foot) be moving away from the wall when:

a. 5 m from the wall b. $4\sqrt{3}$ m from the wall

Solution: At any time t, let $\overline{OA} = x, \overline{OB} = y$ so that $x^2 + y^2 = 13^2$



$$\Rightarrow 2x\frac{dx}{dt} + 2y\frac{dy}{dt} - 0 \text{ and } \frac{dy}{dt} = 3 \text{ m sec}$$

a. When $x = 5$ m $\Rightarrow 5^2 + y^2 = 13$

$\Rightarrow y = 12$m

$$\Rightarrow 2(5)\frac{dx}{dt} + 2(12)(3) = 0$$

$$\Rightarrow \frac{dx}{dt} = \frac{-72}{10} = -7.2 \text{ m/sec}$$

$\therefore$ The base (lower end) is moving 7.2 m/sec.

b. When $x = 4\sqrt{3}$ m $\Rightarrow \left(4\sqrt{3}\right)^2 + y = 169 \Rightarrow y^2 = 169 - 48 = 21$

$\therefore y = 11\text{m} \Rightarrow \frac{dx}{dt} = \frac{-y}{x}\frac{dy}{dt} = \frac{-11}{4\sqrt{3}}(3) = \frac{-11\sqrt{3}}{4}$ m/sec

$\therefore$ The base (lower end is moving) $\frac{11\sqrt{3}}{4}$ m/sec

52. A point moves along the curve $3y = x^2$ in such a way that the y-value is increasing at the rate of 4 unit per second. At what rate is x changing for each of the following values when:

a. $x = \frac{1}{2}$ b. $x = 1$ c. $x = 8$

Solution: $3y = x^2 \Rightarrow 3\frac{dy}{dt} = 2x\frac{dx}{dt} \Rightarrow \frac{dx}{dt} = \frac{3}{2x}\frac{dy}{dt}$

a. When $x = \frac{1}{2} \Rightarrow \frac{dx}{dt} = \frac{3}{2}(2)(4) = 12$ unit/sec $\leftarrow \frac{dy}{dt} = 4\frac{\text{unit}}{\text{sec}}$

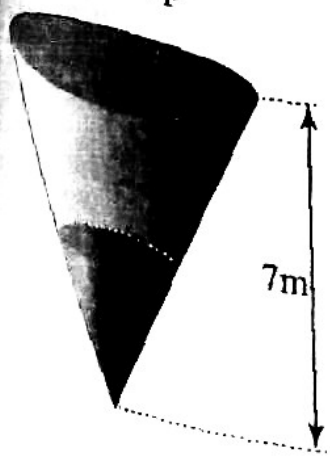
b. When $x = 1 \Rightarrow \frac{dx}{dt} = \frac{3}{2(1)}(4) = 6$ unit/sec

c. Exercise left for you.

53. A water tank in the shape of an inverted circular cone with top radius of 3m and height 7m. If water is flowing into the tank at the rate of 3m^3/min. Find the rate at which the water level is rising when the water is

a. 4m deep

b. 5m deep



Solution: Using similarity of triangle

$\frac{r}{h} = \frac{3}{7} \Rightarrow r = \frac{3}{7}h \Rightarrow h = \frac{7}{3}r$, Thus,

a.

$$\Rightarrow \frac{dv}{dt} = \frac{9}{49}\pi h^2 \frac{dh}{dt} \Rightarrow \frac{dh}{dt} = \frac{49}{9\pi h^2}\frac{dv}{dt}$$

$$\therefore \frac{dh}{dt} = \frac{49}{9\pi(4)^2}\left(\frac{3m^3}{\min}\right) = \frac{49}{48\pi}\text{ m/min}$$

b. When $h = 5m \Rightarrow \frac{dh}{dt} = \frac{49}{9\pi(25)}(3\text{m}^3/\text{min}) = \frac{49}{75\pi}\text{ m/min}$

54. A conical tank full of water is height 20 ft and radius 5 ft at the top. If the water is flowing out at the bottom at the rate of 2 ft^3/min. What is the rate at which the water level is falling when the water level is 16 ft above the bottom?

a. $\frac{8}{\pi}$ ft/min b. $\frac{1}{8\pi}$ ft/min c. $\frac{1}{18\pi}$ ft/min d. $\frac{18}{\pi}$ ft/min

Solution: h = height water level at any time (t)

$$\Rightarrow \frac{5}{20} = \frac{r}{h} \Rightarrow \frac{r}{h} = \frac{1}{4} \Rightarrow r = \frac{1}{4}h \text{ and } \frac{dv}{dt} = 2\text{ ft}^3/\text{min}$$

Volume of water level $v = \frac{1}{3}\pi r^2 h \left(\text{sincer} = \frac{1}{4}h\right)$

$$\Rightarrow v = \frac{1}{3}\pi\left(\frac{1}{4}h\right)^2 h = \frac{1}{48}\pi h^3 \Rightarrow \frac{dv}{dt} = \frac{1}{16}\pi h^2 \frac{dh}{dt}$$

$$\Rightarrow \frac{dh}{dt} = \frac{16}{\pi h^2}\left(\frac{dv}{dt}\right) = \frac{16}{\pi(16)^2}\left(2\text{ ft}^3/\text{min}\right) = \frac{1}{8\pi}\text{ ft/min}$$

∴ Answer: B

55. The radius of a right circular cylinder of height 10 cm increase at rate of 3 cm/min. What is the rate of change of the volume of cylinder when the radius is 8 cm?

A. 8 πcm^3/min C. 480πcm^3/min

B. 320πcm^3/min D. 640πcm^3/min

Solution: We have $\frac{dv}{dt} = 3$ cm/min , $v = \pi r^2 h$

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$\frac{dv}{dt} = 2\pi hr\frac{dr}{dt} = 20\text{ cm}\pi(8\text{ cm})(3\text{ cm/min}) = 480\pi\text{ cm}^3/\text{min}.$

∴ **Answer: C**

Solved Problems

56. Which of the following is the set of critical number of $f(x) = |x^2 - 4|$.

A. {0} B. {−2, 2} C. {−2, 0, 2} D. {0, 2}

Solution: $f(x) = |x^2 - 4| = \begin{cases} x^2 - 4, \text{ if } x < -2 \text{ and } x > 2 \\ 0 \text{ if } x = -2 \text{ and } x = 2 \\ 4 - x^2 \text{ if } -2 < x < 2 \end{cases}$

$\Rightarrow f'(x) = \begin{cases} 2x \text{ if } x < -2 \text{ and } x < 2 \\ \nexists \text{ if } x = -2 \text{ and } x = 2 \Rightarrow f'(x) = 0 \text{ when } x = 0. \\ -2x \text{ if } -2 < x < 2 \end{cases}$

and $f'(x)$ does not exist when $x = -2$ and $x = 2$.

∴{−2, 0, 2} is set of critical number. ∴**Answer: C**

57. Which one of the following is true about $f(x) = xe^{-x}$?

A. f has a local min at $x = 1$

B. f is strictly increasing on $[1, \infty)$

C. the graph of f is concave down on $(-\infty, 2)$

D. $(2, f(2))$ is not an inflection point

Solution: $f(x) = xe^{-x} \Rightarrow f'(x) = e^{-x}(1 - x)$. Then

i) $f'(x) > 0 \Rightarrow 1 - x > 0 \Rightarrow x < 1 \Rightarrow f$ is st. increasing on $[1, \infty) \Rightarrow x = 1$ is local max.

ii) $f''(x) = e^{-x}(x - 2) \Rightarrow f''(x) > 0 \Rightarrow x - 2 > 0 \Rightarrow x > 2$

$\Rightarrow$The graph of f is concave up ward on $(2, \infty)$ and

$f''(x) < 0 \Rightarrow x < 2 \Rightarrow$The graph concave down on $(-\infty, 2)$.

∴**Answer: C**

58. The largest interval where $f(x) = \frac{x}{x^2 + 1}$ is increasing.

A. $[0, \infty)$ B. $(-\infty, 1]$ C. $(1, \infty)$ D. $[-1, 1]$

Solution: $f'(x) = \frac{x^2 + 1 - 2x^2}{(x^2 + 1)^2} \Rightarrow \frac{1 - x^2}{(x^2 + 1)^2}$, then

$f'(x) > 0 \Rightarrow 1 - x^2 > 0 \Rightarrow -1 < x < 1 \Rightarrow f$ increasing on $[-1, 1]$.

∴ Answer: D

59. On which of the following intervals is the graph of $f(x)=\dfrac{x}{x^2+9}$ increasing…(UEE).

A. $(-\infty, \infty)$ B. $[-3, 3]$ C. $(-\infty, -3]$ D. $[3, \infty)$

Solution: $f'(x)=\dfrac{x^2+9-2x^2}{(x^2+9)^2}=\dfrac{9-x^2}{(x^2+9)^2}$, then

$f'(x)<0 \Rightarrow 9-x^2<0 \Rightarrow -3<x<3 \Rightarrow f$ is increasing on $[-3, 3]$.

∴ Answer: B

60. Find the maximum possible area of rectangle with diagonal.

a. $d=8$ b. $d=6$

Solution: $x^2+y^2=8^2 \Rightarrow y^2=64-x^2$

Area $A=xy \Rightarrow A=x\sqrt{64-x^2} \Rightarrow A(x)=\sqrt{64x^2-x^4}$

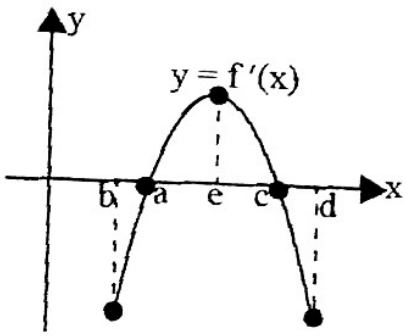
Let $g(x)=64x^2-x^4 \Rightarrow g'(x)\ 128x-4x^3$

$g'(x)=0$ when $x=0$ and $x=\sqrt{32} \Rightarrow y=\sqrt{32}$

∴ Area, $A=xy=\sqrt{32}\sqrt{32}=32$ is maximum area.

b. Exercise left for you.

61. Suppose $f(x)$ is differentiable at all $x \in (-\infty, \infty)$. Then from the graph of $y=f'(x)$ given below. **Find**:



i) The interval on which the function f is increasing and decreasing.

ii) The concave structure and inflection point.

iii) Local extreme value of f

Solution: Recall, 1st derivative test.

i) Since $f'(x)>0$, for all x in (a, c) for which f is increasing on $[a, c]$.

Also $f'(x)<0$ for all x in (b, a) and (c, d) so that f is strictly decreasing on interval $[b, a]$ and $[c, d]$.

ii) Because the graph of f' is increasing on interval (b, e) so that the graph of f is *concave up ward*.

$\Rightarrow f''(x)>0$ for all x in (b, e). **Similarly**, the graph of f' is decreasing on interval (e, d) so that the graph of f is *concave down ward* on (e, d).

$\Rightarrow f''(x)<0$ for all x in (e, d).

Inflection point $(e, f(e))$ since $f''(e)=0$.

iii) Since $f'(a) = 0$ and $f''(a) > 0 \Rightarrow (a, f(a))$ is local min. and $f'(c) = 0$ and $f''(c) < 0 \Rightarrow (c, f(c))$ is local maximum.

62. Water is flowing into vertical cylindrical tank of radius 2m at the rate of $3m^3$/min. How fast is the water level rising.

A. $\frac{2}{3\pi}$m/min B. $\frac{3}{2\pi}$m/min C. $\frac{3}{4\pi}$m/min D. $\frac{4}{3\pi}$m/min

Solution: $v = \pi r^2 h \Rightarrow v = 4\pi h$, since $r = 2 \Rightarrow \frac{dv}{dt} = 4\pi \frac{dh}{dt}$

$$\Rightarrow \frac{dh}{dt} = \frac{1}{4\pi}\left(\frac{dv}{dt}\right) = \frac{3}{4\pi} \text{ m/min since } \frac{dv}{dt} = 3m^3/m$$

63. The minimum area of triangle formed by the x-axis and the y-axis and a ling through the point (2, 1) is:

A. 4 unit sq. B. 6 unit sq. C. 8 units sq. D. 12 units sq.

Solution: Let $(x, 0)$, $(0, y)$ and (2, 1) point on the line. So that:

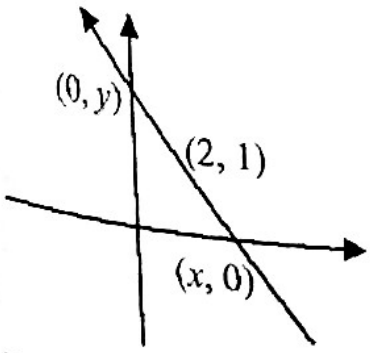
$$\frac{0-1}{x-2} = \frac{y-1}{0-2} \text{ ...from defn of slope of line}$$

$$\Rightarrow (x-2)(y-1) = 2 \Rightarrow xy - x - 2y = 0 \Rightarrow xy - x = 2y$$

$$\Rightarrow x(y-1) = 2y \Rightarrow x(y-1) = 2y \Rightarrow x = \frac{2y}{y-1}.$$

Then, we optimize area

$$A = \frac{1}{2}xy \Rightarrow A(y) = \frac{1}{2}\left(\frac{2y}{y-1}\right)y = \frac{y^2}{y-1} \Rightarrow A'(y) = \frac{y^2 - 2y}{(y-1)^2}$$



$A'(y) = 0 \Rightarrow y^2 - 2y = 0 \Rightarrow y = 2$ and $x = 4$.

$$\therefore \frac{1}{2}xy = \frac{1}{2}(4)(2) = 4 \text{ units sq.}$$

∴ Answer: A

Supplementary Exercise on Unit – 13

1. Find: i) critical number
   ii) absolute maximum value
   iii) absolute minimum value
   iv) the range of function, on the given interval

a. $f(x) = x^3 - 12x, [0, 4]$ e. $f(x) = x + 2\cos x, [0, 2\pi]$

b. $f(x) = 3x^{\frac{2}{3}} - 2x, [-1, 1]$ f. $f(x) = 3x^4 - 24x^2, [-3, 3]$

c. $f(x) = x^2 - 2x, [-1, 3]$ g. $f(x) = 17 - 15x + 9x^2 - x^3, [0, 6]$

d. $f(x) = \sqrt{12 - x - x^2}, [-4, 3]$ h. $f(x) = 3 - |x - 2|, [1, 4]$

2. Find: i) the interval in which f is st. increasing
   ii) the interval in which f is st. decreasing
   iii) local extreme value

a. $f(x) = 3x^3 - 36x + 8$ f. $f(x) = 2x^3 + 3x^2 - 12x + 8$

b. $f(x) = \dfrac{x}{x^2 + 16}$ g. $f(x) = x^3 e^{-x}$

c. $f(x) = x^3 - 3x^2 + 16x + 3$ h. $f(x) = \dfrac{3x + 5}{x - 7}$

3. Find the value of a, b and k for which the function will have extreme value at indicated point.

a. $f(x) = ax^3 - bx^2, \left(1, \dfrac{1}{2}\right)$ c. $f(x) = \dfrac{x^2 - 2x + 3k}{x + 2}$, at $x = 1$

b. $f(x) = x^4 + kx^3 + 10, x = 3$ d. $f(x) = \dfrac{x}{x^2 + k}, x = 3$

4. Determine the interval on which the graph has:
   i) concave upward ii) concave downward
   iii) locate the inflection , in each of the following.

a. $f(x) = x^5$ c. $f(x) = \dfrac{4}{x^2 + 1}$ e. $f(x) = x^4 e^{2x}$

b. $f(x) = x^6$ d. $f(x) = 5x^3 - 3x^2$

5. Find two positive number whose:
   a. sum 12, and product is maximum
   b. sum 15 and product is maximum
   c. product is 24 and whose sum is minimum
   d. sum: $x + y = 2$ and xy^2 is minimum or maximum

6. Find the length and width of rectangle with
   a. perimeter 18 m that maximize the area
   b. perimeter 120 m that maximize the area
   c. are $32m^2$ that minimize the perimeter
   d. area 81 m^2 that minimize the perimeter

7. Find the point on the graph of the function closest to the given point.

   a. $f(x)=x^2,(0,2)$ c. $f(x)=x^{\frac{1}{2}},(4,0)$

   b. $f(x)=4-x^2,(0,1)$

8. A cylindrical can is to be made to hold 1.5L of oil. Find the dimensions that will minimize the cost of the metal to manufacture the can.

9. The combined perimeter of a circle and a square is 24m. Find the dimension of the circle and square that product a minimum total area.

10. The radius r of circle is increasing at a rate of 2 cm/sec. Find the rate of change of the area when: a) $r=1$ cm b. $r=5$ cm

11. A snow ball is melting at the rate of 1 cm^3/min. If it is always a sphere,

    a. How fast is the radius changing when the snow ball is 4 cm.

    b. How fast is the surface area changing of the snow ball is 4 cm.

12. Find the dimension of the largest rectangle that can be inscribed in a semi circle of radius $r=6$ cm.

13. A ladder of length 1^{st}, leans against house wall. If the top of the ladder slides down ward at the rate of 3 ft/sec. Find the rate at which its lower end is moving when it is left from the wall.

14. A spherical balloon is inflated at the rate of 3 cm^3/min. How fast is the radius of the balloon increasing when the radius: a) $r=2$ cm b) $r=6$ cm

15. Water is poured into a conical paper cup at the rate of 2 m^3/min If the cup is 6 m tall and the top of the cup has radius of 2m, how fast does the water level rise when the water is 4m.

Unit Fourteen

Integral Calculus

There are two kinds of integration. These are indefinite integral and the definite integral. Integration as reverse process of differentiation.

- **Integration** is the process of finding the original function $f(x)$ from its derivative $f'(x)$ less commonly called the **anti-derivative** which is denoted by elongated 's' $\int$.
- $x^3+1 \xrightarrow[\text{anti-derivative}]{\text{derivative}} 3x^2$ donated by $\int 3x^2dx = x^3 + c$
- $x^3+4 \xrightarrow[\text{anti-derivative}]{\text{derivative}} 3x^2$ denoted by $\int 3x^2dx = x^3 + c$
- $F(x)+c \xrightarrow[\text{anti-derivative}]{\text{derivative}} f(x)$ denoted by

$$\int f(x)dx = F(x)+c$$

Thus, x^3, $x^3 + 1$, . . . and $x^3 + c$ are all *the set of anti-derivative* or *integration of* $3x^2$ write as $\int 3x^2dx = x^3 + c$. the anti-derivative of $f(x)$ is not unique. The expression $\int f(x)dx$ is called an indefinite integral, where

i) $\int f(x)dx$ is the integral of $f(x)$ w.r.t x

ii) $f(x)$ is called integrand

iii) dx is called the variable of integration

iv) If $F'(x) = f(x)$ then $\int f(x)dx = F(x) + c$

The process of anti-differentiation is commonly called **integration.**

Basic Integration Rule

A. **Integration of constant**

$$\int kdx = kx + c \Rightarrow \frac{d}{dx}(kx + c) = k \text{ and } \int odx = k \text{ Example:}$$

a. $\int 4dx = 4x + c$ b. $\int cdx = cx + c$

c. $\int \pi dx = \pi x + c$

B. Integration of power function

$$\int x^n dx = \frac{x^{n+1}}{n+1} + c \Rightarrow \frac{d}{dx}\left(\frac{x^{n+1}}{n+1} + c\right) = x^n .$$

Example:

a) $\int x^4 dx = \frac{x^5}{5} + c$ c) $\int x^{\pi} dx = \frac{x^{\pi+1}}{\pi+1} + c$

b) $\int x^{\frac{1}{2}} dx = \frac{2}{3} x^{\frac{3}{2}} + c$

C. Integration of $\left(\frac{1}{x}\right)$

$$\int \frac{dx}{x} = \ln|x| + c \text{ , and } \int \frac{kdx}{x} = k \ln|x| + c .$$

Example: a) $\int \frac{dx}{2x} = \frac{1}{2} \ln|x| + c$ b) $\int \frac{3}{x} dx = 3 \ln|x| + c$

D. Integral of exponential function

i) $\int e^x dx = e^x + c$ ii) $\int e^{mx} dx = \frac{e^{mx}}{m} + c$

iii) $\int a^x dx = \frac{a^x}{\ln a} + c$ iv) $\int a^{mx} dx = \frac{a^{mx}}{m \ln a} + c$

Integral of trigonometric function:

i) $\int \sin x dx = -\cos x + c$ vi) $\int \sec x \tan x dx = \sec x + c$

ii) $\int \cos x dx = \sin x + c$ vii) $\int \csc x \cot x dx = -\csc x + c$

iii) $\int \sec^2 x dx = \tan x + c$ iv) $\int \csc^2 x dx = \cot x + c$

viii) $\int \sin(kx) dx = -\frac{1}{k} \cos(kx) + c$

x) $\int \cos(kx) dx = \frac{1}{k} \sin(kx) + c$

Illustrative Example

2. Find: a) $\int \sin 4x dx$ b) $\int \sin(5x-2)dx$

c) $\int \sin(3-5x)dx$ d) $\int \cos(6x+1)dx$

e) $\int \sec^2(3x+1)$ f) $\int \sec(2x)\tan 2x dx$

g) $\int \csc^2(9x)dx$

Solution: a) $\int \sin 4x dx = \frac{-1}{4}\cos 4x + c$

b) $\int \sin(5x-2)dx = -\frac{1}{5}\cos(5x-2)+c$

c) $\int \sin(3-5x)dx = -\frac{1}{5}\cos(3-5x)+c$

d) $\int \cos(6x+1)dx = \frac{1}{6}\sin(6x+1)+c$

e) $\int \sec^2(3x+1) = \frac{1}{3}\tan(3x+1)+c$

f) $\int \sec(2x)\tan(2x)dx = \frac{\sec(2x)}{2}+c$

g) $\int \csc^2(9x)dx = -\frac{\cot 9x}{9}+c$

Properties of the indefinite integrals

i) $\int f'(x)dx = f(x)+c$ ii) $\frac{d}{dx}\int f(x)dx = f(x)$

iii) $\int f(x)\pm g(x)dx = \int f(x)dx \pm \int g(x)dx \leftarrow$ integrating term by term

Illustrative Example

3. Find: a) $\int \left(x^{-\frac{3}{4}} + x\right)dx$ b) $\int \sqrt[4]{x^5}dx$

c) $\int\left(x\sqrt{x}-4\sqrt{x}+\frac{5}{\sqrt{x}}\right)dx$

d) $\int\frac{(x-2)}{\sqrt{x^3}}\,dx$

e) $\int\frac{5-e^x}{e^{2x}}dx$

f) $\int\left(\frac{x^3+3x}{x^2}\right)dx$

g) $\int\frac{(e^{2x}+2e^x+1)}{e^x}$

h) $\int e^x(1-e^x)^2\,dx$

Solution: *Using the property it is the same as*:

a) $\int x^{\frac{-3}{4}}dx+\int xdx=\frac{x^{-\frac{3}{4}+1}}{\left(-\frac{3}{4}+1\right)}+\frac{x^2}{2}+4x^{\frac{1}{4}}+\frac{x^2}{2}+c$

b) $\int x^{\frac{5}{4}}dx-\frac{x^{\frac{5}{4}+1}}{\frac{5}{4}+1}+c=\frac{4}{9}x^{\frac{9}{4}}+c=\frac{4}{9}\sqrt[4]{x^9}+c$

c) $\int x^{\frac{3}{2}}dx-4\int x^{\frac{1}{2}}dx+5\int x^{-\frac{1}{2}}dx=\frac{2}{5}x^{\frac{5}{2}}-\frac{8}{3}x^{\frac{3}{2}}+10x^{\frac{1}{2}}+c$

d) $\int x^{-\frac{1}{2}}dx-\int 2x^{-\frac{3}{2}}dx=2x^{\frac{1}{2}}-4x^{-\frac{1}{2}}+c$

e) $\int 5e^{-2x}dx-\int e^{-x}dx=\frac{-5}{2}e^{-2x}+e^{-x}+c\leftarrow$ integrating term by term

f) $\int xdx+\int\frac{3}{x}dx=\frac{x^2}{2}+3\ln|x|+c\leftarrow$ integrating term by term

g) $\int e^x+\int 2dx+\int e^{-x}dx=e^x-e^{-x}+2x+c$

h) $\int e^x(1-2e^x+e^x)dx=\int e^x dx-\int e^{2x}dx=e^x-\frac{e^{2x}}{2}+c$

Finding a function from its slope function and a point (Initial value problem)

Illustrative Example

4. Find the curve (function) whose slope at the point (x, y) is $3x^2 + 4$ if the curve is passing through the point $(-1, 3)$.

Solution: We have slope function: $\frac{dy}{dx} = 3x^2 + 4$

$\Rightarrow dy = (3x^2 + 4)\,dx \Rightarrow$

Since (-1, 3) on the curve, so that $3 = (-1)^3 + 4(-1) + c$

$\Rightarrow c + 3 + 5 = 8 \Rightarrow y = x^3 + 4x + 8 \leftarrow$ the original function

5. Find $f(x)$ satisfying the given function:

a. $f'(x) = x + \sin x$ and $f(0) = 4$

b. $f'(x) = x^2 + \frac{1}{x}$ and $f(1) = -\frac{5}{3}$

Solution: $f(x) = \int f'(x)dx = \int (x + \sin x)dx = x^2 - \cos x + c$

$\Rightarrow f(0) = 0 - \cos 0 + c = 4 \Rightarrow c = 5 \Rightarrow f(x) = x^2 - \cos x + 5$

b) $f(x) = \int f'(x)dx = \int \left(x^2 + \frac{1}{x}\right) dx = \frac{x^3}{3} + \ln|x| + c$

$\Rightarrow f(1) = \frac{1}{3} + 0 + c = -\frac{5}{3} \Rightarrow c = \frac{-5-1}{3} = \frac{-6}{3} = -2$

$\therefore f(x) = \frac{x^3}{3} + \ln|x| - 2$

Techniques of integration

There are four. These are:

- Integration by substitution .
- Integration by partial fraction.
- Integration by parts.
- Integration by tabular method.

A. **Integration by substitution**

The method of substitution *orginated* by examining the rule of derivative backward.

Remember that: $\frac{d}{dx}[F(g(x))] = F'(g(x))g'(x)$

$\Rightarrow \int F'(g(x))g'(x)dx = F(g(x)) + c$

Let $u = g(x) \Rightarrow du = g'(x)\,dx$

$\Rightarrow \int F'(u)du = F(u) + c \leftarrow$ It is called u-substitution.

Illustrative Example

6. Find:

a. $\int 6\sqrt[4]{4-3x}\,dx$ b. $\int \frac{3}{\sqrt{4-5x}}dx$ c. $\int (2x-3)^6\,dx$

d. $\int \frac{3x^2+4x}{x^3+2x^3+7}\,dx$ e. $\int \frac{x\,dx}{\sqrt{3x^2+1}}$ f. $\int \frac{x+2}{\sqrt{x-2}}\,dx$

g. $3x^2\sqrt{6x^3+11}\,dx$ h. $\int x^2\sqrt{x-1}\,dx$ i. $\int x^5\sqrt{x^2-1}\,dx$

j. $\int \sqrt{2x-3}\,dx$ k. $\sqrt[3]{1-2x}$ l. $\int (\pi-5x)^e\,dx$

Solution: a) Let $u = 4 - 3x \Rightarrow du = -3dx \Rightarrow dx = \frac{-du}{3}$

$$\therefore \int 6\sqrt[4]{4-3x}\,dx = -\int 6\sqrt[4]{u}\,\frac{du}{3} = -2\int u^{\frac{1}{4}}du$$

$$= \frac{-8}{3}[u]^{\frac{5}{4}} + c = \frac{-8}{5}\left(\sqrt[4]{4-3x}\right)^5 + c$$

b) Let $u = 4 - 5x \Rightarrow du = -5dx \Rightarrow dx = \frac{-du}{5}$

$$\therefore \int \frac{3}{\sqrt[6]{4-5x}}\,dx = \frac{-1}{5}\int \frac{3}{\sqrt[6]{u}}\,du$$

$$= -\frac{3}{5}\int u^{\frac{-1}{6}}du = \frac{-18}{25}[u]^{\frac{5}{6}} + c = -\frac{18}{25}\left(\sqrt[6]{4-5x}\right)^5 + c$$

c) Let $u = 2x - 3 \Rightarrow du = 2dx \Rightarrow dx = \frac{du}{2}$

$$\therefore \int (2x-3)^6\,dx = \int u^6\,\frac{du}{2} = \frac{1}{2}\left[\frac{u^7}{7}\right] = \frac{1}{14}(2x-3)^7 + c$$

d) Let $u = x^3 + 2x^2 + 7 \Rightarrow du = (3x^2 + 4x)\,dx$

$$\therefore \int \frac{(3x^2+4x)dx}{x^3+2x^2+7} = \int \frac{du}{u} = \ln|u| + c = \ln|x^3+2x^2+7| + c$$

e) Let $u = 3x^2 + 1 \Rightarrow du = 6x\,dx \Rightarrow x\,dx = \dfrac{du}{6}$

$$\therefore \int \frac{x\,dx}{\sqrt{3x^2+1}} = \frac{1}{6}\int \frac{du}{\sqrt{u}} = \frac{1}{6}\int u^{-\frac{1}{2}}du = \frac{1}{3}u^{\frac{1}{2}} + c$$

$$= \frac{1}{3}\sqrt{3x^2+1} + c$$

f) Let $u = \sqrt{x-2} \Rightarrow u^2 = x-2 \Rightarrow x = u^2 + 2 = dx = 2u\,du$

$$\therefore \int \frac{x+2}{\sqrt{x-2}}dx = 2\int \frac{(u^2+2)+2}{u}u\,du = 2\int (u^2+4)du$$

$$= 2\left(\frac{u^3}{3} + 4u\right) + c = \frac{2\sqrt{x-2}^3}{3} + 8\sqrt{x-2} + c$$

g) Let $u = 6x^3 + 11 \Rightarrow du = 18x^2\,dx \Rightarrow x^2dx = \dfrac{du}{18}$

$$\therefore \int 3x^2\sqrt{6x^3+11}\,dx = \frac{1}{6}\int \sqrt{u}\,du = \frac{1}{9}u^{\frac{3}{2}} + c = \frac{1}{9}\left(\sqrt{6x^3+11}\right)^3 + c$$

h) Let $u = \sqrt{x-1} \Rightarrow u^2 = x-1 \Rightarrow x = u^2 + 1 \Rightarrow dx = 2u\,du$

$$\therefore \int x\sqrt{x-1}\,dx = \int (u^2+1)u(2u\,du)$$

$$= 2\int (u^4 + u^2)\,du = \frac{2}{5}u^5 + \frac{2}{3}u^3 + c$$

$$= \frac{2}{5}\left(\sqrt{x-1}\right)^5 + \frac{2}{3}\left(\sqrt{x-1}\right)^3 + c$$

i) Let $u = \sqrt{x^2+1} \Rightarrow u^2 = x^2 - 1 \Rightarrow x^2 = u^2 + 1 \Rightarrow 2x\,dx = 2u\,du$

$$\therefore \int (x^2)^2\sqrt{x^2-1}\,x\,dx = \int (u^4 + 2u + 1)u^2\,du = \frac{u^7}{7} + \frac{2u^4}{4} + \frac{u^3}{3} + c$$

$$= \frac{\sqrt{x^2+1}^7}{7} + \frac{1}{2}\sqrt{x^2-1}^4 + \frac{\sqrt{x^2-1}^3}{3} + c$$

Let $u = 2x-3 \Rightarrow du = 2dx \Rightarrow dx = \dfrac{du}{2}$

$$\therefore \int \sqrt{2x-3}\, dx = \int \sqrt{u}\, \frac{du}{2} = \frac{1}{3}u^{\frac{3}{2}} + c = \frac{1}{3}\sqrt{2x-3} + c$$

Exercise left for you.

Let $u = \pi - 5x \Rightarrow du = -5dx \Rightarrow dx = \dfrac{-du}{5}$

$$\therefore \int (\pi - 5x)^e\, dx = \int u^e\, \frac{du}{5} = \frac{-u^{e+1}}{5e+5} = \frac{-(\pi-5x)^{e+1}}{5e+5} + c$$

7. Find: a) $\int (x^2+2x+1)^6\, dx$ b) $\int (9x^2-12x+4)^5$

c) $\int (25x^2-10x+1)\, dx$

Solution: a) $x^2 + 2x + 1 = (x+1)^2 \Rightarrow$ let $u = x+1 \Rightarrow du = dx$

$$\therefore \int (x^2+2x+1)^6\, dx = \int (x+1)^{12}\, dx$$

$$= \int u^{12} du = \frac{u^{13}}{13} + c = \frac{(x+1)^{13}}{13} + c$$

b) $9x^2 - 12x + 4 \Rightarrow (3x-2)^2 \Rightarrow u = 3x-2 \Rightarrow du = 3\, dx$

$$\therefore \int (9x^2-12x+4)^5\, dx = \int (3x-2)^{10}\, dx = \int u^{10}\, \frac{du}{3}$$

$$= \frac{u^{11}}{33} + c = \frac{(3x-2)^{11}}{33} + c$$

c) $25x^2 - 10x + 1 = (5x-1)^2 \Rightarrow u = 5x-1 \Rightarrow du = 5dx$

$$\therefore \int (25x^2-10x+1)^8\, dx = \int (5x-1)^{16}\, x$$

$$= \int u^{16}\, \frac{du}{5} = \frac{u^{17}}{85} + c = \frac{(5x-1)^{17}}{85} + c$$

Substitution involving exponential

8. Find:

a. $\int \dfrac{dx}{1+e^{-x}}$ b. $\int \dfrac{e^x - e^{-x}}{e^x + e^{-x}} dx$ c. $\int \dfrac{e^{\frac{1}{x}}}{x^2} dx$

d. $\int \frac{e^{\sqrt{x}}}{\sqrt{x}}\,dx$ e. $\int e^x\sqrt{e^x+1}\,dx$

f. $\int 2^x(1-2^{x+1})^6\,dx$ g. $\int 3^{x\sqrt{x}}\sqrt{x}\,dx$

Solution: $\int \frac{dx}{1+e^{-x}} = \int \frac{e^x dx}{1+e^x}$, let $u = 1 + e^x \Rightarrow du = e^x dx$

$$\therefore \int \frac{du}{u} = \ln|u| + c = \ln|1+e^x| + c$$

b) Let $u = e^x + e^{-x} \Rightarrow du = (e^x - e^{-x})\,dx$

$$\Rightarrow \int \frac{(e^x - e^{-x})}{e^x + e^{-x}}\,dx = \int \frac{du}{u} = \ln|u| + c = \ln(e^x + e^{-x}) + c$$

c) Let $u = \frac{1}{x} \Rightarrow du = \frac{-1}{x^2}dx = \frac{dx}{x^2} = -du$

$$\Rightarrow \int \frac{e^{\frac{1}{x}}}{x^2}dx = -\int e^u du = -e^u + c = -e^{\frac{1}{x}} + c$$

d) Let $u = \sqrt{x} \Rightarrow du = \frac{dx}{\sqrt{x}}$

$$\therefore \int \frac{e^{\sqrt{x}}}{\sqrt{x}}dx = \int e^u du = e^u + c = e^{\sqrt{x}} + c$$

e) Let $u = e^x + 1 \Rightarrow du = e^x dx$

$$\therefore \int e^x\sqrt{e^x+1}\,dx = \int \sqrt{u}\,du = \frac{2}{3}u^{\frac{3}{2}} + c = \frac{2}{3}\sqrt{e^x+1}^3 + c$$

f) Let $u = 1 - 2^{x+1} \Rightarrow du = -2^x(2\ln 2)\,dx \Rightarrow \frac{-du}{2\ln 2} = 2^x dx$

$$\therefore \int (1-2^{x+1})^6(2^x dx) = -\int u^6 \frac{du}{2\ln 2} = -\left(\frac{u^7}{7\ln 4}\right) + c$$

$$= \frac{(1-2^{x+1})^7}{7\ln 4} + c$$

g) Let $u = x\sqrt{x} = x^{\frac{3}{2}} \Rightarrow du = \frac{3}{2}x^{\frac{1}{2}}dx = \frac{3}{2}\sqrt{x}\,dx$

$$\Rightarrow \int 3^{x\sqrt{x}}\sqrt{x}\,dx = \frac{2}{3}\int 3^u\,du = \frac{2(3^u)}{3\ln 3} = \frac{2(3^{x\sqrt{x}})}{3\ln 3} + c$$

Substitution involving natural logarithm

Illustrative Example

9. Find:

a. $\int \frac{\ln x}{x}dx$ b. $\int \frac{(\ln x)^3}{x}dx$ c. $\int \frac{(\ln x)}{x}dx$ d. $\int \frac{dx}{x(\ln x)^2}$

Solution: Here a) Let $u = \ln x \Rightarrow du = \frac{1}{x}dx$

$$\therefore \int \frac{\ln x}{x}dx = \int u\,du = \frac{u^2}{2} + c = \frac{(\ln x)^2}{2} + c$$

b) Let $u = \ln x \Rightarrow du = \frac{1}{x}dx$

$$\therefore \int \frac{(\ln x)^3}{x}dx = \int u^3 du = \frac{u^4}{4} + c = \frac{(\ln x)^4}{4} + c$$

c) Let $u = \ln x \Rightarrow du = \frac{dx}{x}$

$$\therefore \int \frac{(\ln x)^n}{x}dx = \int u^n du = \frac{u^{n+1}}{n+1} + c = \frac{(\ln x)^{n+1}}{n+1} + c$$

d) Let $u = \ln x \Rightarrow du = \frac{dx}{x}$

$$\therefore \int \frac{dx}{x(\ln x)^2} = \int \frac{du}{u^2} = \int u^{-2}du = -\frac{1}{u} = c = -\frac{1}{\ln x} + c$$

Note: **Substitution involving trigonometric**

- $\int \tan x\,dx = -\ln|\cos x| + c$, use $u = \cos x$
- $\int \cot x\,dx = \ln|\sin x| + c$, use $u = \sin x$

- $\int \sin kx\,dx = -\dfrac{\cos kx}{k} + c$, use $u = kx$
- $\int \cos kx\,dx = \dfrac{\sin kx}{k} + c$, use $u = kx$

Illustrative Example

10. Find:

a. $\int \sin(4-5x)\,dx$ b. $\int \sin x\,dx$ c. $\int e^x \sin(e^x)dx$

d. $\int \sin^3 x \cos x\,dx$ e. $\int \cos^n x \sin x\,dx$ f. $\int \dfrac{1}{x^2}\cos\left(\dfrac{1}{x}\right) - 1\,dx$

g. $\int \dfrac{e^x\,dx}{\cos^2(e^x+3)}$ h. $\int \cos^4 x \sin x\,dx$ i. $\int x^2 \sin(x^3)dx$

Solution: a) Let $u = 4 - 5x \Rightarrow du = -5dx \Rightarrow dx = -\dfrac{du}{5}$

$$\int \sin(4-5x)dx = -\int \sin u \frac{du}{5} = \frac{\cos u}{5} + c = \frac{\cos(4-5x)}{5} + c$$

b) Let $u = 1 + 3\cos x \Rightarrow du = -3\sin x\,dx \Rightarrow \sin x\,dx = -\dfrac{du}{3}$

$$\therefore \int \frac{\sin x}{1+3\cos x} = -\int \frac{du}{3u} = -\frac{1}{3}\ln|u| + c = -\frac{1}{3}\ln|1+\cos x| + c$$

c) Let $u = e^x \Rightarrow du = e^x dx \Rightarrow \int \sin u\,du = -\cos u + c = -\cos(e^x) + c$

d) Let $u = \sin x \Rightarrow du = \cos x\,dx \Rightarrow \int u^3 du = \dfrac{u^4}{4} + c = \dfrac{\cos^4 x}{4} + c$

e) Let $u = \cos x \Rightarrow du = -\sin x\,dx$

$$\therefore \int \cos^n x \sin x\,dx = \int u^n du = -\frac{u^{n+1}}{n+1} + c = \frac{\cos^{n+1} x}{n+1} + c$$

f) Let $u = \dfrac{1}{x} - 1 \Rightarrow du = -\dfrac{1}{x^2}dx \Rightarrow \dfrac{dx}{x^2} = -\dfrac{du}{0}$

$$\therefore \int \frac{1}{x^2}\cos\left(\frac{1}{x} - 1\right)dx = -\int \cos u\ du = -\sin u + c = -\sin\left(\frac{1}{x}\right) + c$$

g) Let $u = e^x + 3 \Rightarrow du = e^x dx$

$$\therefore \int \frac{e^x dx}{\cos^2(e^x+3)} = \int \frac{du}{\cos^2 u} = \int \sec^2 u du = \tan u + c$$

$$= \tan(e^x + 3) + 3$$

h) Exercise left for you. Use, $u = \cos x$

i) Let $u = x^3 \Rightarrow du = 3x^2 dx \Rightarrow x^2 dx = \frac{du}{3}$

The integrals of $\sin^2 x$ and $\cos^2 x$

$\sin^2 x$ and $\cos^2 x$ arise frequently in application of integrals.

Recall: i) $\sin^2 x = \frac{1}{2}(1 - \cos 2x)$

ii) $\cos^2 x = \frac{1}{2}(1 + \cos 2x)$

11. Find:

a. $\int \sin^2 x\, dx$ b. $\int \cos^2 x dx$ c. $\int \sin^2 3x\, dx$

d. $\int \sin^3 x dx$ e. $\int \cos^5 x\, dx$ f. $\int \sin^4 x\, dx$

Solution: Use identity property of trigonometry.

a) $\int \sin^2 x dx = \int \frac{(1-\cos 2x)}{2} dx = \frac{1}{2}x - \frac{\sin 2x}{4} + c$

b) $\int \cos^2 x dx = \int \frac{1+\cos(2x)}{2} dx = \frac{1}{2}x + \frac{\sin 2x}{4} + c$

c) $\int \sin^2 3x = \int \frac{1-\cos 6x}{2} dx = \frac{1}{2}x - \frac{\sin 6x}{12} + c$

d) $\int \sin^3 x dx = \int \sin^2 x \sin x\, dx = \int (1-\cos^2 x)\sin x\, dx$

Let $u = \cos x \Rightarrow du = -\sin x\, dx$ and $\int \sin^3 x dx = \int (1-\cos^2 x)\sin x\, dx$

$$\Rightarrow -\int (1-u^2)du = -u + \frac{u^3}{3} + c = -\cos x + \frac{\cos^3 x}{3} + c$$

12. $\int \frac{dx}{x+\sqrt{x}}$ is equal to:

A. $\frac{\ln(x+1)}{2}+c$ C. $\frac{\ln\left(1+\sqrt{x}\right)+c}{2}$

B. $2\ln\left(\sqrt{x}+1\right)+c$ D. $\ln\sqrt{x+\sqrt{x}}+c$

Solution: Let $u=\sqrt{x}\Rightarrow u^2=x\Rightarrow dx=2udu$

$$\therefore\int\frac{dx}{x+\sqrt{x}}=\int\frac{2udu}{u^2+u}=2\int\frac{du}{u+1}$$

$$=2\ln|u+1|+c=2\ln\left(\sqrt{x}+1\right)+c$$ **∴Answer: B**

13. $\int\frac{\log_3^x}{x}dx$ is equal to:

A. $(\ln 3)(\ln x)^2+c$ B. $\frac{(\ln x)^2}{\ln 3}+c$ C. $\left(\frac{\ln x}{\ln 3}\right)^2+c$ D. $(\ln 3)\ln x+c$

Solution: $\log_3^x=\frac{\log_e^x}{\log_e^3}=\frac{\ln x}{\ln 3}$ thus $\int\frac{\log_3^x}{x}dx=\int\frac{\ln x}{x\ln 3}dx$

Let $u=\ln x\Rightarrow du=\frac{1}{x}dx$

$$\therefore\int\frac{udu}{\ln 3}=\frac{1}{\ln 3}u^2+c=\frac{(\ln x)^2}{\ln 3}+c$$ **∴ Answer: B**

14. $\int 2^x\sqrt{1+2^x}\,dx$ is equal to:

A. $\frac{2}{3\ln 2}\left(\sqrt{1+2^x}\right)^3+c$ C. $\frac{3}{2\ln 2}\left(\sqrt{1+2^x}\right)^3+c$

B. $\frac{2\ln 2}{3}(1+2^x)^3$ D. None

Solution: $u=1+2^x\Rightarrow du=2^x\ln 2\,dx\Rightarrow\frac{du}{\ln 2}=2^x dx$

$$\Rightarrow\int\sqrt{u}\frac{du}{\ln 2}=\frac{1}{\ln 2}\left(\frac{2}{3}\right)u^{\frac{3}{2}}+c=\frac{2}{3\ln 2}\left(\sqrt{1+2^x}\right)+c\therefore$$

Answer: A

15. $\int \frac{2^{1-3x}}{3^{x+1}} dx$ is equal to:

A. $\frac{2(24)^x}{3\ln 24} + c$ C. $\frac{-2(24)^x}{3\ln 24} + c$

B. $\frac{-2(24)^{-x}}{3\ln 24} + c$ D. $\frac{2\ln 24(24)^{-x}}{3} + c$

Solution: $\frac{2^{1-3x}}{3^{x+1}} = \frac{2^1}{2^{3x}} \cdot \frac{1}{3.3^x} = \frac{2}{3} \cdot \frac{1}{(24)^x} = \frac{2}{3}.(24)^{-x}$

$\therefore \int \frac{2}{3}(24)^{-x} dx = \frac{-2(24)^{-x}}{3\ln 24} + c$ ∴ **Answer: B**

16. $\int \frac{x^2+2x+3}{x^3+3x^2+9x} dx$ is equal to:

A. $\frac{1}{3}\ln|x^2+2x+3|+c$ B. $\frac{1}{3}\ln|x^3+3x^2+9x|+c$

C. $\ln|x+3| - \ln|x+1| + c$

Solution: Let $u = x^3 + 3x^2 + 9x \Rightarrow du = (3x^2 + 6x + 9)\,dx$

$\therefore \int \frac{x^2+2x+3}{x^3+3x^2+9x} = \int \frac{du}{3u} = \frac{1}{3}\ln|u| + c = \frac{1}{3}\ln|x^3+3x^2+9x| + c$

∴ **Answer: B**

17. $\int \frac{dx}{x\log_{10}^x}$ is equal to:

A. $\ln 10 \ln|\ln x| + c$ B. $\frac{\ln|\ln x|}{\ln 10} + c$ C. $\frac{\ln x}{\ln 10} + c$ D. $(\ln 10)\ln x + c$

Solution:

$u = \log_{10}^x = \frac{\log_e^x}{\log_e^{10}} = \frac{\ln x}{\ln 10} \Rightarrow du = \frac{dx}{x\ln 10} = (\ln 10)du = \frac{dx}{x}$

$\therefore \int \frac{dx}{x\log_{10}^x} = \int \frac{(\ln 10)du}{u} = (\ln 10)\ln|u| + c = (\ln 10)\ln|\ln x| + c \therefore$

Answer: A

18. $\int \cos^3 x\,dx$ is equal to . . .

A. $\cos x - \frac{\cos^3 x}{3} + c$ C. $\sin x - \frac{\cos^2 x}{2} + c$

B. $\sin x - \frac{\sin^3 x}{3} + c$ D. None

Solution: $\int \cos^3 x\,dx = \int \cos^2 x \cos x\,dx = \int (1 - \sin^2 x)\cos x\,dx$

let $u = \sin x \Rightarrow du = \cos x\,dx \Rightarrow$

$$\int (1-u^2)du = u - \frac{u^3}{3} + c = \sin x - \frac{\sin^3 x}{3} + c$$

B. **Integration by partial fraction**

This section examines a method for decomposing rational function into simpler rational function to apply the basic integration rule. This method is called **method of partial fraction**.

Linear factor rule:

a. **Integration of distinct linear fraction**

$$\frac{N(x)}{(ax+b)(cx+d)} = \frac{A}{ax+b} + \frac{B}{cx+d} \Leftrightarrow \mathrm{A}\,(cx+d) + \mathrm{B}\,(ax+b) = \mathrm{N}\,(x)$$

Illustrative Example

19. Find:

a. $\int \frac{dx}{x(x+2)}$ b. $\int \frac{dx}{x^2-5x+6}$ c. $\int \frac{(4x+1)}{x^2-3x+2}dx$

d. $\int \frac{x}{x^2+4x+3}dx$ e. $\int \frac{12}{x^2-9}dx$ f. $\int \frac{4x+5}{x^2+x-2}$

Solution: a) $\frac{1}{x(x+2)} = \frac{A}{x} + \frac{B}{x+2} \Rightarrow A(x+2) + Bx = 1,$

multiplying both side by $x\,(x+2)$. **To find for A**: set $x = 0 \Rightarrow$

$2A=1 \Rightarrow A = \frac{1}{2}$. **To find for B**: set $x = -2 \Rightarrow -2B=1,\ B = -\frac{1}{2}$

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$$\therefore \int \frac{dx}{x(x+2)} = \frac{1}{2}\int \frac{dx}{x} - \frac{1}{2}\int \frac{dx}{x+2}$$

$$= \frac{1}{2}\ln|x| - \frac{1}{2}\ln|x+2| + c = \frac{1}{2}\ln\left|\frac{x}{x+2}\right| + c$$

b) $$\frac{1}{x^2-5x+6} = \frac{1}{(x-2)(x-3)} = \frac{A}{x-2} + \frac{B}{x-3}$$

$$\Leftrightarrow A(x-3) + B(x-2) = 1$$

To find for A: set $x = 2 \Rightarrow A(2-3) + 0 = 1 \Rightarrow A = -1$.

To find for B: set $x = 3 \Rightarrow 0 + B(3-2) = 1 \Rightarrow B = 1$

$$\therefore \int \frac{dx}{(x-2)(x-3)} = \int \frac{-dx}{x-2} + \int \frac{dx}{x-3}$$

$$= -\ln|x+2| + \ln|x-3| + c = \ln\left|\frac{x-3}{x-2}\right| + c$$

c) $$\frac{4x+1}{x^2-3x+2} = \frac{4x+1}{(x-2)(x-1)} = \frac{A}{x-2} + \frac{B}{x-1}$$

$\Leftrightarrow A(x-1) + B(x+2) = 4x + 1$.

To find for A: let $x = 2 \Rightarrow A(2-1) + 0 = 4(2) + 1 \Rightarrow A = 9$.

To find for B: set $x = 1 \Rightarrow 0 + -B = 4 + 1 \Rightarrow B = -5$

$$\therefore \int \frac{4x+1}{(x-2)(x-1)} = 9\int \frac{dx}{x-2} + -5\int \frac{dx}{x-1} = 9\ln|x-2 \quad -5\ln$$

$$\int |x-1| + c = \frac{9}{5}\ln\left|\frac{x-2}{x-1}\right| + c$$

d) $$\frac{x}{(x+3)(x+1)} = \frac{A}{x+3} + \frac{B}{x+1} \Leftrightarrow A(x+1) + B(x+3) = x \leftarrow$$

Multiply by ℓcm.

To find A: set $x = -3 \Rightarrow A(-3+1) + 0 = -3 \Rightarrow A = \frac{2}{3}$.

To find B: set $x = -1 \Rightarrow 0 + 2B = -1 \Rightarrow B = -\frac{1}{2}$

$$\therefore \int \frac{x}{x^2+4x+3} = \frac{2}{3}\int \frac{dx}{x+3} + -\frac{1}{2}\int \frac{dx}{x+1}$$

$$= \frac{2}{3}\ln|x+3| - \frac{1}{2}\ln|x+1| + c$$

e) $\dfrac{12}{(x-3)(x+3)} = \dfrac{A}{x-3} + \dfrac{B}{x+3} \Rightarrow A(x+3) + B(x-3) = 12.$

Multiplying by $x^2 - 9$. **To find A**: set $x = 3 \Rightarrow 6A = 12 \Rightarrow A = 2$.

To find B: set $x = -3 \Rightarrow 0 - 6B = 12 \Rightarrow B = -2$.

$$\therefore \int \frac{12}{x^2-9}dx = \int \frac{2}{x-3}dx - \int \frac{2}{x+3}dx$$

$$= 2\ln|x-3| - 2\ln|x+3| + c = 2\ln\left|\frac{x-3}{x+3}\right| + c$$

f) $\int \dfrac{4x+5}{(x+2)(x-2)}dx \leftarrow$ Exercise left for you.

b) **Integration of repeated linear factor**

$$\int \frac{N(x)}{(ax+b)^k} = \frac{A}{ax+b} + \frac{B}{(ax+b)^2} + \ldots + \frac{C}{(ax+b)^k}$$

Illustrative Example

20. Find: a) $\int \dfrac{2x+3}{x(x+1)^2}dx$ b) $\int \dfrac{x^2-2x-3}{x^2(x^2+5x+6)}$

Solution: $\dfrac{2x+3}{x(x+1)^2} = \dfrac{A}{x} + \dfrac{B}{x+1} + \dfrac{C}{(x+1)^2}$

$\Leftrightarrow A(x+1)^2 + Bx(x+1) + cx = 2x + 3$. **To find A**: set $x = 0 \Rightarrow A(0+1)^2 + 0 + 0 = 0 + 3$

$\Rightarrow A = 3$. **To find C**: set $x = -1$

$\Rightarrow 0 + 0 - c = 2(-1) + 3 \Rightarrow c = -1$.

To find B: set any convenient number, let $x = 1$.

$\Rightarrow A(1+1)^2 + B(2) + c = 2(1) + 3$

$\Rightarrow 4A + 2B + c = 5 \Rightarrow 12 + 2B - 1 = 5 \Rightarrow B = -3$.

$$\therefore \int \frac{2x+3}{x(x+1)^2}dx = \int \left(\frac{3}{x} - \frac{3}{x+1} + \frac{-1}{(x+1)^2}\right)dx$$

$$= 3\ln|x| - 3\ln|x+3| + \frac{1}{x+1} + c$$

b) $$\frac{x^2-2x-3}{x^2(x+2)(x+3)} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x+2} + \frac{D}{x+3}$$

$\Rightarrow Ax(x+2)(x+3) + B(x+2)(x+3) + c(x^2)(x+3) + D(x^2)(x+2) = x^2 - 2x - 3.$

To find B: set $x = 0 \Rightarrow 6B = 0 - 0 - 3 \Rightarrow B = -\frac{1}{2}$

To find C: set $x = -2 \Rightarrow 0 + 0 + 4c + 0 = 4 + 4 - 3 = 5 \Rightarrow c = \frac{5}{4}$

To find D: set $x = -3 \Rightarrow 0 + 0 + 0 + 9D = 9 + 6 - 3 = 12$

$\Rightarrow D = \frac{12}{9} = \frac{4}{3}.$

To find A: set any x, let $x = -1$

$\Rightarrow -2A + 2B + 2C + D = 1 + 2 - 3 = 0.$

$$\Rightarrow -2A + 2\left(-\frac{1}{2}\right) + 2\left(\frac{5}{4}\right) + \frac{4}{3} = 0 \Rightarrow A = \frac{17}{12}$$

$$\therefore \int \frac{x^2-2x-3}{x^2(x+2)(x+3)}dx$$

$$= \frac{17}{2}\int \frac{dx}{x} - \frac{1}{2}\int \frac{dx}{x^2} + \frac{5}{4}\int \frac{dx}{x+2} + \frac{4}{3}\int \frac{dx}{x+3}$$

$$= \frac{17}{12}\ln|x| + \frac{1}{2x} + \frac{5}{4}\ln|x+2| + \frac{4}{3}\ln|x+3| + c$$

Improper rational expression

For $\frac{N(x)}{D(x)}$ if $N(x) \geq D(x)$, then we *first perform long division.*

Illustrative Example

21. Find: a) $\int \frac{x+4}{x-1}dx$ b) $\int \frac{x+2}{4x-5}dx$ c) $\int \frac{x}{3x+4}dx$

d) $\int \frac{x^2-4}{x^2-9}dx$ e) $\int \frac{x^3}{x^2-3x+2}dx$ f) $\int \frac{x^2-x-2}{x^2-9}dx$

Solution: Since the degree of numerator is equal to degree of denominators so we perform division.

a) $\int \frac{x+4}{x-1}dx = \int \left(1+\frac{5}{x-1}\right)dx = x + 5 \ln|x-1| + c \leftarrow$ by long division.

b) $\int \frac{x+2}{4x-5}dx = \int \left(\frac{1}{4}+\frac{13}{4(4x-5)}\right)dx = \frac{1}{4}x+\frac{13}{16}\ln|4x-5|+c$

c) $\int \frac{x}{3x+4}dx \int \left(\frac{1}{3}+\frac{-4}{3(3x+4)}\right)dx = \frac{1}{3}x-\frac{4}{9}\ln|3x+4|c$

d) $\int \frac{x^2-4}{x^2-9}dx = \int \left(1+\frac{5}{x^2-9}\right)dx$ and $\frac{5}{x^2-9} = \frac{A}{x-3}+\frac{B}{x+3}$

$\Leftrightarrow$A $(x+3)$ + B $(x-3) = 5$, set $x = 3$ and $-3 \Rightarrow A = \frac{5}{6}, B = \frac{-5}{6}$

$$\therefore \int \frac{x^2-4}{x^2-3x+2}dx = \int \left(1+\frac{5}{6(x-3)}+\frac{-5}{6(x+3)}\right)dx$$

$$= x+\frac{5}{6}\ln|x-3|-\frac{5}{6}\ln|x+3|+c$$

e) $\int \frac{x^3}{x^2-3x+2}dx = \int \left(x+3+\frac{7x-6}{(x-2)(x-1)}\right)dx$, again

$\frac{7x-6}{(x-2)(x-1)} = \frac{A}{x-2}+\frac{B}{x-1} \Leftrightarrow$ A $(x-1)$ + B $(x-2) = 7x-6$,

let $x = 2 \Rightarrow A = 8$ and let $x = 1 \Rightarrow B = -1$.

$\therefore \int \frac{x^3}{x^2-3x+2}dx = \int \left(x+3+\frac{8}{x-2}+\frac{-1}{x-1}\right)dx = \frac{x^2}{2} + 8\ln|x-2| - \ln|x-1| + c$

f) Exercise left for you.

ii) Quadratic factors of the form $\frac{N(x)}{ax^2+bx+c}$ with $b^2-4ac<0$.

Use: $\frac{N(x)}{ax^2+bx+c}=\frac{Bx+c}{ax^2+bx+c}$

Illustrative Example

22. Find: a) $\int \frac{x}{x^4-1}dx$ b) $\int \frac{8x^3+13x}{(x^2+2)^2}dx$

Solution: $\frac{x}{x^4-1}=\frac{x}{(x^2-1)(x^2+1)}=\frac{A}{x-1}+\frac{B}{x+1}+\frac{Cx+D}{x^2+1}$

$\Leftrightarrow A(x^2+1)(x+1)+B(x-1)(x^2+1)+(Cx+D)(x-1)(x+1)=x$

To find A: set $x=1\Rightarrow 4A+0+0=1\Rightarrow A=\frac{1}{4}$

To find B: set $x=-1\Rightarrow 0-4B+0=-1\Rightarrow B=\frac{1}{4}$

To find D: set any x, let $x=0\Rightarrow A-B-D=0\Rightarrow$ since $A=\frac{1}{4}$, $B=\frac{1}{4}$, so that $\frac{1}{4}-\frac{1}{4}-D=0\Rightarrow D=0$

To find C: set any x, other than, -1, 1 and 0 let $x=2$.

$\Rightarrow 15A+5B+(2C+D)3=2\Rightarrow \frac{15}{4}+\frac{5}{4}+6c+0=2\Rightarrow c=-\frac{1}{2}$

$$\therefore \int \frac{x}{x^4-1}dx=\frac{1}{4}\int \frac{dx}{x-1}+\frac{1}{4}\int \frac{dx}{x+1}+-\frac{1}{2}\int \frac{x}{x^2+1}dx$$

$$=\frac{1}{4}\ln\left|\frac{x^2-1}{x^2+1}\right|+c$$

b) $\frac{8x^3+13x}{(x^2+2)^2}=\frac{Ax+B}{x^2+2}+\frac{Cx+D}{(x^2+2)^2}$ ← Multiplying by $(x^2+2)^2$

$\Rightarrow 8x^3+13x=(Ax+B)(x^2+2)+(Cx+D)$ ← Expanding $\Rightarrow 8x^3$

$13x = Ax^3 + 2Ax + Bx^2 + 2B + Cx + D \leftarrow$ Equate $\Rightarrow A = 7, B = 0,$
and $2Ax + Cx + 13x \Rightarrow 2A + c = 13 \Rightarrow 16 + c = 13 \Rightarrow c = -3$ and $D = 0$

$$\int \frac{8x^3 + 13x}{(x^2+2)^2}dx = \int \frac{8x}{x^2+2}dx + \int \frac{-3x}{(x^2+2)}dx$$

$$= 4\ln(x^2+2) + \frac{3}{2(x^2+2)} + c$$

C. Integration by parts

Integration by parts is method which is the reverse part of the product rule of **differentiation.**

Let u and v be functions of $x \Rightarrow u = u(x)$ and $v = v(x)$.

Then $\frac{d}{dx}(uv) = u\frac{dv}{dx} + v\frac{du}{dx}$ solving for $u\frac{dv}{dx}$, gives

$\Rightarrow u\frac{dv}{dx} = \frac{d}{dx}(uv) - v\frac{du}{dx} \leftarrow$ integrating both side

$\Rightarrow \int u\frac{dv}{dx}(dx) = \int \frac{d}{dx}(uv)dx - \int v\frac{du}{dx} \leftarrow$ so that yield

$$\boxed{\int udv = uv - \int vdu}$$

- This formula express the original integral in terms of another integral.

Depending on the choices of **u** and **dv** it may be easier to evaluate the 2nd integral than the original one.

General approach to choose u and dv

i) $\int \ln x dx$, use $u = \ln x$, $dv = dx$

ii) $\int x^n \ln x dx$ use $u = \ln x$, $dv = x^n dx$

iii) $\int x^n e^x dx$ use $u = x^n$, $dv = e^x dx$

iv) $\int x^n \sin x dx$: use $u = x^n m$ $dv = \sin x\, dx$

v) $\int e^x \sin x dx$: use $u = \sin x$, $dx = e^x dx$

i) **Type** $\int \ln x dx$; Let, $u = \ln x$, $dv = dx$

Illustrative Example

23. Find a) $\int \ln x dx$ b) $\int \log_3^x dx$ c) $\int \log(4x+3)dx$

Solution: Use $\int udv = uv - \int vdu$ and choice u=ln x, dv= dx

a) Let $u = \ln x \Rightarrow du = \frac{dx}{x}$ and $dv = dx \Rightarrow v = \int dx = x$

$\therefore \int udv = uv - \int vdu$ gives $\int \ln x dx = x \ln x - \int dx =$ ($x \ln x - x$) + c

b) We have $\log_3^x = \frac{\ln x}{\ln 3}$, so that $\int \log_3^x dx = \int \frac{\ln x}{\ln 3} dx$

$= \frac{1}{\ln 3}\int \ln x dx = \frac{1}{\ln 3}(x \ln x - x) + c$

c) We have log $(4x + 3)$ = $\frac{\ln(4x+3)}{\ln 10} \Rightarrow \int \log^{(4x+3)} dx = \int \frac{\ln(4x+3)}{\ln 10} dx$

let $u = 4x+3 \Rightarrow du = 4dx \Rightarrow dx = \frac{du}{4}$, $\frac{1}{4\ln 10}\int \ln u du$

$= \frac{1}{4\ln 10}(u \ln u - u) + c = \frac{1}{4\ln 10}(4x + 3) \ln (4x + 3) + c$

Note:

i) $\int \ln x dx = (x \ln x - x) + c$

ii) $\int \log_a^x dx = \frac{1}{\ln a}(x \ln x - x) + c$

ii) Type two $\int x^n \ln x dx$. For this type use $u = \ln x$, $dv = x^n$

24. Find **Illustration Example**

a. $\int x \ln x dx$ b. $\int x^2 \ln x dx$ c. $\int (\ln x)^2 dx$

d. $\int x^3 \ln x dx$ e. $\int x \ln(4x) dx$ f. $\int x \ln(nx) dx$

g. $\int x^n \ln x dx$ h. $\int x \log_3^x dx$ i. $\int x^3 \log_4^x dx$

Solution: Use $\int udv = uv - \int vdu$. Then a) let $u = \ln x$

u = ln x and dv = x dx

$$\Rightarrow du = \frac{dx}{x} \text{ and } dv = x\,dx \Rightarrow v = \int xdx = \frac{x^2}{2}$$

$$\therefore \int udv = uv - \int vdu \text{ substituting}$$

$$\int x \ln x dx = \frac{x^2}{2}\ln x - \int \frac{x}{2}dx = \frac{x^2}{2}\ln x - \frac{x^2}{4} + c$$

b. Let $u = \ln x \Rightarrow du = \frac{dx}{x}$ and $dv = x^2dx \Rightarrow v = \int x^2 dx = \frac{x^3}{3}$.

$$\therefore \int udv = uv - \int vdu \text{ substitute}$$

$$\Rightarrow \int x^2 \ln x dx = \frac{x^3}{3}\ln x - \int \frac{x^2}{3}dx = \frac{x^3}{3}\ln x - \frac{x^3}{9} + c$$

c. Let $u = (\ln x)^2 \Rightarrow du = \frac{2\ln x}{x}dx$, $dv = dx \Rightarrow v = \int dx = x$, Then

$\int udv = uv - \int vdu$ will be

$$\int (\ln x)^2 dx = x(\ln x)^2 - \int x\left(\frac{2\ln x}{x}\right)dx$$

$$= x(\ln x)^2 - 2(x \ln x) + 2x + c$$

d. Let $u = \ln x \Rightarrow du = \frac{dx}{x}$, $dv = x^3dx \Rightarrow v = \int x^3 dx = \frac{x^4}{4}$, thus

$\int udv = uv - \int udv$ gives

$$\int x^3 \ln x\,dx = \frac{x^4}{4}\ln x - \int \frac{x^4}{4}\left(\frac{dx}{x}\right) = \frac{x^4}{4}\ln x - \frac{x^4}{16} + c$$

e. $\int x \ln(4x)$ Exercise left for you.

f. Let $u=(\ln x)\Rightarrow du=\frac{1}{x}dx$, $dv=x\,dx\Rightarrow v=\int xdx=\frac{x^2}{2}$,

then $\int udv=uv-\int vdu$

$$\Rightarrow\int x\ln(\ln x)dx=\frac{x^2}{2}\ln(nx)-\int\frac{x^2}{2}\left(\frac{1}{x}\right)dx=\frac{x^2}{2}\ln(nx)-\frac{x^2}{4}+c$$

g. Let $u=\ln x\Rightarrow du=\frac{1}{x}dx$, $dv=x^n dx$

$\Rightarrow v=\int x^n dx=\frac{x^{n+1}}{n+1}$, then $\int udv=uv-\int vdu$,

$$\Rightarrow\int x^n\ln xdx=\frac{x^{n+1}}{n+1}(\ln x)-\frac{1}{n+1}$$

$$\int x^n dx=\frac{x^{n+1}}{n+1}\ln x-\frac{x^{n+1}}{(n+1)^2}+c,n\neq-1$$

Observe that:

i) $\int x^3\ln xdx=\frac{x^4}{4}\ln x-\frac{x^4}{16}+c$

ii) $\int x^5\ln xdx=\frac{x^6}{6}\ln x-\frac{x^6}{36}+c$

h) $\int x\log_3^x dx=\int\frac{x\ln x}{\ln 3}dx=\frac{1}{\ln 3}\left(\frac{x^2}{2}(\ln x)-\frac{x^2}{4}\right)+c$

$$=\frac{x^2\ln x}{2\ln 3}-\frac{x^2}{4\ln 3}+c$$

i) $\int x^3\log_4^x dx=\int\frac{x^3\ln x}{\ln 4}dx=\frac{1}{\ln 4}\int x^3\ln xdx=\frac{x^4\ln x}{4\ln 4}-\frac{x^4}{16\ln 4}+c$

iii) Part $\int x^n e^x dx$ and $\int x^n \cos x dx$

Illustrative Example

25. Find

a. $\int xe^{-x}dx$ b. $\int x\cos(nx)dx$ c. $\int x\sin(kx)dx, k>0$

Solution: Let $u = x \Rightarrow du = dx$, $dv = e^{-x}dx \Rightarrow v = \int e^{-x}dx = -e^{-x}$.

Then $\int x\sin(kx)dx, k>0$

$$\Rightarrow \int xe^{-x}dx = -xe^{-x} - \int -e^{-x}dx = -xe^{-x} - e^{-x} + c$$

b. Let $u = x \Rightarrow du = dx$ and $dv = \cos(nx)\,dx$

$$\Rightarrow v = \int \cos nxdx = \frac{\sin(nx)}{n}.$$

$$\therefore \int udv = uv - \int vdu \Rightarrow \int x\cos(nx)dx$$

$$= \frac{x\sin(nx)}{n} - \int \frac{\sin(nx)}{n}dx = \frac{x\sin(nx)}{n} + \frac{\cos(nx)}{n^2} + c$$

Observe:

i) $\int x\cos 2xdx = \frac{x\sin 2x}{2} + \frac{\cos 2x}{4} + c$

ii) $\int x\cos(5x)dx = \frac{x\sin(5x)}{5} + \frac{\cos(5x)}{25} + c$

c. Let $u = x \Rightarrow du = dx$, $dv = \sin$

$$(kx)dx \Rightarrow v = \int \sin(kx)dx = -\frac{\cos(kx)}{k}$$

$\therefore \int udv = uv - \int vdu$ *is the same as*

$$\int x\sin(kx)dx =$$

$$\frac{-x\cos(kx)}{k} - \int -\frac{\cos(kx)}{k}dx = \frac{-x\cos(kx)}{k} + \frac{\sin(kx)}{k^2} + c$$

Observe that:

i) $\int x\sin(2x)dx = \frac{-x\cos 2x}{2} + \frac{\sin(2x)}{4} + c$

ii) $\int x\sin(3x)dx = \frac{-x\cos(3x)}{3} + \frac{\sin(3x)}{9} + c$

Repeated use of integration by part

Illustrative Example

26. Find: a. $\int e^x \sin x dx$ b. $\int e^x \cos(3x)dx$ c. $\int x^2 e^x dx$

d. $\int e^{3x} \sin x dx$

Solution: a. Let $u = \sin x \Rightarrow du = \cos x\, dx$, $dv = e^x dx$

$\Rightarrow v = \int e^x dx = e^x$. Then $\int udv = uv - \int vdu$ *is the same as*

$\int e^x \sin x dx = e^x \sin x - \int e^x \cos x dx$.

Again apply integration by part for 2nd time.

$\Rightarrow \int e^x \cos x dx$, let $u = \cos x \Rightarrow du = -\sin x\, dx$ let $dv = e^x dx$

$\Rightarrow v = e^x$. Then $\int e^x \cos x = e^x \cos x + \int e^x \sin x dx$, collecting

$\Rightarrow \int e^x \sin x dx = e^x \sin x - \left(e^x \cos x + \int e^x \sin x dx\right)$

$= e^x \sin x - e^x \cos x - \int e^x \sin x dx \leftarrow$ *adding to the left part*

$\Rightarrow 2\int e^x \sin x dx = e^x \sin x - e^x \cos x + c$

$\Rightarrow \int e^x \sin x dx = \frac{1}{2} e^x (\sin x - \cos x) + c$

b. $u = \cos 3x \Rightarrow du = -3 \sin 3x\, dx$, $dv = e^x dx \Rightarrow v = e^x$ then

$\int udv = uv - \int vdu \Rightarrow \int e^x \cos(3x)dx$

$= e^x \cos 3x - \int e^x (-3)\sin 3x dx = e^x \cos 3x + 3\int e^x \sin 3x dx$

Again apply integration by part 2nd time for $\int e^x \sin 3x dx$.

Let $u = \sin 3x \Rightarrow du = 3 \cos 3x\, dx$, $dv = e^x dx \Rightarrow v = e^x$ then

$\int udv = uv - \int udu$.

$\int e^x \sin 3x dx = e^x \sin 3x - 3\int e^x \cos(3x)dx$. Thus collecting

$\Rightarrow \int e^x \cos(3x)dx = e^x \cos 3x + 3\int e^x \sin(3x)dx$

$= e^x \cos(3x) + 3\left[e^x \sin(3x) - 3\int e^x \cos(3x)dx\right]$

$\Rightarrow 10\int e^x \cos(3x)dx = e^x \cos(3x) + 3e^x \sin(3x) \Rightarrow \int e^x \cos(3x)dx$

$$= \frac{e^x(\cos 3x + 3\sin(3x))}{10} + c$$

Note: (Keep in touch).

i) $\int e^{ax} \sin(bx)dx = \frac{e^{ax}(\sin(bx) - b\cos(bx))}{a^2 + b^2} + c$

ii) $\int e^{ax} \cos(bx)dx = \frac{e^{ax}(a\cos(bx) + b\sin(bx))}{a^2 + b^2} + c$

Thus, we can see that:

i) $\int e^{3x} \sin x dx = \frac{e^{3x}(3\sin x - \cos x)}{3^2 + 1} + c$

$$= \frac{e^{3x}(3\sin x - \cos x)}{10} + c$$

ii) $\int e^{-x} \sin(2x)dx = e^{-x} \frac{-\sin(2x) - 2\cos(2x)}{.(-1)^2 + (2)^2} + c$

$$= \frac{-e^{-x}(\sin 2x + 2\cos(2x))}{5} + c$$

iii) $\int e^{2x} \cos(3x)dx = \frac{e^{2x}2\cos(3x) + 3\sin(3x)}{2^2 + 3^2} + c$

$$= \frac{e^{2x}(2\cos(3x) + 3\sin(3x))}{13} + c$$

d. Tabular method

Integration of the form

- $\int x^n e^{ax}dx$
- $\int x^n \sin(bx)dx$
- $\int x^n \cos(bx)dx$ we can find by tabular method

To find x^n differentiated until you obtain 0.
And e^{ax}, sin (bx), cos (bx) *integrate repeatedly.*

Illustrative Example

27. Find:

a. $\int x^2 e^x dx$ b. $\int (x^2+4x)e^{3x}dx$ c. $\int x^3 \sin(2x)dx$

d. $\int x^2 \cos(3x)dx$ e. $\int x^2 \sin x dx$ f. $\int xe^{2-x}dx$

Solution:

Derivative of polynomial *Integral (repeated anti derv.)*

x^2 — (+) → e^x

$2x$ — (−) → e^x

2 — (+) → e^x

0 → e^x

e^x since $\int e^x dx = e^x$

$$\therefore \int x^2 e^x dx = x^2e^x - 2xe^x + 2xe^x + 2e^x + c$$

b. *Repeated derv.* *Repeated anti derive. (Integral)*

x^2+4x — (+) → e^{3x}

$2x+4$ — (−) → $\frac{e^{3x}}{3}$ since $\int e^{3x}dx = \frac{e^{3x}}{3}$

2 — (+) → $\frac{e^{3x}}{9}$ since $\int \frac{e^{3x}}{3}dx = \frac{e^{3x}}{9}$

0 → $\frac{e^{3x}}{27}$ since $\int \frac{e^{3x}}{9}dx = \frac{e^{3x}}{27}$

$$\therefore \int (x^2+4x)e^{3x}dx = (x^2+4x)\frac{e^{3x}}{3} - (2x+4)\frac{e^{3x}}{9} + 2\frac{e^{3x}}{27} + c$$

c. *Exercise left for you.*

d. $$\int x^2 \cos(3x)dx = \frac{x^2 \sin 3x}{3} + \frac{2x\cos 3x}{9} - \frac{2\sin 3x}{27} + c$$

e. *Exercise left for you.*

f. $x \xrightarrow{+} e^{2-x}$ since $\int e^{2-x}dx = -e^{2-x}dx = -e^{2-x}$

$1 \xrightarrow{-} -e^{2-x}$ since $-\int e^{2-x}dx = e^{2-x}$

$0 \quad e^{2-x}$

$$\therefore \int xe^{2-x}dx = -xe^{2-x} - e^{2-x} + c$$

Solved problem

28. $\int e^{\sqrt{x}}dx$ is equal to:

A. $2\sqrt{2x} - 2e^{\sqrt{x}} + c$ B. $2\sqrt{x}e^{\sqrt{x}} - 2e^{\sqrt{x}} + c$

C. $\sqrt{x}e^{\sqrt{x}} - e^{\sqrt{x}} + c$ D. none

Solution: Let $y = \sqrt{x} \Rightarrow dy = \dfrac{dx}{2\sqrt{x}} \Rightarrow dx = 2\sqrt{x}\,dy$

$\Rightarrow dx = 2ydy$, since $y = \sqrt{x}$

$\Rightarrow \int e^{\sqrt{x}}dx = \int e^{y}(2ydy) = \int 2ye^{y}dy \leftarrow$ Rewriting

Let $u = 2y \Rightarrow du = 2dy$ and $dv = e^{y}dy \Rightarrow v = e^{y}$

$\therefore \int udv = uv - \int udu \Rightarrow \int 2ye^{y}dy = 2ye^{y} - 2\int e^{y}dy$

$= 2ye^{y} - 2e^{y} + c \Rightarrow 2e^{y}(y-1) + c = 2e^{\sqrt{x}}\left(\sqrt{x} - 1\right) + c$.

∴ Answer: B

29. $\int \dfrac{dx}{1+e^{x}}$ is equal to:

A. $\dfrac{1}{2}x - \dfrac{1}{2}\ln(e^{x}+1) + c$ C. $\dfrac{1}{2}\ln(e^{x}+1) + c$

B. $x - \ln(e^{x} + c) + c$ D. $x + \dfrac{1}{2}\ln(e^{x} + c) + c$

Solution: Let $u = 1 + e^{x} \Rightarrow du = e^{x}dx \Rightarrow dx = \dfrac{du}{e^{x}}$, since e^{x} $u - 1$

$$\Rightarrow \int \frac{du}{u(u-1)} = \int \left(\frac{A}{u} + \frac{B}{u-1}\right) du, \text{ since A = -1 and B = 1}$$

$$\Rightarrow \int \frac{-1}{u} du + \int \frac{1}{u-1} du = -\ln u + \ln (u-1) + c$$

$$= \ln\left(\frac{u-1}{u}\right) + c = \ln\left(\frac{e^x}{e^x+1}\right) + c = x - \ln (e^x + 1) + c \therefore$$

Answer: B

30. $\int x^3 e^{x^2} dx$ is equal to:

A. $\frac{1}{2} e^{x^2} (x^2 + 1) + c$

B. $\frac{1}{2} e^{x^2} (x^2 + 1) + c$

C. $\frac{1}{2} x^2 (e^{x^2} + 1) + c$

D. $\frac{1}{2} x^2 (e^{x^2} - 1) + c$

Solution: Let $u = x^2 \Rightarrow du = 2xdx \Rightarrow xdx = \frac{du}{2}$

$$\int x^3 e^{x^2} dx = \int x^2 e^{x^2} (xdx) = \int ue^u \frac{du}{2} = \frac{1}{2} \int ue^u du$$

Note: $\int \mathbf{x^n e^{ax} dx} = \frac{\mathbf{x^n e^{ax}}}{\mathbf{a}} - \frac{\mathbf{n}}{\mathbf{a}} \int \mathbf{x^{n-1} e^{ax} dx}$

$$\therefore \frac{1}{2} \int ue^u du = \frac{1}{2} \left[ue^u - \int e^u du\right] = \frac{1}{2} ue^u - e^u + c = \frac{1}{2} e^u (u-1) + c$$

$$\Rightarrow \frac{1}{2} e^{x^2} (x^2 - 1) + c$$

∴ Answer: A

31. $\int \frac{\ln x}{\sqrt{x}} dx$ . . . (UEE)

A. $2\sqrt{x} \ln x + c$

B. $\sqrt{x} \ln x - \sqrt{x} + c$

C. $2\sqrt{x} (\ln x - 2)$

D. None

Solution: Let $u = \ln x \Rightarrow du = \dfrac{dx}{x}$ and

$v = x^{-\frac{1}{2}}dx \Rightarrow v = \int x^{-\frac{1}{2}}dx = 2x^{\frac{1}{2}}$, thus

$\int udv = uv - \int vdu$ is substitute

$$\int \frac{\ln x}{\sqrt{x}}dx = \left(2\sqrt{x}\right)\ln x - \int 2(x^{\frac{1}{2}})\frac{dx}{x} = 2\sqrt{x}\,(\ln x - 4) + c$$

∴ **Answer: C**

32. $\int x \sin x dx$ . . . (UEE)

A. $x^2\cos x + c$ C. $x^2\sin x - x\cos x + c$

B. $\sin x - x\cos x + c$ D. $-\sin x + x^2\cos x + c$

Solution: Let $u = x \Rightarrow du = dx$ and $dv = \sin x\, dx \Rightarrow v = -\cos x$

$$\therefore \int udv = uv - \int vdu = \int x\sin xdx = -x\cos x + \int \cos xdx$$

$= \sin x - x\cos x + c$ ∴ **Answer: B**

33. $\int\left(\dfrac{x}{x+1}\right)^2 dx$ . . . (UEE)

A. $x + \ln|x+1| + c$ C. $x - \dfrac{1}{x+1} - 2\ln|x+1| + c$

B. $\dfrac{1}{3}x + \dfrac{1}{3}\ln|x+1| + c$ D. $x - \left(\dfrac{2}{x+1}\right)^2 - \dfrac{1}{x+1} + c$

Solution:

$$\int\left(\frac{x}{x+1}\right)^2 dx = \int\left(1-\frac{1}{x+1}\right)^x dx = \int\left(1 - \frac{2}{x+1} + \frac{1}{(x+1)^2}\right)dx$$

$$\Rightarrow x - 2\ln|x+1| - \frac{1}{x+1} + c$$ ∴ **Answer: C**

34. $\int \dfrac{x^2-1}{x^3+x}dx$ . . . UEE 2011

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A. $\ln|x^2+1|+\ln|x|+c$ C. $\frac{1}{2}\ln(x^2+1)-\ln|x|+c$

B. $\ln(x^2+1)-\ln|x|+c$ D. $2\ln(x^2+1)-\ln|x|+c$

Solution: $\frac{x^2-1}{x(x^2+1)}=\frac{A}{x}+\frac{Bx+c}{x^2+1}\Rightarrow A(x^2+1)+(Bx+c)=x^2-1$

$\Rightarrow Ax^2+Bx^2+Cx+A=x^2-1$ . . . comparing coef. of like term $\Rightarrow$ A =-1 and C = 0 and A + B = 1$\Rightarrow$-1 + B = 1$\Rightarrow$B =2

$$\therefore\int\frac{x^2-1}{x^3+x}=\int\left(\frac{-1}{x}+\frac{2x}{x^2+1}\right)dx=\int\frac{-1}{x}dx+\int\frac{2x}{x^2+1}dx$$

$$=-\ln|x|+\int\frac{2x}{x^2+1}dx$$

Let $u=x^2+1\Rightarrow du=2xdx\Rightarrow\int\frac{du}{u}=\ln u=\ln(x^2+1)$

$\therefore\int\frac{x^2-1}{x^3+x}dx=\ln(x^2+1)-\ln|x|+c$ ∴ Answer: B

35. $\int\frac{x}{x^2+2x+1}dx$ . . .

A. $\ln(x^2+2x+1)+c$ C. $\ln(x^2+2x+1)-\frac{1}{x+1}+c$

B. $\ln|x+1|+\frac{1}{x+1}+c$ D. $\ln(x^2+2x+1)+\frac{1}{x+1}+c$

Solution: $\frac{x}{x^2+2x+1}dx=\int\frac{1}{x+1}dx-\int\frac{1}{(x+1)^2}dx$

$=\ln|x+1|+\frac{1}{x+1}+c$ ∴ Answer: B

36. $\int\frac{\cos\sqrt{x}}{\sqrt{x}}dx$ . . . UEE

A. $\sin\sqrt{x}+\cos\sqrt{x}+c$ C. $2\sin\sqrt{x}+c$

B. $-\sqrt{x}\cos\sqrt{x}+c$ D. $\frac{1}{2}\sqrt{x}\sin x+c$

Solution: Let $u=\sqrt{x} \Rightarrow du=\frac{dx}{2\sqrt{x}} \Rightarrow 2du=\frac{dx}{\sqrt{x}}$

$\Rightarrow 2\int \cos u du = 2\sin u + c = 2\sin\sqrt{x}+c$ ∴ **Answer: B**

37. $\int \frac{e^{2x}-4x}{xe^{2x}}dx$ . . . UEE 2004 /12

A. $\ln|x|+2e^{-2x}+c$ C. $\ln|x|-4e^{2x}+c$

B. $\ln|x|+4e^{-2x}+c$ D. $-\frac{1}{x^2}+2e^{-2x}+c$

Solution: $\int \frac{e^{2x}-4x}{xe^{2x}}dx = \int\left(\frac{1}{x}-4e^{-2x}\right)dx \leftarrow$ *simplifying*

$\Rightarrow \ln|x|+\frac{4}{2}e^{-2x}+c = \ln|x|+2e^{-2x}+c$ ∴ **Answer: A**

38. $\int (2+\sin^2 x)\cot x dx$ is equal to: UEE

A. $2\csc x-\sec^2 x+c$ C. $2\ln|\sin x|-\cos^2 x+c$

B. $2\ln|\sin x|+\frac{1}{2}\sin^2 x+c$ D. $2\csc x+\sin x\cos x+c$

Solution: $\int (2+\sin^2 x)\frac{\cos x}{\sin x}dx = \int\left(\frac{2\cos x}{\sin x}+\sin x\cos x\right)dx$

$\Rightarrow \int \frac{2\cos x dx}{\sin x}+\int \sin x\cos x dx$

Let $u=\sin x \Rightarrow du=\cos x\, dx$

$\Leftrightarrow \int 2\frac{du}{u}+\int u du = 2\ln|u|+\frac{u^2}{2}+c$

$=2\ln|\sin x|+\frac{\sin^2 x}{2}+c$

∴ **Answer: B**

39. $\int \frac{x}{1+e^{-x^2}}$ is equal to:

A. $-\frac{1}{2}(x^2+\ln(e^{-x^2}+1))+c$ C. $-\frac{1}{2}(x+\ln(e^{-x^2}+1))+c$

B. $\frac{(x^2+\ln(e^{-x^2}+1))+c}{2}$ D. $x+\ln(e^{-x^2}+1)+c$

Solution: Let $u=-x^2 \Rightarrow du=-2x\,dx \Rightarrow xdx=-\frac{1}{2}du$

$$\int\frac{x}{1+e^{-x^2}}dx=-\frac{1}{2}\int\frac{1}{1+e^u}du=-\frac{1}{2}\int\left(\frac{1+e^u}{1+e^u}-\frac{e^u}{1+e^u}\right)du$$

$$=-\frac{1}{2}\int\left(1-\frac{e^u}{1+e^u}\right)du=-\frac{1}{2}(u-\ln(e^u+1))+c$$

$$\Rightarrow-\frac{1}{2}(-x^2-\ln(e^{-x^2}+1))+c=\frac{(x^2+\ln(e^{-x^2}+1))+c}{2}\therefore$$

Answer: B

40. $\int\frac{\ln x}{x^2}dx$ is equal to:

A. $\frac{\ln x}{x}-\frac{1}{x}+c$ B. $\frac{-\ln x}{x}-\frac{1}{x}+c$ C. $\frac{-\ln x}{x}+\frac{1}{x}+c$

Solution: It requires integration by parts

let $u=\ln x \Rightarrow du=\frac{1}{x}dx$ and $dv=\frac{1}{x^2}dx \Rightarrow v=\frac{-1}{x}$

$\therefore\int\frac{\ln x}{x^2}dx=\frac{-\ln x}{x}+\int-\frac{1}{x^2}dx=\frac{-\ln x}{x}+\frac{1}{x}+c$ ∴Answer: C

41. $\int\frac{e^x}{(e^x-1)(e^x+4)}dx$ is equal to:

A. $\frac{1}{5}\ln\left(\frac{e^x-1}{e^x+4}\right)+c$ C. $\frac{1}{5}\ln\left(\frac{e^x+2}{e^x+1}\right)+c$

B. $\frac{1}{5}\ln\left(\frac{e^x+2}{e^x+1}\right)+c$ D. None

Solution: Let $y=e^x-1 \Rightarrow e^x=y+1$ and $dy=e^x dx$

$$\Rightarrow \int \frac{1}{y(y+5)}dy = \int \left(\frac{A}{y} + \frac{B}{y+5}\right)y \Leftrightarrow A(y+5) + By \quad 1$$

Let $y = 0 \Rightarrow A(0+5) = 1 \Rightarrow A = \frac{1}{5}$.

Let $y = -5 \Rightarrow A(0) + B(-5) = 1 \Rightarrow B = -\frac{1}{5}$.

$$\therefore \int \frac{1}{5y}dy + \frac{-1}{5}\int \frac{dy}{y+5} \Rightarrow \frac{1}{5}\ln y - \frac{1}{5}\ln(y+5) = \frac{1}{5}\ln\left(\frac{y}{y+5}\right)$$

but $y = e^x - 1$, so that $\frac{1}{5}\ln\left(\frac{e^x - 1}{e^x + 4}\right) + c$ **∴ Answer: A**

- **Definite integrals**
- **Area of the curve**
- **Fundamental theorem of calculus**

- **Definite integral as area of the region**

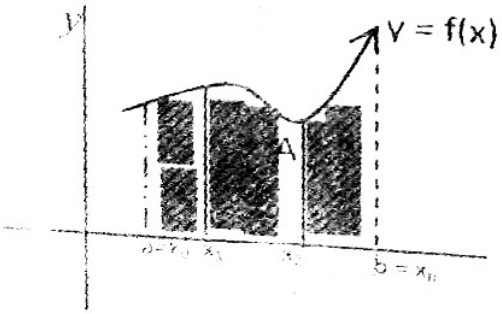
The area under a curve can also be found by integration. In geometry we have formula to find the area of figures such as **triangle, circle, rectangle and trapezoid.**

Now here, we going to determine *the area of the region (R)* by the curve $y = f(x)$ which is continuous on $[a, b]$ above the x-axis and between the line $x = a$ and $x = b$ calculated as follow.

To estimate the area, we will start from the definition of the area of rectangle.

- Define the area of the region divide into thinner and thinner rectangle.
- Divide the interval $[a, b]$ into n equal sub interval, these are: $[x_0, x_1], [x_1, x_2], [x_2, x_3], \ldots [x_{n-1}, x_n]$ with $x_0 = a$ and $x_n = b$

The length of each thinner rectangle is $x_1 - x_0 = x_2 - x_1 = \ldots$ $x_{n-1} x_{n-1} = \Delta x$. Thus the region is divided into ***n*** stripes which is nearly ***n*** *rectangle of the same width.* $\Delta x = \frac{b-a}{n}$



$\Rightarrow b - a = n\,\Delta x$
$\Rightarrow b = a + n\,\Delta x$
$\Rightarrow x_i = a + i\Delta x$

So that,
Area of rectangle = (length)(height).

Thus, $A_i = f(x_i)\Delta x \leftarrow$ area of ith rectangle,
$\Delta x \leftarrow$ length of rectangle
$f(x_i) \leftarrow$ height ith rectangle,

Total area of the region is given by:

$$A = \sum_{i=0}^{n} A_i = A_0 + A_1 + A_2 + \ldots$$

$$A_n = f(x_0)\,\Delta x + f(x_1)\Delta x + \ldots + f(x_n)\,\Delta x$$
$$= \Delta x\,[f(x_0) + f(x_1) + f(x_2) + \ldots + f(x_n)]$$
$$= \left[\sum_{i=0}^{n} f(x_i)\right]\Delta x = \frac{b-a}{n}\left[\sum_{i=0}^{n} f(x_i)\right]$$

As n gets *larger and larger* without bound the area of the region become exact and express as:

$$A = \lim_{n\to\infty}\sum_{i=1}^{n} f(x_i)\Delta x = \lim \frac{b-a}{n}\sum_{i=1}^{n} f(x_i)$$

The definite integral

There is surprising link between the **definite integrals** and **the area** of the curve $y = f(x)$.

The definite integral of function f from ***a*** to ***b*** is the number

$$\lim_{n\to\infty}\sum_{i=1}^{n} f(x_i)\Delta x = \int_a^b f(x)dx = F(b) - F(a)$$

Where:

- $f(x)$ is continuous on $[a, b]$
- a and b are called limit of integration
- a is called the lower limit of integration
- b is called the upper limit of integration

To illuminate the above let we have illustrative example.

Illustrative Example

42. Divide the interval [1, 3] into equal length of:

a. 3 – sub interval b. 5 – sub interval c. n – subs interval

Solution: a) We have $\Delta x = \frac{b-a}{n} = \frac{3-1}{3} = \frac{2}{3}$. Then we use, $x_i = a + i\,\Delta x$, where $a = 1$, $b = 3$, $\Delta x = \frac{2}{3}$, $x_0 = 1 + 0 = 1$ and

$$x_1 = 1 + \frac{2}{3} = \frac{5}{3} \Rightarrow \left[1, \frac{5}{3}\right] \Rightarrow x_2 = 1 + 2\left[\frac{2}{3}\right] = \frac{7}{3} \leftarrow \text{add } \frac{2}{3} \text{ to}$$

each next element. The interval, $[x_0, x_1]$, $[x_1, x_2]$, and $[x_2, x_3]$

$$\Rightarrow \left[1, \frac{5}{3}\right], \left[\frac{5}{3}, \frac{7}{3}\right], \left[\frac{7}{3}, \frac{9}{3}\right] \leftarrow 3 \text{ sub equal interval.}$$

b. [1, 3] into 5 – sub interval $\Delta x = \frac{b-a}{n} = \frac{3-1}{5} = \frac{2}{5}$.

$$\left[1, 1+\frac{2}{5}\right], \left[\frac{7}{5}, \frac{2}{5}+\frac{7}{5}\right], \left[\frac{9}{5}, \frac{2}{5}+\frac{9}{5}\right], \left[\frac{11}{5}, \frac{2}{5}+\frac{11}{5}\right], \left[\frac{13}{5}, \frac{2}{5}+\frac{13}{5}\right]$$

$$= \left[1, \frac{7}{5}\right]\left[\frac{7}{5}, \frac{9}{5}\right], \left[\frac{9}{5}, \frac{11}{5}\right], \left[\frac{11}{5}, \frac{13}{5}\right] \text{ and } \left[\frac{13}{5}, 3\right]$$

c. [1, 3] in to n – sub equal interval.

$$\Rightarrow \Delta x = \frac{b-a}{n} = \frac{3-1}{n} = \frac{2}{n}$$

$$\Rightarrow \left[1, \frac{n+2}{n}\right], \left[\frac{n+2}{n}, \frac{n+4}{n}\right], \ldots \left[\frac{3n-2}{n}, 3\right] \leftarrow n \text{ – sub interval.}$$

43. Estimate the area using the limit sum of rectangle of the graph $f(x) = x^2$ between the line $x = 1$ and $x = 2$ by dividing the interval [1, 2] into equal length of

a.3 – sub interval b. 6 – sub interval c. 10 – sub interval

Solution: We have area (A) = (height) (length)

$$\Delta x = \frac{b-a}{n} = \frac{2-1}{n} = \frac{1}{n} = \frac{1}{3} \longleftarrow n = 3.$$

$A=\sum_{i=1}^{n} A_i=\sum_{i=1}^{n} f(x_i)\Delta x \leftarrow f(x_i)$ the height of the ith rectangle. But

$\Delta x=\frac{b-a}{n}=\frac{2-1}{n}=\frac{1}{n}$ and $b=a+n\,\Delta x$

$\Rightarrow x_i=a+i\,\Delta x=1+\frac{i}{n}$.

$\Rightarrow f(x_i)=f\left(1+\frac{i}{n}\right)=\left(1+\frac{i}{n}\right)^2=1+\frac{2i}{n}+\frac{i^2}{n^2}\leftarrow$ since $f(x)=x^2$

$$A_n=\sum_{i=1}^{n} A_i=\sum_{i=1}^{n} f(x_i)\Delta x=\sum_{i=1}^{n}\left(1+\frac{2i}{n}+\frac{i^2}{n^2}\right)\left(\frac{1}{n}\right)$$

$$=\frac{1}{n}\sum_{i=1}^{n}\left(1+\frac{2i}{n}+\frac{i^2}{n^2}\right)$$

$\Rightarrow A_n=\sum_{i=1}^{n} A_i=\frac{1}{n}\left(\sum_{i=1}^{n}1+\frac{2}{n}\sum_{i=1}^{n} i+\frac{1}{n^2}\sum_{i=1}^{n} i^2\right)\leftarrow\sum$, rule . . .(*)

Since $\sum_{i=1}^{n}1=n$ and $\sum_{i=1}^{n} i=\frac{1}{2}n(n+1)=\frac{n^2+n}{2}$ also

$\sum_{i=1}^{n} i^2=1^2+2^2+3^2+\ldots+n^2=\frac{1}{6}n(n+1)(2n+1)$

$=\frac{2n^3+3n^2+n}{6}$; Then substitute (i) in (*)

$\Rightarrow A_n=\frac{1}{n}\left[n+\frac{2}{n}\left(\frac{n^2+n}{2}\right)+\frac{1}{n^2}\left(\frac{2n^3+3n^2+n}{6}\right)\right]=\frac{7}{3}+\frac{1}{n}+\frac{1}{2n}+\frac{1}{n^2}$

$\therefore A=\frac{7}{3}+\frac{1}{n}+\frac{1}{2n}+\frac{1}{n^2}\leftarrow$ Area of nth sub – interval

a. $n=3$ – sub interval $\Rightarrow A=\frac{7}{3}+\frac{1}{3}+\frac{1}{6}+\frac{1}{9}=\frac{53}{18}\approx 2.94$

b. $n = 6$ – sub interval $\Rightarrow A = \frac{7}{3} + \frac{1}{6} + \frac{1}{12} + \frac{1}{36} = \frac{94}{36} \approx 2.61$

As the number of interval increase without bound then the area:

$$A = \lim_{n\to\infty}\left(\frac{7}{3} + \frac{1}{n} + \frac{1}{2n} + \frac{1}{n^2}\right) = \frac{7}{3} = \int_1^2 x^2 dx$$

Theorem: If the function f is continuous on $[a, b]$ then $\lim_{n\to\infty}\sum_{i=1}^{n} f(x_i)\Delta x$ exist. That is f is integrable so that continuity implies integrability.

Example:

- $\int_{-2}^{2} \frac{1}{x^2-1}\,dx$ is not integrable because $f(x) = \frac{1}{x^2-1}$ is not cont. at $x = -1$ and $x = 1$.
- $\int_{-1}^{3} \frac{1}{x-2}\,dx$ is doesn't exist because not cont. at $x = 2$.

Fundamental theorem of Calculus

This theorem will relate the rectangle and anti-derivative methods for calculating areas, and evaluating definite integrals using anti derivatives.

Theorem **Fundamental theorem of calculus I.**
If f is continuous on $[a, b]$ and F is an anti-derivative of f on $[a, b]$ then $\int_a^b f(x)dx = F(b) - F(a)$, if $F'(x) = f(x)$ for all $x \in [a, b]$.

Illustrative Example

44. Evaluate each definite integral:

a. $\int_1^2 (x^2 - 3)dx$ b. $\int_0^2 (2x^2 - 3x + 2)dx$

Solution: a. $\int_1^2 (x^2 - 3)dx = \left[\frac{x^3}{3} - 3x\right] = \left(\frac{8}{3} - 6\right) - \left(\frac{1}{3} - 3\right) = -\frac{2}{3}$

b. $\int (2x^2 - 3x + 2)dx = \left[\frac{2x^3}{3} - \frac{3x^2}{2} + 2x\right]_0^2$

$$=\left(\frac{16}{3}-\frac{12}{2}+4\right)-(0-0+0)=\frac{10}{3}$$

Properties of Definite integrals

Let f and g be integrable on $[a, b]$ and $C \in [a, b]$:

1. $\int_a^a f(x)dx = 0$
2. $\int_b^a f(x)dx = -\int_a^b f(x)dx$
3. $\int_a^b kf(x)dx = k\int_a^b f(x)dx$
4. $\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx$
5. If $f(x) \geq 0$ on $[a, b]$ then $\int_a^b f(x)dx \geq 0$

- If f is **an even function**, then $\int_{-a}^{a} f(x)dx = 2\int_0^a f(x)dx$.
- If f is **an odd function**, then $\int_{-a}^{a} f(x)dx = 0$.

Example: a. $\int_{-4}^{4} x^2 dx = 2\int_0^4 x^2 dx$

b. $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos x dx = 2\int_0^{\frac{\pi}{2}} \cos x dx$ c. $\int_{-2}^{2} x^3 dx = 0$

d. $\int_{-4}^{4} (x^3 - x)dx = 0$

Illustrative Example

45. Evaluate the integrals

a. $\int_{-3}^{2} |x+1| dx$ b. $\int_{-3}^{4} |x^2 - 4| dx$ c. $\int_{-1}^{2} |x^2 - x| dx$

Solution: a) since $|x + 1| = x + 1$ if $x \geq -1$ and $|x + 1| = -x - 1$ if $x < -1$

$$\Rightarrow \int_{-3}^{2} |x+1| dx = \int_{-3}^{-1} (x+1)dx + \int_{-1}^{2} (-x-1)dx$$

$$=\left(\frac{x^2}{2}+x\right)\Big|_{-3}^{-1}+\left(\frac{-x^2}{2}-x\right)\Big|_{-1}^{2}=-2+\frac{-3}{2}=-\frac{7}{2}$$

b) Since $|x^2-4|=x^2-4$ if $x \le -2$ or $x \ge 2$

and $|x^2-4|=-x^2+4$, if $-2<x<2$. $\Rightarrow \int_{-3}^{4}|x^2-4|\,dx$

$$=\int_{-3}^{-2}(x^2-4)dx+\int_{-2}^{2}(-x^2+4)dx+\int_{2}^{4}(x^2-4)dx$$

$$=\left(\frac{x^3}{3}-4x\right)\Big|_{-3}^{2}+\left(\frac{-x^3}{3}+4x\right)\Big|_{-2}^{2}+\frac{x^3}{3}-4x\Big|_{2}^{4}$$

$= F(2) - F(-3) + G(2) - G(-2) + R(4) - R(2) = 13.$

c) *Exercise left for you. Answer Hint*

$$\int_{-1}^{2}|x^2-x|\,dx=\int_{-1}^{0}(x^2-x)dx+\int_{0}^{1}(-x^2+x)dx+\int_{1}^{2}(x^2-x)dx$$

46. Evaluate the integrals

a. If $f(x)=\begin{cases}x^2+2, x<2\\ 5x-4, x\ge 2\end{cases}$ then find $\int_{-1}^{3}f(x)dx$

b. If $f(x)=\begin{cases}x^2+4, x<2\\ 6x-4, 2\le x\le 3\\ 2x+8, x>3\end{cases}$ then $\int_{0}^{4}f(x)dx$

Solution: a) $\int_{-1}^{3}f(x)dx=\int_{-1}^{2}(x^2+2)dx+\int_{2}^{3}(5x-4)dx$

$$=\left(\frac{x^3}{3}+2x\right)\Big|_{-1}^{2}+\left(\frac{5x^2}{2}-4x\right)\Big|_{2}^{3}$$

$$F(2)-F(-1)+G(3)-G(2)=\frac{35}{2}$$

b) $\int_{0}^{4}f(x)dx=\int_{0}^{2}(x^2+4)dx+\int_{2}^{3}(6x-4)dx+\int_{3}^{4}(2x+8)dx$

$$= \left(\frac{x^3}{3} + 4x\right)\Big|_0^2 + (3x^2 - 4x)\Big|_2^3 + (x^2 + 8x)\Big|_3^4$$

$$= F(2) - F(0) + G(3) - G(2) + R(4) - R(3) = \frac{110}{3}$$

Change of variables for Definite integrals

If $u = g(x)$ has a continuous derivative on the closed interval $[a, b]$ then $\int_a^b f(g(x))g'(x)dx = \int_{g(a)}^{g(b)} f(u)du$

Illustrative Example

47. Find

a. $\int_{-1}^{1} \frac{2x+1}{e^{x^2+x}} dx$

b. $\int_{\frac{\pi^2}{4}}^{\pi} \frac{\cos\sqrt{x}}{\sqrt{x}} dx$

c. $\int_1^4 \frac{\sqrt{1+\sqrt{x}}}{\sqrt{x}} dx$

d. $\int_1^2 x^2(x^3-3)^3 dx$

e. $\int_0^{\pi} \sin^2 x dx$

f. $\int_{-1}^{0} \frac{x+2}{(x^2+4x-1)^4} dx$

g. $\int_{-1}^{1} \frac{1}{1+e^{-x}} dx$

h. $\int_{\frac{\pi}{2}}^{\pi} x\sin x dx$

i. $\int_e^{e^3} x\ln x dx$

Solution: Let $u = x^2 + x \Rightarrow du = (2x + 1)\,dx$ so that when $x = 1 \Rightarrow u = 2$ and $x = -1 \Rightarrow u = 0$.

$$\therefore \int_{-1}^{1} \frac{2x+1}{e^{x^2+x}} = \int_0^2 \frac{du}{e^u} = -e^u\Big|_0^2 = -e^2 - (-1) = 1 - e^2$$

b) Let $u = \sqrt{x} \Rightarrow du = \frac{dx}{2\sqrt{x}}$ so that when $x = \pi^2 \Rightarrow u = \pi$ and

$$x = \frac{\pi^2}{4} \Rightarrow u = \frac{\pi}{2}$$

$$\therefore \int_{\frac{\pi^2}{4}}^{\pi^2} \frac{\cos\sqrt{x}}{\sqrt{x}}dx = \frac{1}{2}\int_{\frac{\pi}{2}}^{\pi} \cos u du$$

$$= \frac{\sin u}{2}\Bigg|_{\frac{\pi}{2}}^{\pi} = \frac{\sin\pi - \sin\left(\frac{\pi}{2}\right)}{2} = -\frac{1}{2}$$

c) Let $u = \sqrt{1+\sqrt{x}} \Rightarrow u^2 = 1+\sqrt{x} \Rightarrow \sqrt{x} = u^2 - 1$

$\Rightarrow 2udu = \dfrac{dx}{2\sqrt{x}} \Rightarrow dx = 4\sqrt{x}\, udu = 4\,(u^2 - 1)u\, du$ so that when

$x = 4 \Rightarrow u = \sqrt{3}$ and when $x = 1 \Rightarrow u = \sqrt{2}$.

$$\therefore \int_1^4 \frac{\sqrt{1+\sqrt{x}}}{\sqrt{x}}dx = \int_{\sqrt{2}}^{\sqrt{3}} \frac{4u^2(u^2-1)}{u^2-1}du = 4\int_{\sqrt{2}}^{\sqrt{3}} u^2 du$$

$$= \frac{u^2}{3}\Bigg|_{\sqrt{2}}^{\sqrt{3}} = \frac{9\sqrt{3} - 2\sqrt{2}}{3}$$

d) Let $u = x^3 - 3 \Rightarrow du = 3x^2\, dx \Rightarrow x^2 dx = \dfrac{du}{3}$. So that when $x = 2 \Rightarrow u =$ 5 and when $x = 1 \Rightarrow u = -2$.

$$\therefore \int_1^2 (x^3-3)^3 x^2 dx = \int_{-2}^{5} u^3 \frac{du}{3} = \frac{1}{3}\left[\frac{u^4}{4}\right]_{-2}^{5} = \frac{u^4}{12}\Bigg|_{-2}^{5} = \frac{609}{12}$$

e) **Recall**: $\cos(2x) = \cos(x + x) = \cos^2 x - \sin^2 x = 1 - 2\sin^2 x$

$$\Rightarrow \sin^2 x = \frac{1}{2}(1 - \cos 2x).$$

$$\therefore \int_0^{\pi} \sin^2 x dx = \int_0^{\pi} \frac{1}{2}(1-\cos 2x) = \left(\frac{1}{2}x - \frac{\cos 2x}{4}\right)\Bigg|_0^{\pi} = \frac{\pi}{2}$$

f) Let $u = x^2 + 4x - 1 \Rightarrow du = 2\,(x + 2)\, dx \Rightarrow \dfrac{du}{2} = (x + 2)dx$

$$\int_{-1}^{0}\frac{x+2}{(x^2+4x-1)}dx=\frac{1}{2}\int_{-4}^{-1}\frac{1}{4}du=-\frac{1}{2}\frac{u^{-3}}{3}\Big|_{-4}^{-1}=\frac{65}{384}$$

g) Since $\int\frac{1}{1+e^{-x}}dx=\int\frac{1}{1+\frac{1}{e^x}}dx=\int\frac{1}{\frac{e^x+1}{e^x}}dx=\int\frac{e^x dx}{1+e^x}$

$$\therefore\int_{-1}^{1}\frac{1}{1+e^{-x}}dx=\int_{-1}^{1}\frac{e^x}{1+e^x}dx=\ln(1+e^x)\Big|_{-1}^{1}$$

$$=\ln(1+e)-\ln\left(\frac{1+e}{e}\right)=1$$

h) **Using inetegration by parts:** Let $u=x\Rightarrow du=dx$ and $dv=\sin x\,dx\Rightarrow v=-\cos x$.

$$\int udv=uv-\int vdu\Rightarrow\int x\sin xdx=-\cos x+\int\cos xdx$$

$$\therefore\int_{\frac{\pi}{2}}^{\pi}x\sin xdx=(-x\cos x)\Big|_{\frac{\pi}{2}}^{\pi}+(\sin x)\Big|_{\frac{\pi}{2}}^{\pi}=\pi\,(0)+(0-1)=\pi-1$$

i) **Using integration by parts:** Let $u=\ln x\Rightarrow du=\frac{dx}{x}$ and $dv=xdx$

$$\Rightarrow xdx\Rightarrow v=\frac{x^2}{2}\text{ so that }\int udv=uv-\int vdu$$

$$\Rightarrow\int x\ln xdx=\frac{x^2\ln x}{2}-\frac{1}{2}\int xdx$$

$$\therefore\int_{e}^{e^3}x\ln xdx=\frac{x^2\ln x}{2}\Big|_{3}^{e^3}-\left(\frac{x^2}{4}\right)\Big|_{e}^{e^3}$$

$$=\left(\frac{3e^6}{2}-\frac{e^2}{2}\right)-\left(\frac{e^6}{4}-\frac{e^2}{4}\right)=\frac{5e^6-e^2}{4}$$

48. If $\int_1^7 f(x)dx=4$ then **find**

a) $\int_0^2 f(3x+1)dx$ b) $\int_{\sqrt{3}}^3 f(x^2-2)dx$

***Solution*: Note:** $\int_1^7 f(x)dx = F(7) - F(1) = 4$, so that:

a) Let $u = 3x + 1 \Rightarrow du = 3\,dx \Rightarrow dx = \frac{du}{3}$. Then

$$\therefore \int_0^2 f(3x+1)dx = \int_1^7 f(u)\frac{du}{3} = \frac{1}{3}[F(7) - F(1)] = \frac{1}{3}(4) = \frac{4}{3}$$

b) Let $u = x^2 - 2 \Rightarrow du = 2x\,dx \Rightarrow x\,dx = \frac{du}{2}$ and when $x = 3 \Rightarrow u = 7$ and when $x = \sqrt{3} \Rightarrow u = 1$.

$$\therefore \int_{\sqrt{3}}^3 f(x^2-1)xdx = \int_1^7 f(u)\frac{du}{2} = \frac{1}{2}[F(7) - F(1)] = \frac{4}{2} = 2$$

Solved Problem

49. $\int_0^2 x^2 \ln(1+x^3)dx$ . . . UEE

A. $\frac{2}{3}\ln 2 - \frac{2}{3}$ B. $6\ln 3 - \frac{10}{3}$ C. $6\ln 3 - \frac{8}{3}$ D. $18\ln 3 - 24$

***Solution*:** Let $u = 1 + x^3 \Rightarrow du = 3x^2dx \Rightarrow x^2dx = \frac{du}{3}$ and when $x = 2 \Rightarrow u = 9$ and when $x = 0 \Rightarrow u = 1$.

$$\therefore \int_0^2 x^2 \ln(1+x^3)dx = \int_1^9 \ln u \frac{du}{3} = \frac{1}{3}(u\ln u - u)\Big|_1^9$$

$$= \frac{1}{3}((9)\ln 9 - 9) - \frac{1}{3}(0-1) = 6\ln 3 - \frac{8}{3}$$ ∴ **Answer: C**

50. If f' is continuous on [0, 1] such that $f(0) = 3$ and $f(1) = -1$ then $\int_0^1 f'(x)\sqrt{1+f(x)}dx$ . . . UEE

A. −12 B. $\frac{14}{3}$ C. $\frac{4\sqrt{2}-2}{3}$ D. $\frac{-16}{3}$

...let $1-x=1-f(x) \Rightarrow du=f'(x)\,dx$ and when $x=1$ $\Rightarrow u=1-f(1)=0$ and $x=0 \Rightarrow u=1-f(0)=4$

$$\int_0^1 f'(x)\sqrt{1-f(x)}dx=\int_4^0 \sqrt{u}du=\frac{2}{3}u^{\frac{3}{2}}\Big|_4^0=0-\frac{16}{3}=\frac{16}{3}$$

...swer: D

$\int_0^1 e^{-x}(1+e^{2x})dx$ ...

... $e-\frac{1}{e}$ B. $\frac{1}{e}-e$ C. $e-\frac{1}{e}-1$ D. $e+\frac{1}{e}$

...lution: $\int_0^1 e^{-x}(1+e^{2x})dx=\int_0^1 (e^{-x}+e^{x})dx=-e^{-x}-e^{x}\Big|_0^1$

$=(-e^{-1}+e)-(-1+1)=e-\frac{1}{e}$ ∴ **Answer: A**

$\int_0^1 (x+1)e^{x}dx$: A. 0 B. 1 C. e D. $\frac{e-1}{e^2}$

Solution: $u=x+1\Rightarrow du=dx$ and $dv=e^{x}dx \Rightarrow v=e^{x}$

$\int udv=uv-\int vdu \Rightarrow \int_0^1 (x+1)e^{x}dx=(x-1)e^{x}-\int e^{x}dx$

$=((x+1)e^{x}-e^{x})\Big|_0^1=(2e-e)-(1-1)=e$ ∴ **Answer: C**

$\int_0^1 \frac{3x^2+1}{e^{x}+x}dx$: A. -2 B. $1-e^{-2}$ C. e^{-2} D. $\frac{2}{e^2}$

Solution: Let $u=x^3+x \Rightarrow du=(3x^2+1)dx$ when $x=1\Rightarrow u=2$ and $x=0\Rightarrow u=0$, $\int_0^2 \frac{du}{e^{u}}=-e^{-u}\Big|_0^2=-e^{-2}-(-1)=1-e^{-2}$

$\int \frac{1}{e^{x}(1+e^{x})}dx$ ... **UEE 2003/11**

... $\ln(e+1)$ B. $\ln(1+e)-\ln 2$ C. $\ln(2+e)$ D. $\ln 2$

∴ **Answer: B**

Solution: Let $u = 1 + f(x) \Rightarrow du = f'(x)\,dx$ and when $x = 1$
$\Rightarrow u = 1 + f(1) = 0$ and $x = 0 \Rightarrow u = 1 + f(0) = 4$

$$\therefore \int_0^1 f'(x)\sqrt{1+f(x)}dx = \int_4^0 \sqrt{u}du = \frac{2}{3}u^{\frac{3}{2}}\Big|_4^0 = 0 - \frac{16}{3} = \frac{16}{3}$$

∴ **Answer: D**

51. $\int_0^1 e^{-x}(1+e^{2x})dx \ldots$

A. $e - \frac{1}{e}$ B. $\frac{1}{e} - e$ C. $e - \frac{1}{e} - 1$ D. $e + \frac{1}{e}$

Solution: $\int_0^1 e^{-x}(1+e^{2x})dx = \int_0^1 (e^{-x} + e^{x})dx = -e^{-x} + e^{x}\Big|_0^1$

$= (-e^{-1} + e) - (-1 + 1) = e - \frac{1}{e}$ ∴ **Answer: A**

52. $\int_0^1 (x+1)e^{x}dx$: A. 0 B. 1 C. e D. $\frac{e-1}{e^2}$

Solution: $u = x + 1 \Rightarrow du = dx$ and $dv = e^{x}dx \Rightarrow v = e^{x}$

$$\therefore \int udv = uv - \int vdu \Rightarrow \int_0^1 (x+1)e^{x}dx = (x+1)e^{x} - \int e^{x}dx$$

$$= ((x+1)e^{x} - e^{x})\Big|_0^1 = (2e - e) - (1 - 1) = e \therefore \textbf{Answer: C}$$

53. $\int_0^1 \frac{3x^2+1}{e^{x^3}+x}dx$: A. -2 B. $1 - e^{-2}$ C. e^{-2} D. $\frac{2}{e^2}$

Solution: Let $u = x^3 + x \Rightarrow du = (3x^2 + 1)dx$ when $x = 1 \Rightarrow u = 2$
and $x = 0 \Rightarrow u = 0$. $\int_0^2 \frac{du}{e^{u}} = -e^{-u}\Big|_0^2 = -e^{-2} - (-1) = 1 - e^{-2}$

∴ **Answer: B**

54. $\int_0^1 \frac{1}{e^{-x}(1+e^{x})}dx \ldots$ **UEE 2003/11**

A. $\ln(e-1)$ B. $\ln(1+e) - \ln 2$ C. $\ln(2+e)$ D. $\ln 2$

Solution: $\int_0^1 \frac{1}{e^{-x}(1+e^x)}dx = \int_0^1 \frac{e^x}{1+e^x}dx = \ln(1+e^x)\Big|_0^1$

$= \ln(1+e) - \ln(2)$ ∴ **Answer: B**

55. $\int_1^3 \frac{\cos\left(\frac{\pi}{x}\right)}{x^2}dx$ is equal to:

A. $\frac{-\sqrt{3}}{4\pi}$ B. $\frac{-\sqrt{3}}{3\pi}$ C. $\frac{\sqrt{3}}{\pi}$ D. $\frac{-\sqrt{3}}{2\pi}$

Solution: $u = \frac{\pi}{x} \Rightarrow du = \frac{-\pi}{x^2}dx \Rightarrow \frac{-du}{\pi} = \frac{dx}{x^2}$,

and when $x = 1 \Rightarrow u = \pi$ and $x = 3 \Rightarrow u = \frac{\pi}{3}$.

$\therefore \int_{\pi}^{\frac{\pi}{3}} \cos u \frac{du}{3} = -\frac{\sin u}{\pi}\Big|_{\pi}^{\frac{\pi}{3}} = \frac{-1}{\pi}\left[\frac{\sqrt{3}}{2} - 0\right] = \frac{-\sqrt{3}}{2\pi}$ ∴ **Answer: D**

56. If $f(x) = \begin{cases} \sin x, x \le 0 \\ 3x\sqrt{x^2+1}, x > 0 \end{cases}$ then $\int_{-\frac{\pi}{2}}^{\sqrt{8}} f(x)dx$ ... **UEE 2004**

A. 17 B. 25 C. 27 D. $3\sqrt{8}$

Solution:

$$\int_{-\frac{\pi}{2}}^{0} f(x)dx + \int_0^{\sqrt{8}} f(x)dx = \int_{-\frac{\pi}{2}}^{0} \sin x dx + \int_0^{\sqrt{8}} 3x\sqrt{x^2+1}\,dx$$

Let $u = x^2 + 1 \Rightarrow du = 2xdx \Rightarrow xdx = \frac{du}{2}$, when $x = \sqrt{8}$

$\Rightarrow u = 9$ and $x = 0 \Rightarrow u = 1$. Hence

$$\int_0^{\sqrt{8}} 3x\sqrt{x^2+1}\,dx = \int_1^9 \frac{3}{2}\sqrt{u}du = u^{\frac{3}{2}}\Big|_1^9 = 27 - 1 = 26$$

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$$\Rightarrow \int_{-\frac{\pi}{2}}^{0} \sin x dx + \int_{0}^{\sqrt{8}} 3x\sqrt{x^2+1}\, dx = -\cos x \Big|_{-\frac{\pi}{2}}^{0} + 26$$

$$= -1 + 26 = 25 \qquad \therefore \textbf{Answer: B}$$

57. $\int_1^2 x\left(\frac{1}{x^2} + 2\ln x\right)dx$ is equal to:

A. $2\ln 2 - 1$ B. $5\ln 2 + \frac{5}{2}$ C. $5\ln 2 + \frac{5}{2}$ D. $5\ln 2 - \frac{3}{2}$

Solution: Implies $\int_1^2 \frac{dx}{x} + 2\int_1^2 \frac{dx}{x} + 2\int_1^2 x\ln x dx \leftarrow$ integration by parts $\left(\ln x + x^2 \ln x - \frac{x^2}{2}\right)\Big|_1^2 = \ln(2) + 4\ln 2 - 2 - \left(0 - 0 - \frac{1}{2}\right)$

$$= 5\ln 2 - \frac{3}{2}$$

58. If $f(1) = 3, f(0) = 2$, then $\int_0^1 \frac{f'(x)}{f(x)} dx$

A. $\ln 3$ B. $\ln 2$ C. $\ln\left(\frac{3}{2}\right)$ D. $\ln 6$

Solution: Let $u = f(x) \Rightarrow du = f'(x)\, dx = \ln 3 - \ln 2 = \ln\left(\frac{3}{2}\right)$

$\therefore$ **Answer: C**

59. If $f(0) = 5, f(1) = 3$, then $\int_0^1 e^x (f(x) + f'(x))dx$ is

A. $3e - 5$ B. $2e$ C. $5 - e$ D. $-2 + e$

Solution:

$$\int_0^1 e^x (f(x) + f'(x))dx = e^x f(x)\Big|_0^1 = e^1 f(1) - e^0 f(0) = 3e - 5$$

$\therefore$ **Answer: A**

Fundamental theorem of Calculus (2)

i) If the function F is defined on $[a, b]$ by $F(x) = \int_a^x f(t)dt$ then $F'(x) = f(x)$ for all $x \in [a, b]$. That is $\frac{d}{dx}\int_a^x f(t)dt = f(x)$.

Example

a. If $F(x) = \int_0^x \cos t dt$, then **find** $F'(x)$:

Solution:

$$\int_0^x \cos t dt = \sin t \Big|_0^x = \sin x$$

$$\therefore F'(x) = \frac{d}{dx}\int_0^x \cos t dx = \frac{d}{dx}\sin x = \cos x$$

b. If $F(x) = \int_{\frac{\pi}{2}}^{x^3} \cos t dx$, then find $F'(x)$

Solution: $F(x) = \int_{\frac{\pi}{2}}^{x^3} \cos t dt = \sin t \Big|_{\frac{\pi}{2}}^{x^3} = \sin(x^3) - 1$

$\Rightarrow F'(x) = \cos(x^3)(3x^2) = 3x^2 \cos(x^3)$.

c. If $F(x) = \int_1^{x^2+1} \frac{1}{t} dt$ then $F'(x)$ . . . UEE

A. $\ln(x^2 + 1)$ B. $2x \ln(x^2 + 1)$ C. $\frac{2x}{x^2+1}$ D. $\frac{1}{x^2+1} + c$

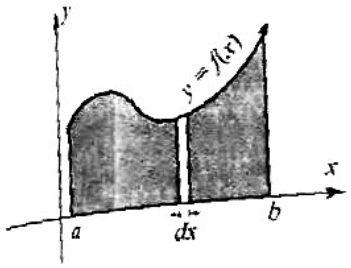
Solution: $F(x) = \int_1^{x^2+1} \frac{1}{t} dt = \ln t \Big|_1^{x^2+1} = \ln(x^2 + 1)$

$$\Rightarrow F'(x) = \frac{2x}{x^2+1}$$ ∴ **Answer: C**

Application of Integral Calculus

A. **Area between a curve and axis**

1. If $y = f(x)$ is non-negative and continuous on $[a, b]$ then the area of the region bounded by the graph of $y = f(x)$ and by the line $x = a$ and $x = b$ is given by: Area (A) = $\int_a^b f(x)dx$.



Note: Area = (height)(width) ⟶

$dA = f(x)dx \rightarrow A = \int_a^b f(x)dx$.

Illustrative Example

60. Find the area A of the region between the graph of f the x-axis from $x = a$ to $x = b$ when:

a. $f(x) = x^3 - 9x$; $x = -2$ and $x = 2$

b. $f(x) = \frac{1}{x}, x = -6$ and $x = -2$

c. $f(x) = \ln x$, $x = e$ and $x = e^3$

d. $f(x) = 18 - 2x^2$, $x = -4$ and $x = 2$

e. $f(x) = x^3$, $x = -1$ and $x = 2$

f. $f(x) = 2x^2 - 3x + 2$, $x = 0$ and $x = 2$

g. $g(x) = \frac{x}{x^2 + 1}$, $x = 0$ and $x = 3$

Solution: a. $A = A_1 + A_2$;

$$A_1 = \int_{-2}^{0} (x^3 - 9x)dx = \left(\frac{x^4}{4} - \frac{9x^x}{2}\right)_{-2}^{0} = 14$$

But on $(0, 2), f(x) < 0 \Rightarrow A_2 = \int_0^2 (-x^3 + 9x)dx$

$$= \left(\frac{-x^4}{4} + \frac{9x^2}{2}\right)_0^2 = \frac{-16}{4} + \frac{36}{2} = 14$$

$\therefore A = A_1 + A_2 = 14 + 14 = 28$ unit square

b. **Since $f(x) < 0$ when $x \in (-6, -2)$** so that $f(x) = -\frac{1}{x}$.

$$\therefore A = \int_{-6}^{-2} -\frac{1}{x}dx = -\int_{-6}^{-2} \frac{1}{x}dx = -\ln|x|\Big|_{-6}^{-2}$$

$$= -\ln|2| + \ln|-6| = \ln 6 - \ln 2 = \ln\left(\frac{6}{2}\right) = (\ln 3) \text{ unit sq.}$$

c. $A = \int_{e}^{e^3} (\ln x)dx = (x\ln x - x)\Big|_{e}^{e^3} = (3e^3 - e^3) - (e - e) = 2e^3$

d. **Since $f(x) < 0$ on [-4, -3] $\Rightarrow A = A_1 + A_2$**

$$\therefore A = \int_{-4}^{-3} (-18 + 2x^2)dx + \int_{-3}^{2} (18 - 2x^2)dx$$

$$= \left(-18x + \frac{2x^3}{3}\right) + \left(18x - \frac{2x^3}{3}\right)\Big|_{-3}^{2} = (F(-3) - F(-4)) + (6(2) - G(-3))$$

$$= [F(-3) - F(-4)] + [G(2) - F(-3)] = \frac{20}{3} + \frac{200}{3} = \frac{220}{3}$$

e. $f(x) < 0$ on $[-1, 0] \Rightarrow A = A_1 + A_2$

$$\therefore A = \int_{-1}^{0} -x^3dx + \int_{0}^{2} x^3dx = \frac{-x^4}{4}\Big|_{-1}^{0} + \frac{x^4}{4}\Big|_{0}^{2} = \frac{1}{4} + 4 = \frac{17}{4}$$

f. **Since $f(x) > 0$ on interval [0, 2].**

$$\therefore A = \int_{0}^{2} f(x)dx = \int_{0}^{2} (2x^2 - 3x + 2)dx = \left[\frac{2x^3}{3} - \frac{3x^2}{2} + 2x\right]_{0}^{2}$$

$$= \left(\frac{16}{3} - 6 + 4\right) - (0 - 0 + 0) = \frac{10}{3}$$

g. $f(x) > 0$ on $[0, 3] \Rightarrow A = \int_{0}^{3} \frac{x}{x^2 + 1}dx$

$$\Rightarrow A = \frac{1}{2}\int_{0}^{3} \frac{2x}{x^2 + 1}dx = \frac{1}{2}\ln(x^2 + 1)\Big|_{0}^{3} = \frac{1}{2}\ln(10)$$

61. Find the area bounded by the graph of:

a. $f(x) = |x^2 - 4|$, x – axis and the line $x = -3$ and $x = 4$.

b. $f(x) = |x + 3|$, x – axis and the line $x = -4$ and $x = 1$

Solution:

$$|x^2-4|=\begin{cases}x^2-4 \text{ if } x\le -2 \text{ or } x\ge 2\\ -x^2+4 \text{ if } -2<x<2\end{cases} \Rightarrow A=\int_{-3}^{4}|x^2-4|\,dx$$

$\therefore$ Area, (A) = $A_1 + A_2 + A_3$ =

$$\int_{-3}^{-2}(x^2-4)dx+\int_{-2}^{2}(4-x^2)dx+\int_{2}^{4}(x^2-4)dx$$

$$=\left(\frac{x^3}{3}-4x\right)\Big|_{-3}^{-2}+\left(4x-\frac{x^3}{3}\right)\Big|_{-2}^{2}+\left(\frac{x^3}{3}-4x\right)\Big|_{2}^{4}=\frac{71}{3}$$

b. $|x+3|=x+3$ if $x\ge -3$ and $|x+3|=-x-3$ if $x<-3$

$$\therefore A=\int_{-4}^{1}|x+3|\,dx=\int_{-4}^{-3}(-x-3)dx+\int_{-3}^{4}(x+3)dx=25$$

62. Find the area A of the entire region R bounded by:
   a. $f(x)=x^3-x$ and x-axis
   b. $f(x)=x^3-x^2-2x$ and x-axis

Solution: $f(x)=x^3-x<0$ on $(0, 1) \Rightarrow f(x)=-x^3+x$ and x^3-x

$$=x(x-1)(x+1)=0\Rightarrow x=0,-1,x=1$$

$$\therefore A=\int_{-1}^{0}(x^3-x)dx+\int_{0}^{1}(-x^3+x)dx=\frac{1}{4}+\frac{1}{4}=\frac{2}{4}=\frac{1}{2}$$

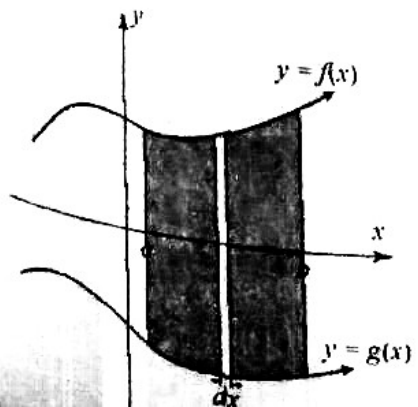
b. $x^3-x^2-2x=0\Rightarrow x(x^2-x-2)=x(x-2)(x+1)=0\Rightarrow x=0, 2, -1$
since $f(x)=x^3-x^2-2x<0$ on $(0, 2) \Rightarrow f(x)=-x^3+x^2+2x$.

$$\therefore A=\int_{-1}^{0}(x^3-x^2-2x)dx+\int_{0}^{2}(-x^3+x^2+2x)dx$$

$$=\frac{x^4}{4}-\frac{x^3}{3}-x^2\Big|_{-1}^{0}+\left(\frac{-x^4}{4}+\frac{x^3}{3}+x^2\right)\Big|_{0}^{2}=\frac{5}{12}+\frac{8}{3}=\frac{37}{12}$$

The Area between two curves

The area A bounded by two continuous curve $y=f(x)$ and $y=g(x)$ on $[a, b]$ with $f(x)\ge g(x)$, for all x in $[a, b]$ is given by:



$$A=\int_{a}^{b}(f(x)-g(x))dx$$

- $f(x)$ is top curve
- $g(x)$ is bottom curve

$$\therefore A=\int_{-2}^{2}[4-x^2-(3x^2-12)]dx=\int_{-2}^{2}(-4x^2+16)dx$$

$$\Rightarrow A=\frac{-4x^2}{3}+16x\Big|_{-2}^{2}=\left(\frac{-32}{3}-32\right)-\left(\frac{32}{3}-32\right)=\frac{128}{3}$$

c) **The intersection of g and f is $f(x)=g(x)$**, then $\Rightarrow x^2+2x+1=2x+5\Leftrightarrow x^2-4=0\Rightarrow x=-2$and $x=2$ since $g(x)\geq f(x)$ for all when $x\in[-2,2]$.

$$\therefore A=\int_{-2}^{2}[g(x)-f(x)]dx=\int_{-2}^{2}[2x+5-(x^2+2x+1)]dx$$

$$=\int_{-2}^{2}(4-x^2)dx$$

$$\therefore A=\left(4x-\frac{x^3}{3}\right)\Big|_{-2}^{2}=\left(8-\frac{8}{3}\right)-\left(-8+\frac{8}{3}\right)=\frac{32}{3}\text{ unit sq.}$$

d) **The interaction point of $f(x)$ and $g(x)$ is $f(x)=g(x)$** $\Rightarrow x^2-4x+3=-x^2+2x+3\Leftrightarrow 2x^2-6x=0\Rightarrow x=0;\ x=3$ since $g(x)\geq f(x)$ for all $x\in[0,3]$.

$$\therefore A=\int_{0}^{3}[g(x)-f(x)]dx=\int_{0}^{3}(-x^2+2x+3-x^2+4x-3)dx$$

$$\Rightarrow A=\int_{0}^{3}(-2x^2+6x)dx=\left(\frac{-2x^3}{3}+3x^2\right)\Big|_{0}^{3}=9\text{ unit sq.}$$

e) **The intersection of $g(x)$ and $f(x)$ is $f(x)=g(x)$** $\Rightarrow 3x^2-x^2-10x=-x^2+2x\Leftrightarrow 3x^3-12x=0\Rightarrow x=0,\ x=-2,2$ since $f(x)\geq g(x)$, for all $x\in[-2,0]$ but $g(x)\geq f(x)$ for all $x\in[0,2]\Rightarrow A=\int_{-2}^{0}[f(x)-g(x)]+\int_{0}^{2}[g(x)-f(x)]dx$

$$\Rightarrow A=\int_{-2}^{0}[(3x^3-x^2-10x)-(-x^2+2x)]dx$$

$$+\int_{0}^{2}(-x^2+2x-3x^3+x^2+10x)dx$$

$$\therefore A=\int_{-2}^{0}(3x^3-12x)dx+\int_{0}^{2}(-3x^3+12x)dx$$

$$=\frac{3x^4}{4}-6x^2\Big|_{-2}^{0}+\frac{-3x^4}{4}+6x\Big|_{0}^{2}=12$$

f) **Exercise left for you (Tip)**

g) **The intersection of $f(x)$ and $g(x)$ is $f(x) = g(x)$**

$\Rightarrow x^2 - 3x = 4 \Leftrightarrow x^2 - 3x - 4 = 0 \Rightarrow x = -1$ and $x = 4$ since $g(x) \geq f(x)$ for all $x \in [-1, 4]$.

$$\Rightarrow A = \int_{-1}^{4} [g(x) - f(x)]dx = \int_{-1}^{4} (4 - x^2 + 3x)dx$$

$$= 4x - \frac{x^3}{3} + \frac{3x^2}{2}\Big|_{-1}^{4} = \frac{143}{6}$$

65. Find the area of the region bounded by:

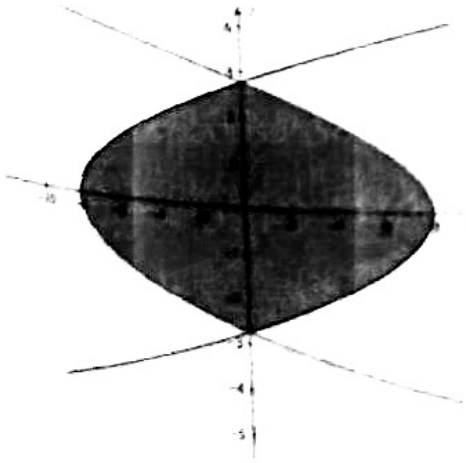
a. $x = 8 - y^2$ and $x = y^2 - 8$ c. $y = \frac{1}{2}x$ and $x = 8$

b. $x = 3 - y^2$ and $y = x - 1$

Solution: Area, $A = \int_c^d x dy$ where $x = f(y)$. Here the role of x and y is reverse to make easy so that to find the area, it is *better to integrate with respect to y*. Then use:

$A = \int_c^d [f(y) - g(y)]dy$ for $f(y) \geq g(y)$, for all $y \in [c, d]$

a) Let $f(y) = 8 - y^2$ and $g(y) = y^2 - 8$. The intersection of $f(y)$ and $g(y)$ is $f(y) = g(y) \Rightarrow 8 - y^2 = y^2 - 8 \Leftrightarrow 2y^2 - 16 = 0$



$\Rightarrow y = -2\sqrt{2}$ and $y = 2\sqrt{2}$ since $f(y) \geq g(y)$ for all $y \in [-2\sqrt{2}, 2\sqrt{2}]$

$$\Rightarrow A = \int_{-2\sqrt{2}}^{2\sqrt{2}} [f(y) - g(y)]dy$$

$$= \int_{-2\sqrt{2}}^{2\sqrt{2}} (8 - y^2 - y^2 + 8)dy = \int_{-2\sqrt{2}}^{2\sqrt{2}} (16 - 2y^2)dy$$

$$= \int_{-2\sqrt{2}}^{2\sqrt{2}} (16 - 2y^2)dy = \frac{128\sqrt{2}}{3}$$

b) Let $f(y) = 3 - y^2$ and $y = x - 1 \Rightarrow x = y + 1 \Rightarrow g(y) = y + 1$. The intersection point is $f(y) = g(y) \Rightarrow 3 - y^2 = y + 1 \Leftrightarrow y^2 + y - 2 = 0$ $\Rightarrow y = -2$ and $y = 1$. Since $f(y) \geq g(y)$ for all $y \in [-2, 1]$

$$\Rightarrow A = \int_{-2}^{1} (f(y) - g(y))dy = \int_{-2}^{1} (3 - y^2 - y - 1)dy$$

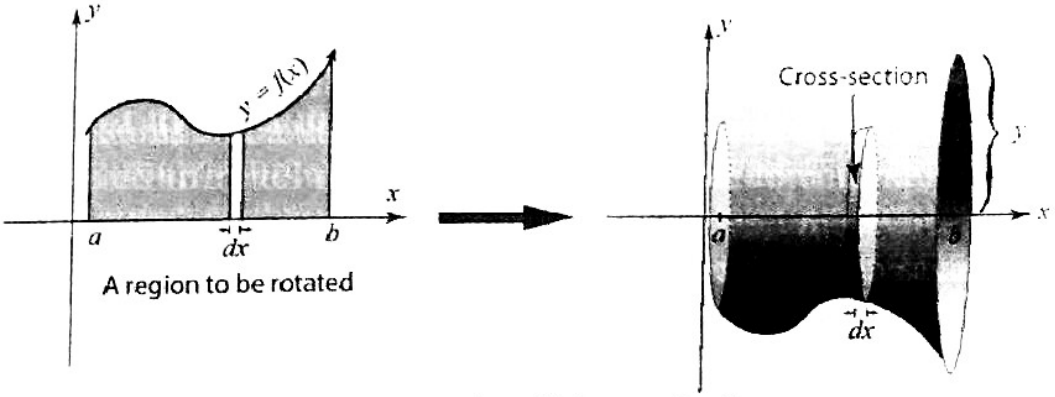
$$\therefore A = \frac{-y^3}{3} - \frac{y^2}{2} + 2y \Big|_{-2}^{1} = \left[\left(-\frac{1}{3} - \frac{1}{2} + 2\right) - \left(\frac{8}{3} - 2 - 4\right)\right] = \frac{9}{2}$$

c) Exercise left for you. (Answer 36 sq. unit)

II. Volume of Revolution

Another important application of integration is *in finding the volume of three dimensional solid*. If the region in the plane is revolved (rotated) about a line the resulting solid is a **called solid revolution** and the line is called **the axis of revolution**. The volume of solid generated by rotating is called **volume of revolution.**

Methods of determining the volume of a solid of revolution about the x-axis



A solid revolution

To find the volume:

- In precalculus (volume of cylinder) = (base area)(height) = $\pi r^2 h$.
- In calculus the volume generated by rotating infinite small rectangle is:
  Δv = (Area of circular cross section)(width of cross section)
  $\Delta v = (\pi y^2)\Delta x \Rightarrow dv = \pi y^2 dx$, but
  $y = f(x) \Rightarrow dv = \pi (f(x)^2)\, dx \leftarrow$ infinite small volume
- The volume of a solid can be obtained by integrating the cross sectional area from one end of the solid to other. **The volume of a**

solid revolution with disk (circular cross section) method is given by:

i) volume $= v = \int_a^b \pi(f(x)^2)dx$ ←Horizontal axis of revolution (about x-axis).

ii) volume $= v = \int_c^d \pi[f(y)]^2 dy$ ← revolution about y-axis (vertical)

Illustrative Example

66. Find the volume of the solid of revolution generated by revolving the region bounded by the given function about: i) the x axis ii) **the y axis**

a. $y = 3x$; $x = 0$ and $x = 1$ c. $y = x^3$, $x = -2$ and $x = 1$

b. $y = -x + 1$, $x = -2$ and $x = 1$ d. $y = x^2 + 2$, $x = -1$ and $x = 2$

Solution: Here

a. i) **volume about x-axis**: $v = \int_a^b \pi[f(x)]^2 dx$

$$\Rightarrow v = \int_0^1 \pi(3x)^2 dx = \int_0^1 \pi 9x^2 dx = \pi[3x^3]_0^1 = 3\pi$$

ii) **volume revol. about y axis**: $v = \int_c^d \pi[f(y)]^2 dy$,

but $f(y) = x$ since $y = 3x \Rightarrow x = \frac{y}{3} \Rightarrow f(y) = \frac{y}{3}$ and

if $x = 0 \Rightarrow y = 0$ and if $x = 1$ then $y = 3$.

$$\therefore v = \int_0^3 \pi(f(y))^2 dy = \int_0^3 \pi\left(\frac{y}{3}\right)^2 dy = \pi\left[\frac{y^3}{27}\right]_0^3 = \pi$$

b. i) **volume revol. about x axis**: $y = -x + 1 \Rightarrow f(x) = -x + 1$

$$\Rightarrow v = \int_{-2}^1 \pi(-x+1)^2 dx = \int_{-2}^1 \pi(x^2 - 2x + 1)dx$$

$$= \pi\left[\frac{x^3}{3} - x^2 + x\right]_{-2}^1 = 9\pi$$

ii) **vol. revolution about y axis**: since $y = -x + 1$

$\Rightarrow x = 1 - y \Rightarrow f(y) = 1 - y$. Then vol $= \int_c^d \pi[f(y)]^2 dy$

when $x = 1 \Rightarrow y = 0$ and when $x = -2 \Rightarrow y = 3$.

$$\therefore v=\int_0^3 \pi(1-y)^2\,dy=\int_0^3 \pi(1-2y+y^2)dy$$

$$=\pi\left[y-y^2+\frac{y^3}{3}\right]_0^3=3\pi$$

c.i) **vol. about *x*-axis** since $y=x^3\Rightarrow f(x)=x^3$

$$\Rightarrow v=\int_{-2}^1 \pi(x^3)^2\,dx=\pi\left[\frac{x^7}{7}\right]_{-2}^1=\pi\left[\frac{1}{7}+\frac{128}{7}\right]=\frac{129\pi}{7}$$

ii) **vol. about *y* – axis** since $y=x^3\Rightarrow x=\sqrt[3]{y}$

$\Rightarrow f(y)=\sqrt[3]{y}$ and when $x=1\Rightarrow y=1$ and $x=-2\Rightarrow y=-8$

$$\therefore v=\int_{-8}^1 \pi[f(y)]^2\,dy=\int_{-8}^1 \pi(y^{\frac{1}{3}})^2\,dy=\frac{3\pi 3}{5}(y^{\frac{5}{3}})\Big|_{-8}^1=\frac{99\pi}{5}$$

d. *Exercise left for you (tip).*

67. Find the volume of the solid formed by revolving the region bounded by the given graph of function about x-axis.

a. $f(x)=\ln x$, $x=1$ and $x=e$

b. $f(x)=\sin x$, $x=\frac{\pi}{4}$ and $x=\frac{\pi}{3}$

c. $f(x)=\cos x$, $x=0$, and $x=\frac{\pi}{3}$

d. $f(x)=3^x$, $x=-2$ and $x=1$

e. $f(x)=e^x+e^{-x}$, $x=-1$ and $x=1$

Solution: a) $v=\int_1^e \pi[f(x)]^2\,dx=\int_1^2 \pi(\ln x)^2\,dx$, let $u=(\ln x)^2$

$$\Rightarrow du=\frac{2\ln x}{x}dx \text{ and } dv=dx\Rightarrow v=x.$$

$$\therefore v=\int_1^e \pi(\ln x)^2\,dx=\pi\left[x(\ln x)^2-2\int \ln x\,dx\right]\leftarrow \text{using}$$

integration by parts.

$$\Rightarrow v=\pi\left[x(\ln x)^2-2(x\ln-x)\right]\Big|_1^e=(e-2)\pi$$

b) $v=\int_{\frac{\pi}{4}}^{\pi}\pi(\sin x)^2\,dx=\int_{\frac{\pi}{4}}^{\frac{\pi}{3}}\pi\sin^2 x\,dx=\int_{\frac{\pi}{4}}^{\frac{\pi}{3}}\pi\frac{(1-\cos 2x)dx}{2}$

Using $\cos 2x=\cos^2 x-\sin^2 x$ and $\cos^2 x+\sin^2 x=1$.

$$\Rightarrow v=\left[\pi\left(\frac{1}{2}x-\frac{\sin 2x}{4}\right)\right]_{\frac{\pi}{4}}^{\frac{\pi}{3}}=\frac{\pi^2}{6}-\frac{\pi^2}{8}+\frac{\pi}{8}=\frac{\pi^2+3\pi}{24}$$

c) *Exercise left for you (tip).*

d) $\text{vol}=\int_{-2}^{1}\pi(3^x)^2\,dx=\int_{-2}^{1}\pi(3^{2x})dx=\pi\left[\frac{3^{2x}}{2\ln(3)}\right]_{-2}^{1}$

$$\therefore v=\frac{\pi}{2}\left[\frac{3^2-3^{-4}}{\ln 3}\right]=\frac{4\pi}{81\ln 3}$$

e) $\text{vol}=\pi\int_{-1}^{1}(e^x+e^{-x})^2\,dx=\pi\int_{-2}^{1}(e^{2x}+2+e^{-2x})dx$

$$\therefore v=\pi\left[\frac{e^{2x}}{2}+\frac{-e^{-2x}}{2}+2x\right]_{-1}^{1}=\pi\left[\frac{e^2}{2}-\frac{e^{-2}}{2}+2\right]-\left(\frac{e^{-2}}{2}-\frac{e^2}{2}-2\right)$$

$$\therefore v=\pi\left[\frac{e^{2x}}{2}-\frac{e^{-2x}}{2}+2x\right]_{-1}^{1}=\pi\left[\frac{e^2}{2}-\frac{e^{-2}}{2}+2-\left(\frac{e^2}{2}-\frac{e^2}{2}-2\right)\right]$$

$=(e^2-e^{-2}+4)\pi$

Volume of sphere

- Verify that the volume of sphere is $\frac{4}{3}\pi r^3$ sphere with radius r is generated by rotating a semicircle; $x^2+y^2=r^2$

$$\Rightarrow y=\sqrt{r^2-x^2}$$

$$\therefore v=\int_{-4}^{4}\pi\left(\sqrt{r^2-x^2}\right)^2dx=\pi\int_{-r}^{r}(r^2-x^2)dx$$

$$\Rightarrow v=\pi\left(r^2x-\frac{x^3}{3}\right)_{-r}^{r}=\pi(2)\left[r^2x-\frac{x^3}{3}\right]_0^{r}=2\pi\left(r^3-\frac{r^3}{3}\right)=\frac{4\pi r^3}{3}$$

Illustrative Example

68. Find the volume of hemisphere generated by circle:

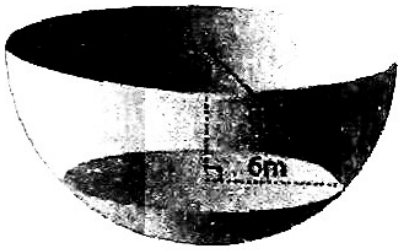
a. $x^2+y^2=9$ b. $x^2+y^2=1$

Solution: a) Since $r=3$, $\Rightarrow v=\frac{2}{3}\pi r^3=\frac{2}{3}\pi(9)=6\pi$

b) Since $r=1\Rightarrow v=\frac{2}{3}\pi r^3=\frac{2}{3}\pi$

69. A bowl shaped like hemisphere of radius 10 m contains some water. If the radius of surface of water is 6m, then find the volume of water.

Solution: A hemisphere generated by rotating the quarter of circle whose equation $x^2+y^2=10^2$.



Hemispherical bowl

$\Rightarrow x=\sqrt{100-y^2}, 0\le x\le 10,$ and $0\le y\le 10$

- To find the limit of integration: since radius of water, 6 m, so let $x=6\Rightarrow 36+y^2=100$

$\Rightarrow y=\sqrt{100-36}=8\text{m}$ and when $x=0\Rightarrow y=10$.

∴ Volume of water,

$$v=\int_8^{10}\pi\left(\sqrt{100-y^2}\right)^2dy=\int_8^{10}\pi(100-y^2)dy$$

$$\Rightarrow v=\pi\left[100y-\frac{y^3}{3}\right]_8^{10}=\pi\left[\left(1000-\frac{1000}{3}\right)-\left(800-\frac{512}{3}\right)\right]$$

$$=\frac{112\pi\text{m}^3}{3}$$

70. A bowl shaped like hemisphere of radius 8m contains some water to depth of the water is 3m, then find:

a. Volume of water b. the radius of the surface of water

Solution: A hemisphere generated by rotating the quarter of the circle whose equation $x^2+y^2=8^2\Rightarrow x=\sqrt{64-y^2}, 0\le y\le 8$ and $0\le x\le 8$.

To find the limit of integration:

∴ volume of water: $\int_5^8\pi\left(\sqrt{64-y^2}\right)^2dy=\int_5^8\pi(64-y^2)dy$

$\Rightarrow$vol. of water

$$\pi\left[64y-\frac{y^3}{3}\right]_5^8=\left(512-\frac{512}{3}-\left(320-\frac{125}{3}\right)\right)\pi=63\pi\text{m}^3$$

The radius of the surface of water is obtained when

$y=5\Rightarrow x=\sqrt{64-25}=\sqrt{39}\leftarrow$ radius of water surface.

71. Find the volume of frustum of a right circular cone has radii R and r, and height h in terms of h, R and r.

Solution: Let (0, r) and (h, R) point on the line to be rotated. The equation of the line to be rotated is $\frac{y-r}{x-0}=\frac{\text{R}-r}{h}$

$\Rightarrow y=\frac{R-r}{h}x+r$; Now the frustum generated by rotating the graph of $y=\left(\frac{R-r}{h}\right)x+r$ on [0, h] about the x-axis

$$v=\int_0^h\pi\left(\frac{R-r}{h}x+r\right)^2dx=\frac{\pi h}{3(R-r)}[R^3-r^3]$$

$$\therefore v=\frac{\pi h}{3(R-r)}((R-r)(R^2+Rr+r^2))=\frac{\pi h}{3}[R^2+Rr+r^2]$$

72. Find the volume of the cone, from #71.

Solution: Set $\frac{y}{x}=\frac{R}{h}\Rightarrow y=\frac{R}{h}x,\Rightarrow v=\int_0^h\pi\left(\frac{R}{h}x\right)^2dx$

$$\Rightarrow v=\pi\left[\frac{R^2}{h^2}\frac{x^3}{3}\right]_0^h=\frac{\pi R^2(h^3)}{3h^2}=\frac{1}{3}\pi R^2h$$

Volume of a solid or revolution generated by two curves (Washer cross sections)

If r and R are the inner and outer radii of the washer circular cross section and Δx is the width of cross-section then volume generated by two curves is given by:

Volume of cross section = $\pi(\text{R}^2-r^2)\Delta x$

- Let the outer radius: $\text{R}=f(x)$

- Let the inner radius $r = g(x)$ then the volume of solid revolution is given by:

$$v = \int_a^b \pi[f(x)]^2 - \left([g(x)]^2\right)dx \text{ if } f(x) \geq g(x)$$

73. Find the volume of solid generated by revolving the region enclosed by the graphs of the equations about the x-axis.
a. $f(x) = x^2 + 2$ and $g(x) = 3$
b. $f(x) = x^2$ and $g(x) = 4x - x^2$
c. $f(x) = \sqrt{x}$ and $g(x) = x^3$

Solution: First determine the intersection of f and g (set $f(x) = g(x)$)

a) $f(x) = g(x) \Leftrightarrow x^2 + 2 = 3 \Rightarrow x^2 = 1 \Rightarrow x = -1$ and $x = 1$

$$\Rightarrow \text{vol} = \int_{-1}^{1} \pi[(g(x))^2 - (f(x))^2]dx = \int_{-1}^{1} \pi(3^2 - (x^2+2)^2)dx$$

$$\Rightarrow \text{vol} = \int_{-1}^{1} \pi(5 - x^4 - 4x^2)dx = \pi\left[5x - \frac{x^5}{5} - \frac{4x^3}{3}\right]_{-1}^{1} = \frac{104\pi}{15}$$

b) **Intersection**: $f(x) = g(x) \Rightarrow x^2 = 4x - x^2 \Rightarrow 4x - 2x^2 = 0 \Rightarrow 2x(2 - x) = 0$
$\Rightarrow x = 0$ and $x = 2$, since $g(x) \geq f(x)$, for all $x \in [0, 2]$

$$\therefore v = \int_0^2 \pi[(4x - x^2)^2 - (x^2)^2]dx = \int_0^2 \pi(16x^2 - 8x^3)dx$$

$$v = \pi\left(\frac{16}{3}x^3 - 2x^4\right)\Big|_0^2 = \pi\left(\frac{128}{3} - 32\right) = \frac{32}{3}\pi$$

c. **The intersection of f and g**: $f(x) = g(x) \Leftrightarrow \sqrt{x} = x^3 \Rightarrow x = x^6$
$\Rightarrow x^6 - x = 0 \Rightarrow x(x^5 - 1) = 0 \Rightarrow x = 0$ and $x = 1$ since $f(x) \geq g(x)$ on $[0, 1]$.

$$\therefore v = \int_0^1 \pi\left(\left(\sqrt{x}\right)^2 - (x^3)^2\right)dx = \int_0^1 \pi(x - x^6)dx$$

$$= \pi\left[\frac{x^2}{2} - \frac{x^7}{7}\right]_0^1 = \frac{5\pi}{14}$$

74. Find the volume of solid of revolution about the y-axis generated by the region enclosed by the graphs of the given function.
a. $y = x^2$ and $y = 2x$ rotates about y-axis.
b. $y = e^x$, y-axis and $y = e$, rotates about y-axis.

Solution: a) Intersection of $y = x^2$ and $y = 2x$ is $x^2 = 2x$

$\Rightarrow x^2 - 2x = 0 \Leftrightarrow x(x-2) = 0 \Rightarrow x = 0$ and $x = 2$

Since about y-axis, so that y-axis so that $y = x^2 \Rightarrow x = \sqrt{y}$ and $y = 2x \Rightarrow x = \dfrac{y}{2}$, also limit of integration when $x = 0 \Rightarrow y = 0$ and $x = 2 \Rightarrow y = 4$.

$$\therefore v = \int_0^4 \pi\left(\sqrt{y}\right)^2 - \left(\frac{y}{2}\right)^2 dy = \int_0^4 \pi \frac{1}{2}\left(y - \frac{y^2}{4}\right)dy$$

$$\therefore v = \pi\left[\frac{y^2}{2} - \frac{y^3}{12}\right]_0^4 = \pi\left[8 - \frac{64}{12}\right] = \frac{8\pi}{3}$$

b) The intersection of $y = e^x$ and $y = e$ is $e^x = e \Rightarrow x = 1$ and when $x = 0 \Rightarrow y = 1$, since about rotation about y-axis, so that, $y = e^x \Rightarrow x = \ln y$. $\therefore$ vol $= \int_1^e \pi(\ln y)^2 dy = y(\ln y)^2 - 2\int \ln y dy$

$$= \pi(y(\ln y)^2 - 2y\ln y + 2y)\Big|_1^e = (e-2)\pi \text{ cubic unit}$$

Work done by variable force

If a variable force is applied to an object calculus is needed to determine the work done because the amount of force changes as the object changes position, for example the force required to compress the spring. Work done by a variable force as the object is moved from $x = a$ to $x = b$ is $W = \int_a^b F(x)dx$.

Motion of a particle in a straight line

We know that the velocity $(v) = \dfrac{\Delta s}{\Delta t} \Rightarrow v(t) = s'(t) = \dfrac{ds}{dt}$

$$\Rightarrow a(t) = v'(t) = \frac{dv}{dt} \Rightarrow a(t)dt = dv \Rightarrow v = \int a(t)dt$$

Example: A particle moves along straight line with velocity $v(t) = t^2 - 3t$, then find the displacement if particle has its position 4m when the time $t = 0$.

Solution: $s(t) = \int v(t)dt = \int (t^2 - 3t)dt = \dfrac{t^3}{3} - \dfrac{3t^2}{2} + c$ and $s(0) = 4$

$$\Rightarrow s(t)=\frac{t^3}{3}-\frac{3t^2}{2}+c \Rightarrow s(0)=0-0+c=4 \Rightarrow c=4$$

$$\therefore s(t)=\frac{t^3}{3}-\frac{3t^2}{2}+4$$

Solved Problem

75. The volume of the solid obtained when the graph of $y = \sin x$, on $[0, \pi]$ is rotated about the x-axis is:

A. $\frac{1}{2}\pi^2$ B. $\frac{1}{2}\pi+1$ C. $\frac{1}{2}\pi^2-1$ D. $\frac{1}{2}\pi$

Solution: $v=\int_0^{\pi}\pi\left(\frac{1}{2}-\frac{\cos 2x}{2}\right)dx=\pi\left[\frac{1}{2}x-\frac{\sin 2x}{4}\right]_0^{\pi}=\frac{1}{2}\pi^2$

∴Answer: A

76. The volume of solid obtained by revolving the region bounded by the graph of $f(x)=x^2$ and $g(x)=4$ about the x-axis . . . (UEE)

A. $\frac{256\pi}{5}$ B. $\frac{512\pi}{15}$ C. $\frac{32\pi}{3}$ D. $\frac{16}{5}\pi$

Solution: The intersection of f and g is $f(x) = g(x) \Rightarrow x^2 = 4$

$\Leftrightarrow x = -2$ and $x = 2$. $v=\int_{-2}^{2}\pi(4^2-(x^2)^2)dx=\int_{-2}^{2}\pi(16-x^2)$

$$=\pi\left(16x-\frac{x^5}{5}\right)\Big|_{-2}^{2}=\frac{256\pi}{5}, \therefore \textbf{Answer: A}$$

77. The area of the region enclosed by the graph of $y=x^3-x$ and $y=2x$.

A. $\frac{7}{2}$ B. $\frac{3}{4}$ C. $\frac{9}{2}$ D. $\frac{61}{105}$

Solution: The intersection of $y=x^3-x$ and $y=2x$ is solving $x^3-x=2x \Rightarrow x^3-3x=0 \Rightarrow x(x^2-3)=0 \Rightarrow x=0, x=-\sqrt{3}$ and $x=\sqrt{3}$. Area of region (A) = $|A_1| + |A_2|$

$$\Rightarrow A=\int_{-\sqrt{3}}^{0}(x^3-x-2x)dx+\int_{0}^{\sqrt{3}}(2x-x^3+x)dx$$

$$=\int_{-\sqrt{3}}^{0}(x^3-3x)dx+\int_{0}^{\sqrt{3}}(-x^3+3x)dx$$

$$=\left(\frac{x^4}{4}-\frac{3x^2}{2}\right)\Bigg|_{\sqrt{3}}^{0}+\left(\frac{-x^4}{4}+\frac{3x^2}{2}\right)\Bigg|_{0}^{\sqrt{3}}$$

$$=\left(0-\left(\frac{9}{4}-\frac{9}{2}\right)\right)+\left(\frac{9}{4}+\frac{9}{2}\right)=\frac{9}{2}$$

78. The area of the region enclosed by the graph of $y=\sin x$ and $y=0$ on $x\in[0,2\pi]$.

A. 4 B. 2 C. 3 D. 0

Solution: The intersection will be $\sin x=0\Rightarrow x=0,\pi,2\pi$.

$$\therefore A=\int_0^{\pi}\sin x dx+\int_{\pi}^{2\pi}(0-\sin x)dx=-\cos x\Big|_0^{\pi}+\cos x\Big|_{\pi}^{2\pi}$$

$\Rightarrow A=-(-1-1)+(1-(-1))=4$ ∴ **Answer: A**

79. The area of the region bounded by the graph of $f(x)=\sqrt{2-x}$, $g(x)=-x$ and x-axis.

Solution: The intersection of $f(x)$ and $g(x)$ is $f(x)=g(x)$

$\Rightarrow\sqrt{2-x}=-x\Leftrightarrow 2-x=x^2\Rightarrow x^2+x-2=0\Rightarrow x=-2$, a and $f(x)$ and x-axis intersects when $\sqrt{2-x}=0\Rightarrow x=2$

$$\therefore A=\int_{-2}^{0}\left(\sqrt{2-x}+x\right)dx+\int_0^2\sqrt{2-x}\,dx$$

$$=-\frac{2}{3}\sqrt{2-x}^3\Big|_{-2}^{0}+\frac{x^2}{2}\Big|_{-2}^{0}=\frac{-2}{3}\sqrt{2-x}^3\Big|_0^2$$

$$=\frac{-2\sqrt{8}}{3}+\frac{16}{3}+-2+\frac{2\sqrt{8}}{3}=\frac{10}{3}$$

80. The area of the region bounded between the graph of $y=x^2$ and $y=\sqrt{x}$ and is . . . UEE 2003

A. 3 sq. unit B. $\frac{1}{3}$ sq. unit C. $\frac{2}{3}$ sq. unit D. 1 sq. unit

Solution: Intersection of x^2 and $\sqrt{x}$ is $x^2 = \sqrt{x} \Rightarrow x = 0$ and $x = 1$.

$$\therefore A = \int_0^1 \left(\sqrt{x} - x^2\right) dx = \left(\frac{2}{3}\sqrt{x}^3 - \frac{x^3}{3}\right)\Big|_0^1 = \frac{2}{3} - \frac{1}{3} = \frac{1}{3}$$

Answer: B

81. $\int_0^1 \frac{(x-1)^2}{x+1} dx$ equal to . . . UEE 2004

A. $-\frac{3}{2} + \ln(6)$ B. $\frac{-5}{2} + \ln(15)$ C. $\frac{-5}{2} + 4\ln 2$ D. $\frac{5}{2} + \ln 16$

Solution:

$$\int_0^1 \frac{(x+1)^2}{x+1} dx = \int_0^1 \frac{x^2 - 2x + 1}{x+1} dx = \int_0^1 \left((x-3) + \frac{4}{x+1}\right) dx$$

$$= \left(\frac{x^2}{2} - 3x + 4\ln|x+1|\right)\Big|_0^1 = \frac{1}{2} - 3\ln + 4\ln 2 - 0 = -\frac{5}{2} + 4\ln 2$$

Answer: C

82. The area of the region enclosed by the graph of $y = x$, $y = \frac{1}{x}$ and $y = 4$ is equal to:

A. $\frac{1}{2}(15 - \ln 4)$ C. $\frac{1}{2}(15 - \ln 8)$

B. $\frac{1}{2}(15 - \ln 16)$ C. $\frac{1}{2}(15 + \ln 8)$

Solution: The intersection $y = x$, and $y = \frac{1}{x}$ is (1, 1), (4, 4) and $\left(\frac{1}{4}, 4\right)$, simply to integrate, use with respect to dy:

$$\therefore A = \int_1^4 \left(y - \frac{1}{y}\right) dy = \left(\frac{y^2}{2} - \ln y\right)\Big|_1^4 = 8 - \ln 4 - \frac{1}{2}$$

$$= \frac{1}{2}(15 - 2\ln 4) = \frac{1}{2}(15 - \ln 16)$$

∴Answer: B

83. $\int_0^1 \frac{1}{(x+1)(x+2)}dx$ is equal to:

A. $\log\left(\frac{2}{3}\right)$ B. $\log\left(\frac{4}{3}\right)$ C. 1 D. $\log\left(\frac{3}{4}\right)$

Solution: $\frac{1}{(x+1)(x+2)} = \frac{A}{x+1} + \frac{B}{x+2} \Leftrightarrow A(x+2) + B(x+1) =$

$1, A = 1, B = -1. \int_0^1 \frac{1}{(x+1)(x+2)}dx = \int_0^1 \frac{dx}{x+1} - \int_0^1 \frac{dx}{x+2}$

$= \ln\left|\frac{x+1}{x+2}\right|\Big|_0^1 = \ln\left(\frac{4}{3}\right)$ ∴ **Answer: B**

84. $\int_0^1 \frac{dx}{\sqrt{x+3}-\sqrt{x}}$ is equal to:

A. $\frac{6-2\sqrt{3}}{3}$ B. $\frac{6+2\sqrt{3}}{3}$ C. $9-2\sqrt{3}$ D. $6-2\sqrt{3}$

Solution: Rationalizing the denominator, it gives

$\int_0^1 \frac{1\left(\sqrt{x+3}+\sqrt{x}\right)}{\left(\sqrt{x+3}-\sqrt{x}\right)\left(\sqrt{x+3}+\sqrt{x}\right)} = \int_0^1 \frac{\left(\sqrt{x+3}+\sqrt{x}\right)}{3}dx$

$= \frac{2}{9}\sqrt{x+3}^3 + \frac{2}{9}\sqrt{x}^3\Big|_0^1 = 2 - \frac{2\sqrt{3}}{3} = \frac{6-2\sqrt{3}}{3}$ ∴ **Answer: A**

85. $\int_1^2 \log x dx$ is equal to:

A. $2\log 4 - 1$ B. $\log 4 + 1$ C. $\log 4 - 1$
D. $\log 4 + 4$

Solution: We know that $\log x = \frac{\ln x}{\ln 10}$ let $dv = dx$

$\Rightarrow v = \int dx = x$ and $u = \frac{\ln x}{\ln 10} \Rightarrow du = \frac{1}{x\ln 10}$.

$\therefore \int_1^2 \log x dx = \int_1^2 \frac{\ln x}{\ln 10} = \left(\frac{x\ln x}{\ln 10} - \frac{x}{\ln 10}\right)\Big|_1^2 = \frac{2\ln 2 - 1}{\ln 10}$

$$\Leftrightarrow \frac{2\ln 2-1}{\ln 10}=\frac{\ln 4-1}{\ln 10}=\log 4-1$$ ∴ **Answer: C**

Supplementary Exercise on Unit 14

1. **Find**

a. $\int \sqrt{3}dx$ b. $\int x^{\sqrt{2}}dx$ c. $\int \sqrt{3-4x}dx$

d. $\int \sqrt{2x-1}dx$ e. $\int (3x-1)^4 dx$ f. $\int (3x+1)\sqrt[3]{3x+1}dx$

2. **Find**

a. $\int e^{4x}dx$ b. $\int e^{3x-4}dx$ c. $\int 3^{4x}dx$

d. $\int \frac{e^{2x}-e^{3x}}{e^{4x}}dx$ e. $\int \frac{5-e^x}{e^{2x}}dx$ f. $\int \frac{2^{1-3x}}{3^{x+1}}dx$

3. **Find**

a. $\int \frac{3}{x}dx$ b. $\int \frac{1}{4x-1}dx$ c. $\int \frac{3}{2-5x}dx$

d. $\int \frac{\sqrt{2}}{1-7x}dx$ e. $\int \frac{(e^x+1)^2}{e^x}dx$ f. $\int \left(\frac{1}{\sqrt{x+1}}-e^{2x}\right)dx$

g. $\int \left(\frac{1}{e^x}+\sqrt{e^x}\right)dx$ h. $\int \frac{1}{x^2}dx$ i. $\int \frac{1}{(x+3)^2}dx$

j. $\int \frac{1}{(x-4)^3}dx$ k. $\int \frac{1}{(2-3x)^2}dx$

4. **Evaluate**

a. $\int \frac{e^{\sqrt{x+1}}}{\sqrt{x+1}}dx$ b. $\int x^3 e^{x^4}dx$ c. $\int \frac{x+1}{\sqrt{x+3}}dx$

d. $\int \frac{6x}{\sqrt{x^2+4}}dx$ e. $\int (4x+1)(4x^2+2x+7)^6 dx$

f. $\int \frac{4x-1}{1-2x+4x^2}$ g. $\int (4x^2+12x+9)^4 dx$

h. $\int (16x^2-24x+9)^5 dx$ i. $\int 3x^2\sqrt{x^2-2}dx$

5. **Evaluate the integral**

a. $\int \frac{(\ln x)^2}{x} dx$ b. $\int \frac{1}{x(\ln x)^3} dx$ c. $\int \frac{1}{x \ln x} dx$

6. **Evaluate the integral**

a. $\int \frac{1}{4x^2-9} dx$ b. $\int \frac{x+2}{x^2+4x+3} dx$ c. $\int \frac{5-x}{2x^2+x-1} dx$

d. $\int \frac{x^2+12x+12}{x^3-4x} dx$ e. $\int \frac{x^2-1}{x^3+x} dx$ f. $\int \frac{x}{16x^4-1} dx$

7. **Evaluate the integral**

a. $\int x^2 e^x dx$ b. $\int x^3 e^{-x} dx$ c. $\int x^2 e^{2x} dx$

d. $\int x \ln(x+1) dx$ e. $\int \ln 3x dx$ f. $\int (\ln x)^2 dx$

g. $\int x^2 \cos x dx$ h. $\int x^3 \cos 2x dx$ i. $\int x^3 \sin x dx$

8. **Evaluate the integral**

a. $\int e^x \sin 2x dx$ b. $\int e^{2x} \cos x dx$ c. $\int e^{-x} \sin x dx$

9. Find

a. $\int_0^3 |2x-3| dx$ b. $\int_1^4 (3-|x-3|) dx$ c. $\int_0^3 |x^2-4| dx$

d. $\int_{-1}^1 x(x^2+1)^3 dx$ e. $\int_1^2 2x^2 \sqrt{x^3+1} dx$ f. $\int_0^4 \frac{1}{\sqrt{2x+1}} dx$

g. $\int_1^2 (x-1)\sqrt{2-x} dx$ h. $\int_0^{\frac{\pi}{2}} \cos\left(\frac{2x}{3}\right) dx$ i. $\int_0^1 \frac{1}{(2x+1)^2} dx$

10. Find

a. $\int_0^1 \frac{3}{2x^2+5x+2} dx$ b. $\int_1^5 \frac{x-1}{x^2(x+1)} dx$ c. $\int_2^3 \frac{1}{x^2-1} dx$

d. $\int_0^1 x^2 e^x dx$ e. $\int_0^{\pi} x \sin 2x dx$ f. $\int_0^{\frac{\pi}{2}} x \cos x dx$

g. $\int_0^1 e^x \sin x dx$ h. $\int_0^2 e^{-x} \cos x dx$ i. $\int_1^2 x^2 \ln x dx$

11. Find

a. $\int_{-\pi}^{\pi} \sin^2 x dx$ b. $\int_0^{\frac{\pi}{2}} \frac{\cos x}{1+\sin x} dx$ c. $\int_0^{\frac{\pi}{2}} \cos^2 x dx$

d. $\int_0^1 \frac{1-x}{1+x} dx$ e. $\int_0^{\frac{1}{2}} \frac{1}{\sqrt{3-2x}} dx$ f. $\int_0^2 \frac{x^2}{\sqrt{x^3+1}} dx$

g. $\int_0^1 \frac{1}{x+\sqrt{x}} dx$ h. $\int_0^2 x^2 e^x dx$ i. $\int_1^2 x \ln x dx$

j. $\int_1^2 x^2 \sqrt{x^3+8} dx$ k. $\int_0^1 x^3 \sqrt{1+3x^4} dx$ l. $\int_0^{\frac{\pi}{2}} \cos^3 x dx$

12. Find the area of the region R bounded by the graph of each of the following (in square unit).

a. $f(x) = 6 - x^2$ and $g(x) = x$, $\frac{125}{6}$

b. $f(x) = \sin x$ and $g(x) = \cos x$, on $\left[\frac{\pi}{4}, \frac{5\pi}{4}\right]$

c. $f(x) = -x^2 + 2x$ and $g(x) = 3x^3 - x^2 - 10x$

d. $f(x) = x^3 - 4x$ and $g(x) = 0$

e. $f(x) = x^2 + 4x + 1$ and $g(x) = x + 1$

f. $f(x) = 3^{2x}$, $x = 0$, $x = 1$ and $y = 0$

13. Find the area of the region bounded by:

a. $x = y^2 - 6$ and $y = x$

b. $y = \sqrt{x+2}$ and $y = x$

c. $x = y^2 + 2y + 1$ and $x = 2y + 5$

14. Find the volume generated by the following graph about the x-axis (in cubic unit).

a. $f(x) = -x + 2$, $x = 0$ and $y = 0$

b. $f(x) = \Rightarrow 4 - x^2$, $y = 0$ and $x = 0$

c. $f(x) = 5x - x^2$, x – axis

d. $f(x) = x^3$ and $g(x) = 4x$

e. $f(x) = |x|$ and $g(x) = x^2$, from $x = -3$ to $x = 3$

15. Find the volume generated by the rotating the graph about the y-axis.

a. $y = x^2 + 2$ and the line $y = 6$, rotates about y-axis.

b. $y = x^2 - 1$ and the line $y = 2$, rotates about y-axis.

c. $y = e^{-x}$ the y-axis and the line $y = e$ rotates about y-axis.

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Unit Fifteen

Three Dimensional Geometry

Here, we will discuss coordinate systems in three dimensional space and some basic facts of 3D.

Coordinate axes and coordinate planes in space

- The three mutually perpendicular lines, x-axis, y-axis and z-axis are called *three coordinate axis*.
- Taken as a pair, the axes determine three coordinate planes: The ***xy*-plane, *xz*-plane** and the ***yz*-plane**. These three coordinate planes intersect at origin. These three coordinate planes separate three-space into eight parts called **octants**. The set of points with three positive coordinate forms the **first octant** the remaining octants have no standard numbering.

Coordinates of a point in space

Recall a point P in plane:

- The point P(-3, 4) means that p is 3 unit to the left of y-axis and 4 units above the x-axis.

In three dimensional system, a point P (x, y, z) indicates:

- x = directed distance from yz plane to P
- y = directed distance from xz plane to P
- z = directed distance from xy plane to P.

The standard equation determined by the coordinate axes

It is possible to describe some geometric figures in space by means of equation. In xy-plane $x = 0$, in x-axis, $z = 0$, $y = 0$ in y-axis, $x = 0$, $z = 0$ and z-axis, $x = y = 0$.

Illustrative Example

1. Give the equation of the:
   a. xy-plane b. xz-plane c. yz-plane
   d. x-axis e. y-axis f. z-axis

   Solution:

   a. xy-plane consists of all points of the form $(x, y, 0)$
   b. xz-plane consists of all points of the form $(x, 0, z)$

c. yz-plane consists of all points of the form $(0, y, z)$
d. x-axis consists of all points of the form $(x, 0, 0)$
e. y-axis consists of all points of the form $(0, y, 0)$
f. z-axis consists of all points of the form $(0, 0, z)$

Distance between two points in space

The distance between the point $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$ is

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

Illustrative Example

2. Find the distance between the points:
   a. A (1, -2, 1) and B (6, -2, -2) b. A (2, -1,3) and B (1,0, -2)

   Solution: Let d = distance b/n A and B, then:

   a. $d = \sqrt{(6-1)^2 + (-2+2)^2 + (-2-1)^2} = \sqrt{25+0+9} = \sqrt{34}$

   b. $d = \sqrt{(1-2)^2 + (0-(-1))^2 + (-2-3)^2} = \sqrt{1+1+25}$

   $= \sqrt{27} = 3\sqrt{3}$

3. Find the distance from the point P (-3,1, -2) to the
   a. $x = 1$ b. xy-plane c. xz-plane d. yz-plane
   e. z-axis f. y-axis g. x-axis h. $z = 0$

   Solution: We know that, xy-plane is $(x, y, 0)$ or $z = 0$.

   a. $x = 1$, means the yz-plane meeting the x-axis at the point $(1, 0, 0)$ is $d = \sqrt{(-3-1)^2 + (1-0)^2 + (-2-0)^2} = \sqrt{21}$

   b. xy-plane contain (-3, 1, 0) set $z = 0$

   $\Rightarrow d = \sqrt{(-3-(-3)^2) + (1-1)^2 + (-2-0)^2} = \sqrt{4} = 2$

   c. xz-plane contains (-3, 0, -2) set $y = 0$

   $\Rightarrow d = \sqrt{(-3-(-3))^2 + (1-0)^2 + (-2-(-2))^2}$

   $= \sqrt{0+1+0} = \sqrt{1} = 1$

   d. *Exercise left for you*. (Hint, set $x = 0$).

   e. z-axis contains (0, 0, -2), thus distance between

   $d = \sqrt{(3-0)^2 + (1-0)^2 + (-2-(-2))^2} = \sqrt{9+1} = 10$

   f. y – axis contains (0, 1, 0) ← Exercise left for you.

   g. x – axis contains (-3, 0, 0).

$d = \sqrt{(-3-(-3))^2+(1-0)^2+(-2-0)^2} = \sqrt{0+1+4} = \sqrt{5}$

h. $z = 0$ is point on xy plane $\Rightarrow$(-3, 1, 0)

so that $d = \sqrt{0+1+4} = \sqrt{5}$

4. Find the possible coordinate of points on the x-axis which are $\sqrt{6}$ units from the point (3, -1, 2).

Solution: x-axis contains $(x, 0, 0)$. The distance b/n $(x, 0, 0)$ and (3, -1, 2) is $\sqrt{6} \Rightarrow d = \sqrt{(x-3)^2+(-1)^2+(0-2)^2} = \sqrt{6}$

$\Rightarrow (x-3)^2 + 1 + 4 = 6 \Leftrightarrow (x-3)^2 = 1 \Rightarrow x - 3 = 1$ or $x - 3 =$

give $x = 4$ and $x = 2$.

∴ The coordinate points are (4, 0, 0) and (2, 0, 0).

5. Identify the type of triangle, isosceles, equilateral, right triangle if the vertices are given:

a. A (4, 5, 2), B (1, 7, 3) and C (2, 4, 5)

b. A (0, 7, 10), B (-1, 6, 6) and C(-4, 9, 6)

Solution: Here

a. $\overline{AB} = \sqrt{(4-1)^2+(5-7)^2+(2-3)^2} = \sqrt{9+4+1} = \sqrt{1}$

$\overline{BC} = \sqrt{(2-1)^2+(4-7)^2+(5-3)^2} = \sqrt{1+9+4} = \sqrt{1}$

$\overline{CA} = \sqrt{(4-2)^2+(5-4)^2+(2-5)^2} = \sqrt{4+1+9} = \sqrt{1}$

∴ΔABC is equilateral triangle.

b. Exercise left for you.

6. Find the set of all points on xz-plane which is equidistant from the point (1, -1, 5) and (5, 1, -1).

Solution: The set of all point on xz-plane is $(x, 0, z)$. Hence $(x, 0, z)$ is equidistant from the point (1, -1, 5) and (5, 1, -1)

Thus $(x-1)^2 + (0+1)^2 + (z-5)^2 = (x-5)^2 + (0-1)^2 +$

$1)^2 \Rightarrow x^2 - 2x + 1 + 1 + z^2 - 10z + 25 = x^2 - 10x + 25 + 1 +$

$2z + 1$ cancelling and simplifying similar terms, we get

$8x - 12z = 0 \Rightarrow z = \frac{8}{12}x = \frac{2}{3}x$

$\therefore (x, y, z) = \{(x, 0, z)\} = \left\{\left(x, 0, \frac{2}{3}x\right)\right\}$, let $x = 3t$.

$\therefore$ the set of all point on xz plane is $(3t, 0, 2t)$, $t \in \mathbb{R}$

Equation of sphere

A sphere with center at (x_0, y_0, z_0) and radius r is defined to be the set of all point (x, y, z) such that the distance between (x, y, z) and (x_0, y_0, z_0) is the radius of sphere (r).

> If (x, y, z) is an arbitrary point on the sphere the equatin of sphere is $(x-x_0)^2 + (y-y_0)^2 + (z-z_0)^2 = r^2$

Illustrative Example

7. Find the equation of the sphere:
   a. Center: (0, 2, 5), radius, $r = 2$
   b. Center: (4, -1, 1), radius, $r = 5$
   c. End points of a diameter: (2, 0, 0) and (0, 6, 0)

   Solution: Let (x, y, z) be point on sphere, then:

   a. $(x-0)^2 + (y-2)^2 + (z-5)^2 = 4$
   $\Rightarrow x^2 + y^2 + z^2 - 4y - 10z + 25 = 0$

   b. $(x-4)^2 + (y+1)^2 + (z-1)^2 = 25$
   $\Rightarrow x^2 + y^2 + z^2 - 8x + 2y - 2z - 7 = 0$

   c. Diameter

$$(d) = \sqrt{(0-2)^2 + (6-0)^2 + (0-0)^2} = \sqrt{40} = 2\sqrt{10}$$

$\Rightarrow$radius , r = $\sqrt{10}$

$$\text{center (midpoint)} = \left(\frac{2+0}{2}, \frac{0+6}{2}, \frac{0+0}{2}\right) = (1, 3, 0)$$

$\therefore (x-1)^2 + (y-3)^2 + (z-0)^2 = 10$

$\Rightarrow x^2 + y^2 + z^2 - 2x - 6y = 0$ is equation of sphere.

8. Find the equation of the sphere:
   a. Center: (2, -1, -3), tangent to the xy plane
   b. Center: (2, -1, -3) tangent to the xy-plane
   c. Center: (-3, 2, 4) tangent to the yz plane

   Solution: a) The distance b/n (2, -1, -3) and xy-plane is 3

   $\Rightarrow (x-2)^2 + (y+1)^2 + (z+3)^2 = 3^2$

   $\Rightarrow x^2 + y^2 + z^2 - 4x + 2y + 6z + 5 = 0$

   b) $r = 1 \Rightarrow (x-2)^2 + (y+1)^2 + (z+3)^2 = 1$

   c) Exercise left for you.

9. Find the equation of sphere
   a. Center: (1, 0, -1) and passing through (2, 1, 3)

b. Center: (-1,. 3, 2) and passing through (0, 0, 0)

Solution: as radius

$r=\sqrt{(2-1)^2+(1-0)^2+(3-(-1))^2}=\sqrt{18}$

$\Rightarrow (x-1)^2+(y-0)^2+(z+1)^2=18$

b) Exercise left for you

Determining the distance between the point P (*x*, *y*, *z*) and the sphere

How far is the point P (x, y, z) from the sphere $(x-a)^2+(z-c)^2=r^2$.

10. How far is the sphere $(x-2)^2+(y+1)^2+(z-1)^2=4$ from the point P and which point is closer to sphere.

a. P (4, 0, 3) b. P (-2, 2, 1) c. P (0, 1, 3)

Solution: Center of sphere (2, -1, 1), with radius $r=2$.

a. The distance b/n the center (2, -1, 1) and point P (4, 0, 3).

$\Rightarrow d=\sqrt{(4-2)^2+(0+1)^2+(3-1)^2}=\sqrt{4+1+4}=3$

∴ The distance b/n the sphere and point P is $3-2=1$

b. The distance b/n the center (2, -1, 1) and point P (-2, 2, 1) is

$d=\sqrt{(2-(-2))^2+(-1-2)^2+(1-1)^2}=\sqrt{16+9+0}=5$

∴The distance b/n the sphere and point P is $5-2=3$.

c. Distance b/n (2,-1, 1)and P (0, 1, 3) is

$d=\sqrt{(3-2)^2+(1+1)^2+(3-1)^2}=2\sqrt{3}$

∴ The distance b/n the sphere and point is $2\left(\sqrt{3}-1\right)$. Note:

Therefore, P (4, 0, 3) is closer to sphere than other point.

11. If the center of a sphere is at (1, -2, 3) and radius $r=3$ then determine which points lie inside or outside or on the sphere.

a. P (2, -1, 6) b. P (3, 0, 4) c. P (3, -1, 2)

Solution: Equation of sphere with center (1, -2, 3) and radius, $r=3$ is $(x-1)^2+(x+2)^2+(y-3)^2=9$

a. (2, -1, 6): $(2-1)^2+(-1+2)^2+(6-3)^2\Rightarrow 11>9$

∴(2 -1, 6) is outside the sphere.

b. P (3, 0, 4): $(3-1)^2+(0+2)^2+(4-3)^2\Rightarrow 9=9$

∴ $f(3, 0, 4)$ is on the sphere.

c. P (3, -1, 2): $(3-1)^2+(-1+2)^2+(2-3)^2\Rightarrow 6<9$

∴ P (3, -1, 2) is inside the sphere.

12. Which of the following is a point of intersection of the sphere $(x-1)^2+(y-2)^2+(z+1)^2=6$ and the y-axis …UEE 2002

A. (0, -2, 0) B. (0, 2, 0) C. (0, -4, 0) D. (0, 4, 0)

Solution: The equation on y-axis is $(0, y, 0)$

$\Leftrightarrow(0-1)^2+(y-2)^2+(0+1)^2=6\Rightarrow(y-2)^2=4$
$\Leftrightarrow(0-1)^2+(y-2)^2=4\Leftrightarrow y-2=2$ and $y-2=-2=y=4$ and $y=0$. $\therefore(0,4,0)$ **$\therefore$Answer: D**

13. Find the intersection of the sphere $(x-1)^2+(y+2)^3+(z-3)^2=8$ with: a. x-axis b. z-axis c. y-axis

Solution: a. x-axis $(x,0,0)\Leftrightarrow(x-1)^2+(0+2)^2+(0-3)^2=8$
$\Rightarrow(x-1)^2=8-13=-5$.
$\therefore$ There is no point of intersection with x-axis.
b. z-axis: $(0,0,z)\Rightarrow(0-1)^2+(0+2)^2+(z-3)^2=8\Leftrightarrow(z-3)^2=8-5=3\Rightarrow z-3=\sqrt{3}$ or $z-3=-\sqrt{3}$.
$\therefore$ The sphere intersects at two point on $z=3+\sqrt{3}$ and $z=3-\sqrt{3}$. That is $(0,0,3+\sqrt{3})$ and $(0,0,3-\sqrt{3})$.
c. Exercise left for you.

14. Find the point of intersection of the sphere of the radius 3 unit with center at (-1, 3, 2) and the z-axis . . . UEE 2003/11

Solution: $(x+1)^2+(y-3)^2+(z-2)^2=9$ and z – axis $(0,0,z)$
$\Rightarrow(0+1)^2+(0-3)^2+(z-2)^2=9\Rightarrow(z-2)^2=9-10=-1$
$\Rightarrow(z-2)^2=-1$. $\therefore$ There is no point of intersection.

15. If a sphere with center C (0, 1, 1) intersects the z-axis at P (0, 0, 3) then the radius of the sphere is equal to . . . UEE 2004/12

A. 5 B. 3 C. $\sqrt{3}$ D. $\sqrt{5}$

Solution: Eqn of sphere $(x-0)^2+(y-1)^2+(z-1)^2=r$
$\Rightarrow(0-0)^2+(0-1)^2+(3-1)^2=r^2\Leftrightarrow r^2=5\Rightarrow r\ \sqrt{5}$. $\therefore$ **Ans. D**

16. Suppose A and B are the end-points of a diameter of the sphere whose equation is $x^2+y^2+(z+2)^2=1$. If A (1, 0, -2) then B is equal to . . . UEE 2004/12

A. (0, 1, -2) B. (0, 0, -1) C. (-1, 0, -2) D. (0, -1, -2)

Solution: Center of sphere (0, 0, -2) as the mid-point of AB let the end-point of B (x, y, z) and A = (1, 0, -2)

$$\left(\frac{1+x}{2},\frac{0+y}{2},\frac{z-2}{2}\right)=(0,0,-2)\leftarrow\text{midpoint of AB}$$

$\Rightarrow \frac{1+x}{2}=0 \Rightarrow x = -1$ and $\frac{y+0}{2}=0 \Rightarrow y=0$ and

$\frac{z-2}{2}=-2 \Rightarrow z=-2$. ∴B = (-1, 0, -2). ∴ **Answer: C**

17. If the line $z = 4 - \sqrt{3}y$ for $x = -1$ is tangent to the sphere whose equation is: $(x + 1)^2 + y^2 + z^2 = 4$ then what is the distance of the line from the center of the sphere.

A. 2 B. 3 C. $2\sqrt{3}$ D. 4

Solution: $(-1 + 1)^2 + y^2 + \left(4-\sqrt{3}y\right)^2 = 4$

$\Leftrightarrow y^2 + 16 - 8\sqrt{3}y + 3y^2 = 4 \Rightarrow 4y^2 - 8\sqrt{3}y + 12 = 0$

$\Rightarrow y^2 - 2\sqrt{3}y + 3 = 0 \Rightarrow \left(y-\sqrt{3}\right)^2 = 0 \Rightarrow y = \sqrt{3}$, then

$z = 4 - \sqrt{3}.\sqrt{3} = 4 - 3 = 1$

∴ The distance b/n $\left(-1, \sqrt{3}, 1\right)$ and center (-1, 0, 0) is

$d = \sqrt{(-1+1)^2 + \left(\sqrt{3}-0\right)^2 + (1-0)^2} = \sqrt{4} = 2$ ∴ **Answer: A**

Supplementary Exercise on Unit – 15

1. Find the distance from the point P (2, -1, 3) to the:
   a. $xy-$plane b. $xy-$plane c. $y-$axis
   d. $z-$axis e. $x-$axis f. $x = 0$
2. Find the possible coordinate of points on the $y-$axis which are $\sqrt{29}$ units from the point (4, -2, 3).
3. Find the equation of sphere:
   a. Center: (1, 0, -2) and passing through (3, 1, 2)
   b. Center: (3, -1, 2) and passing through (0, 0, 4)
4. How far is the point P (3, 0, 1) from the sphere $(x + 1)^2 + y^2 + (z - 4)^2 = 9$.

5. If the center of a sphere is at (1, 0, 2) and radius $r = 3$, then determine which points lie inside, or outside or on the sphere.
   a. (-2, 1, 3) b. (3, -1, 4) c. (2, -2, 3)
6. Find the point of intersection of the sphere $(x+1)^2+(y-3)^2+z^2 = 4$ with:
   a. $x-$axis b. $y-$axis c. $z-$axis
7. Find the point of intersection of the sphere of the radius $\sqrt{38}$ unit with center at C (2, 1, 3) with: a. $y-$axis b. $x-$axis
8. If a sphere with center C (-3, 0, 3) intersects the $x-$axis at (1, 0, 0) then the radius of the sphere is equal to _____.
9. For what value of k is the sphere:
   a. $x^2+y^2-4x+kz=-4$ of radius 3 unit
   b. $x^2+y^2+z^2-2x+4y+6z+k=0$ of radius $\sqrt{5}$ unit

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Answer to Review Exercise on Unit – 1

a) Not function b) function c) Not function, since $a \geq 0$

d) Not function, since $A = \{1, 2, 3, 4\}$.

a) odd b) odd c) odd d) even

a) one – to – one correspondence

b) on to

c) one – to – one but no on to

d) one – to – one correspondence

e) one – to – one correspondence

f) one – to – one correspondence

g) on to but not one – to – one

h) on to

a) one – to – one c) one to one correspondence b) one – to – one

a) $\left\{x : x \geq \frac{1}{2}\right\}$ b) $\{x: x \geq 2\}$ c) $\{x: x > 2\}$

a) True b) False c) False d) True

a) $f^{-1}(x) = e^{2(x-3)}$ b) $f^{-1}(x) = \frac{3(e^x - 1)}{e^x - 2}$

c) $f^{-1}(x) = 3 - \sqrt{\frac{x-1}{2}}$ d) $f^{-1}(x) = \ln x - \ln(1 - x)$

$f(g(x)) = 2(3x + k)^2 + 5 = 45$, when $x = 0 \Rightarrow 2(k^2) = 40$

$\Rightarrow k = 2\sqrt{5}$ or $k = -2\sqrt{5}$

$f(x + 1) = 2x^2 - 7x + 4$

a) $\{x \in \mathbb{R}: 0 \leq x \leq 4\}$, b) $\{x \in \mathbb{R}: 0 \leq x \leq 4\}$

a) $f^{-1}(x) = -5 + \sqrt{x}$ b) $f^{-1}(x) = \sqrt{x - 3}$

c) $f^{-1}(x) = \frac{4x + 1}{3x - 1}$ d) $f^{-1}(x) = x^3 + 4$

e) $f^{-1}(x) = x^2 - 2$ for $x \geq 0$ f) $f^{-1}(x) = \sqrt[3]{x + 2}$

Solution for Unit Three

1. a) $\sqrt{38 - 12\sqrt{6}}$ b) $2\sqrt{a^2 + b^2}$ c) $|x|\sqrt{10}$

2. (12, 0) and (-4, 0)

3. $y = -3 \pm \sqrt{119}$ 4. $x - y = 3$ 5. $x = 2$ 6. 4:1 (internally)

7. a) $\tan^{-1}\left(\frac{1}{8}\right)$ b) $\tan^{-1}\left(\frac{ab'-a'b}{aa'+bb'}\right)$ c) $\tan^{-1}\left(\frac{7}{4}\right)$

8. a) (4, 0) b) $x^2 + y^2 - 8x = 34$

9. (2, -4) 10. (9,2),(4,10),(-7,-6) 11. $k = 5$

12. $y + 2 = 0$, $\sqrt{3}x - y - (3\sqrt{3} + 2) = 0$ 13. $k = 0$, center = (-4, 3) and radius 4.

14. a) $(x - 2)^2 + y^2 = 9$ b) $x^2 + (y + 2)^2 = 25$

15. (a) $x^2 + y^2 + 2y - 12 = 0$ 16. a outside, B and C inside 17. $2\sqrt{5}$

18. Externally; (1, 0) 19. a) $5x - 4y = 11$, b) $10x + 3y + 20 = 0$

20. [Take center as (2a, a), radius = r] $x^2 + y^2 - 8x - 4y + 7 = 0$

21. $2y^2 - 9x = 0$ 22. a) latus rectum $4p = 6$

focus $\left(\frac{3}{2}, 0\right)$, vertex (0, 0), directrix $x = -\frac{3}{2}$

b) latus rectum $|4p| = 9$ focus $\left(0, \frac{-9}{4}\right)$, vertex (0, 0) and directrix, $y = \frac{9}{4}$.

c) latus rectum 4P=6, focus $\left(\frac{7}{2}, 3\right)$, vertex (2,3), directrix, x=½

d) latus rectum, $|4P| = 1$, focus $\left(-2, \frac{-9}{4}\right)$, vertex (-2, -2), directrix, $y = -\frac{7}{4}$

e) latus rectum $|4P| = 2$, focus (2, 2), vertex $\left(2, \frac{5}{2}\right)$, directrix, $y = 3$

f) latus rectum $|4P| = 1$, focus $\left(-\frac{7}{4}, 2\right)$, vertex (-2, 2), directrix $x = \frac{-9}{4}$

23. a) $y^2 = 12x$ c) $x^2 = -12y$ e) $(x + 1)^2 = 4(y - 2)$

b) $y^2 = -28x$ d) $(x - 1)^2 = 12(y - 1)$ f) $(x + 4)^2 = 4(y - 1)$

g) $(x - 3)^2 = 4(u + 4)$ 24. a) vertices (0, 5), (0, -5), foci (0, 3), (0, -3)

b) vertices (3, 0), (-3, 0), foci ($\sqrt{5}$, 0), ($-\sqrt{5}$, 0)

c) vertices (0, 3), (0, -3), foci (0, $\sqrt{8}$), (0, $-\sqrt{8}$)

d) vertices (8, -2), (-2, -2), foci (6, -2), (0, -2)

e) vertices (-2, 5), (-2, -5), foci (-2, 4), (-2,-4)

f) vertices: ($1+\sqrt{21}$,3), ($1-\sqrt{21}$,3), foci $\left(0, \frac{\sqrt{55}}{2}\right), \left(0, \frac{-\sqrt{55}}{2}\right)$

g) vertices: (1, 2), (1, -4), foci $\left(1, -1 + \frac{\sqrt{65}}{3}\right), \left(1, -1 - \frac{\sqrt{65}}{3}\right)$

h) vertices: (2, 0), (-2, 0), foci (1, 0), (-1, 0)

i) vertices: (3, 6), (3, 2), foci $(3, 4+\sqrt{3})$, $(3, 4-\sqrt{3})$

j) vertices (-4, 1), (2, 1), foci $(-1-\sqrt{8}, 1)$ $(-1+\sqrt{8}, 1)$

k) vertices (-1, 9),(-1, 1), foci $(-1, 5+2\sqrt{3})$, $(-1, 5-2\sqrt{3})$

25. a) $\frac{1}{9}x^2+\frac{1}{4}y^2=1$

b) $\frac{1}{169}x^2+\frac{1}{144}y^2=1$

c) $\frac{1}{3}x^2+\frac{1}{2}y^2=1$

i) $\frac{x^2}{20}+\frac{y^2}{36}=1$

f) $\frac{(x+2)^2}{16}+\frac{(y-4)^2}{7}=1$

g) $\frac{24(x-2)^2}{25}+\frac{(y-4)^2}{25}=1$

h) $\frac{x^2}{25}+\frac{y^2}{21}=1$

d) $\frac{x^2}{36}+\frac{y^2}{16}=1$

e) $\frac{x^2}{36}+\frac{8y^2}{81}=1$ f)

j) $\frac{(x-1)^2}{25}+\frac{(y-3)^2}{21}$

26. a) $\frac{1}{4}x^2-\frac{1}{5}y^2=1$

b) $4x^2-y^2=4$

c) $9x^2-4y^2=64$

d) $y^2-\frac{9}{4}x^2=36$

e) $\frac{1}{16}(x-6)^2-\frac{1}{9}(y-4)^2=1$

f) $(y-2)^2-4\left(x-\frac{1}{2}\right)^2=3$

g) $4y^2-x^2=64$

h) $\frac{41}{144}(x-4)^2-\frac{41}{225}(y+2)^2=1$

i) $\frac{(y-2)^2}{3}-\frac{(x-7)^2}{12}=1$

27. a) $(y-7)^2-\frac{(x-1)^2}{3}=1$

b) $\frac{x^2}{4}-\frac{y^2}{12}=1$

c) $\frac{7(x-4)^2}{36}-\frac{(y-4)^2}{4}=1$ and $\frac{7}{36}(x-4)^2-\frac{(x-4)^2}{4}=1$

28. a) $3x^2-y^2=3$

b) $\frac{y^2}{9}-\frac{x^2}{7}=1$

29. a) center (0, 0), vertices (±4, 0), foci, $(\pm\sqrt{41}, 0)$, asymptote $y=\pm\frac{5x}{4}$

b) center (0, 0), vertices (0, ±2), foci $(0, \pm\sqrt{29})$, asymptote: $y=\pm\frac{2x}{5}$

c) center (0, 0), vertices $\left(\pm\frac{3}{2},\ 0\right)$, foci $\left(\pm\frac{\sqrt{73}}{2},\ 0\right)$, asymptote;

$$y = \pm\frac{8x}{3}$$

d) center (3, -4), vertices (7, -4), (-1, 4), foci (8, -4), (-2, -4), asymptote,

$$y + 4 = \pm\frac{3(x-3)}{4}$$

e) center (1, -2) foci $(1, -2 \pm 2\sqrt{5})$ vertices (1, 0), (1, -4) asymptotes:

$$y + 2 = \pm\frac{(x-1)}{2}$$

f) center: (-2, 0)

foci $(-2 \pm \sqrt{34}, 0)$

vertices: (1, 0), (-5, 0)

asymptote: $y = \pm\frac{5(x+2)}{3}$

g) center; (1, -1)

vertices $\left(\frac{7}{3}, -1\right), \left(-\frac{1}{3}, -1\right)$

foci $\left(1 \pm \frac{\sqrt{97}}{3}, -3\right)$

asymptote $y + 1 = \frac{\pm 9(x-1)}{4}$

h) center (0, 0)

vertices $(0, \pm 3)$

foci $(\pm 3\sqrt{2}, 0)$

asymptote: $y = \frac{\pm 3x}{4}$

i) center; (3, 4)

vertices: (3, 6), (3, 2)

foci $(3, 4 \pm 2\sqrt{2})$

asymptote $y - 1 = \pm(x - 3)$

h) center (0, 0)

vertices (-2, 2),(-2, -4)

foci $(-2, -1 \pm \sqrt{13})$

asymptote: $y + 2 = \frac{\pm 3(x+2)}{2}$

Solution for Unit Four

1. a) T b) T c) T d) F e) T
2. a) All students study their lesson.
   b) $3 + 6 > 7$ and $4 - 1 \neq 5$
   c) $5 < 3$ and $-3 \geq -5$
   d) $3 < 1$ if and only if $3 \neq 1$
   e) It is false that you will study orelse you will pass your examination.
   f) you will not study if and only if you will pass your examination.
   g) some Ethiopians are not honest.
3. a) Inverse: If a number is not prime then it is not odd
   Converse: If a number is odd then it is prime.

Contra positive: If a number is not odd, then it is not prim.

h) Inverse: If $5 > 3$ then $-3 > -5$

Converse: If $-3 < -5$, then $5 < 3$

Contra positive: If $-3 > -5$, then $5 > 3$

g) Converse: If poverty is eradicated, then every Ethiopia does not work hard.

Contra positive: If poverty is not eradicated then every Ethiopian does not work hard.

a) neither b) tautology c) tautology d) tautology

e) neither f) tautology g) tautology h) tautology

a) valid b) invalid c) valid d) valid e) valid f) valid

Solution for Unit Five

The mean is 4.7, the modes are 1 and 4 and median is 4

the mean is 1.05, the mode is 0 and the median is 0

$\frac{1+2+x+5+y+8}{6} = 4 \Rightarrow 16 + x + y = 24 \Rightarrow x + y = 8$. Thus if the

mode 5, then x = 5 and y = 3 and y = 5

x = 5.4

a) $\frac{a+d+b+d+c+d+2d}{4} = \frac{a+b+c+d+4d}{4} = \frac{a+b+c+d}{4} + d = k + d$

b) $\frac{ac+bc+c^2+dc}{4} = \frac{c(a+b+c+d)}{4} = kc$

Solution from the table reconstruct the c.f table with class inter value

Marks	No. of student (f)	c.f
0 – 10	15	15
10 – 20	20	35
20 – 30	25	60
30 – 40	24	84
40 – 50	10	94
50 – 60	33	127
60 – 70	71	198
70 – 80	51	249

Here N = 249

a) **Calculating the median,** $\frac{N}{2} = 124.5$ ∴ median class is 50 – 60, $\ell_{med} = 50$,

h = 10, f = 33, c = 94

b) **Calculating lower quartile** Q_1, $\frac{N}{4} = 62.25$

∴ lower quartile class is 30 – 40, $\ell = 30$, h = 10, f = 24, c = 60

$$\therefore Q_1 = \ell + \frac{h}{f}\left(\frac{N}{4} - c\right) = 30 + \frac{10}{24}(62.25 - 60) = 30.94 \text{ marks}$$

iii) **Calculation of upper quartile** Q_3, $\frac{3N}{4} = \frac{747}{4} = 186.75$

$\therefore$ upper quartile class is 60 – 40, $\ell = 60$, h = 10, f = 71, c = 127

$$\therefore Q_3 = \ell + \frac{h}{f}\left(\frac{3N}{4} - c\right) = 60 + \frac{10}{71}(186.75 - 127) = 68.41 \text{ marks}$$

7. 13 8. mean = 9.87, median = 9.97

9. $P(5, 5) = \frac{5!}{(5-5)!} = \frac{5!}{0!} = 5! = 120$

10. P(10, 4) = 5040

11. $P(6, 3) = \frac{6!}{3!} = 120$

12. P(4, 1) + P(4, 2) + P(4, 3) + P(4, 4) = 4 + 12 + 24 + 24 = 64

13. $P(6, 4) = \frac{6!}{2!} = 6 \times 5 \times 4 \times 3 = 360$

14. Total number of sample point is $P(30, 2) = \frac{30!}{28!} = 870$

15. Total number of possible schedules is $P(5, 3) = \frac{5!}{2!} = 60$

16. Total number of distinct arrangements is $\frac{9!}{3!4!2!} = 1260$

17. a) Total number of possible partitions would be $\frac{8!}{3!3!2!2!} = 1260$

18. n = 6

19. a) The number of ways selecting 3 person from total of 9 is

$$C(9, 3) = \frac{9!}{3!6!} = 84$$

b) C(4, 2) × C(5, 1) = 30 way

c) The number of ways of selecting 2 men where certain man is selected will leave us to select only 1 man from a total of (4-1) = 3 men and this is done is

C(3, 1) = 3 way and number of ways of selecting a woman from 5 women is C(5, 1) = 5 way

$\therefore$ C(3, 1) × C(5, 1) = 15 way

r = 8

120

Signals way be made by raising the flags 1, 2, 3, 4 and 5 at a time. Hence the sum of all numbers of signals is P(5, 1) + P(5, 2) + P(5, 3) + P(5, 4) + P(5, 5)

= 325 signals

There are 3.4 = 12 books on which 3 are a like 3 others a like, etc.

Number of arrangements = $\frac{(3\cdot 4)!}{3!3!3!3!} = \frac{12!}{(3!)^4} = 369{,}600$

C(8, 2) + C(8, 3) + C(8, 4) + … + C(8, 8) = 247 way

K = ±1

i) 4th term ii) 6th term, iii) 6th term

C(9, 3) a^3b^6

$k = \frac{9}{7}$

C(2n + 1, r) = c(2n+1, r+1) ⇒ r + r + 1 = 2n + 1 ⇒ r = n

r = 6

$\left(\frac{1}{2}\right)^5 (C(5,2) + C(5,3) + C(5,4)$ 32. $\left(\frac{1}{6}\right)$

$\frac{7}{36}$ 34. a) $\frac{1}{3}$ b) $\frac{3}{5}$ c) $\frac{11}{15}$ d) $\frac{2}{5}$ e) $\frac{4}{5}$

Three 6's and two non 6's can occur.

$C\binom{5}{3}\left(\frac{1}{6}\right)^3\left(\frac{5}{6}\right)^2$ 36. a) $\frac{2}{5}$ b) $\frac{1}{5}$ c) $\frac{4}{5}$ d) $\frac{13}{15}$

Answer to Review Exercise for Unit – 6

a) x = 1, y = -2, z = $-\frac{3}{2}$ b) x = 2, y = -1, z = -4

c) x = -2, y = 2 d) x = 3, y = $\frac{1}{2}$, z = $-\frac{1}{5}$

x = 1, y = 3 4. a) x = $-\frac{1}{2}$, y = 3 b) x = $\frac{1}{3}$, y = -8

a. 16 b. 5 c. 4

a. -10 b. 12 c. -5

a. $\begin{pmatrix} 7 & -4 \\ -5 & 3 \end{pmatrix}$ b. $\begin{pmatrix} -7 & -9 \\ -3 & -4 \end{pmatrix}$ c. $\begin{pmatrix} -1 & 2 \\ -5 & 9 \end{pmatrix}$

8. $2x + 4y = 1$ and $x + 3y = 0 \Rightarrow x = \frac{3}{2}$ and $y = -\frac{1}{2}$

9. $33 + 0 + 16m = 1 \Rightarrow m = -2$ and $32 + -2 + 6n = 0 \Rightarrow n = -5$

10. $x = A^{-1}B = \begin{bmatrix} 1 & -5 \\ -2 & 11 \end{bmatrix}\begin{bmatrix} 4 & -1 \\ 2 & 0 \end{bmatrix} = \begin{bmatrix} -6 & -1 \\ 14 & 0 \end{bmatrix}$

11. $(AB)^{-1} = B^{-1}A^{-1} = \begin{pmatrix} 4 & 3 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 2 & -5 \\ -3 & 7 \end{pmatrix} = \begin{pmatrix} -1 & 1 \\ -1 & 2 \end{pmatrix}$

12. a) $x = -2, y = 3$ b) $x = -2, y = 5$

13. $x = \frac{D_x}{D} = \frac{9(4-1)}{6} = \frac{9}{2}, y = \frac{D_y}{D} = \frac{9}{6} = \frac{3}{2}$ and $Z = \frac{D_z}{D} = \frac{-36}{6} = -6$

14. $\det(A) = 4\begin{vmatrix} 2 & 3 \\ 1 & 4 \end{vmatrix} + 0 + 0 = 4(8-3) = 20$ but $\det(A^T) = \det(A)$ and $\det(A^TA) = \det(A^T)\det(A) \Rightarrow \det(A^T)\det(A) = 20 \times 20 = 400$

15. $-3R_1 + R_3$ denote that -3 times the first row added to the third row. $-3R_1 + R_3$ replace either row 1 or row 3.

16. a) $x = -1, y = -2, z = 3$ b) $x = 2, y = -1, z = -2$

17. a) $\{(-2k-2, k, k) | k \in \mathbb{R}\}$ b) $\{(5k+2, -3k+3, k) | k \in \mathbb{R}\}$

Solution for Unit Seven

1. $\vec{a} + 3b,\ 2\vec{a} + 2\vec{b},\ 2b,\ 3\vec{b} - \vec{a}$

2. left for the reader

3. Left for the reader

4. $\sqrt{179}$ unit

5. a) $\overrightarrow{AB}$ b) $\overrightarrow{AC}$ c) $\overrightarrow{AB}$ d) $\overrightarrow{AC}$ e) 0 f) $\overrightarrow{BD}$

6. exercise left for

7. $7, \frac{2}{7}i + \frac{3}{7}j + \frac{6}{7}k$

8. $\frac{-2}{\sqrt{21}}i + \frac{1}{\sqrt{21}}j + \frac{4}{\sqrt{21}}k$

9. a) 90^0 b) 45^0 c) 135^0

10. $t = -1$

11. a) 1, b) -11

12. a) $k = -1$, b) $k = -\frac{1}{2}$

13. a) $5\sqrt{2}$ b) $\theta = \cos^{-1}\left(\frac{1}{\sqrt{7}}\right)$

15. a) 90 b) 300^0

16. $\frac{-2}{\sqrt{30}}, \frac{-5}{\sqrt{30}}, \frac{-1}{\sqrt{30}}$

17. a) $r = (-3, 4) + t(2, 1)$ b) $r = (5, 1) + t(3, -2)$

$\frac{x+3}{2} = y - 4$ $\frac{x-5}{3} = \frac{y-1}{-2}$

a) $2\sqrt{3}$ b) $\sqrt{14}+\sqrt{2}$ c) $2\sqrt{37}$ d) $\frac{1}{\sqrt{6}}i+\frac{1}{\sqrt{6}}j-\frac{2}{\sqrt{6}}k$ f) 1

a) $\frac{3\sqrt{2}}{2}i+\frac{3\sqrt{2}}{2}j$ b) 2j c) $\frac{-5}{2}i+\frac{5\sqrt{3}}{2}j$ d) –i + 0j

$\bar{x}=\left(\frac{-2}{3},1\right)$ 21. a) x = 3 + 2t, y = -2 + 3t, b) x=5–3t, y=-2+6t, z=1+t

a) y = -2x + 8, b) y – x = -1

a) $(x-3)^2+(y-4)^2=4$, b) $(x+1)^2+(y-1)^2=3$

x = 3 + 4t, y = -4 + 3t

a) (6, -3) b) (7, -3) c) $(x-5)^2+(y+2)^2=5$

a) y = -2x – 3, b) (-3, 3) c) y = 2 d) x = -3

a) (4, -2) b) $(x-4)^2+(y+2)^2=6$, c) x = 4

a) $(4\sqrt{2},2\sqrt{2}),(-3\sqrt{3}-1,3+\sqrt{3}),(3-\sqrt{3},-3\sqrt{3}-1)$

b) $\left(\frac{\sqrt{2}}{2},\frac{3\sqrt{2}}{2}\right),\left(-\sqrt{3}-\frac{1}{2},1-\frac{\sqrt{3}}{2}\right),\left(1+\frac{\sqrt{3}}{3},\sqrt{3}+\frac{1}{2}\right)$

c) $y=x+\sqrt{2},\ y+\frac{\sqrt{3}}{2}=\frac{-1}{\sqrt{3}}\left(x+\frac{1}{2}\right),\ y-\frac{1}{2}=-\sqrt{3}\left(x-\frac{\sqrt{3}}{2}\right)$

d) left for reader

e) $x^2+y^2=2$, $x^2+y^2=2$, $x^2+y^2=2$

a) (-3, -4) about origin (-45) about (1, 2)

b) $y=-\frac{1}{2}x-\frac{1}{2}$ about origin, $y=-\frac{1}{2}x+\frac{3}{2}$ about (1, 1)

c) $(x+3)^2+(y+4)^2=4$, about origin (0, 0)
$(x+4)^2+(y-5)^2=4$ about (1, 2)

Solution for Unit Eight

a) 6i b) $\left(2\sqrt{3}+2\sqrt{2}\right)i$ c) $-1+\sqrt{3}i$ d) $4\sqrt{2}-4\sqrt{2}i$

a) (2,-3)
b) $\left(\sqrt{2}+1,\sqrt{3}-4\right)$ c) (0,4) d) (5,0)

a) 30-7i
b) $\left(\sqrt{6}+1+\left(\sqrt{3}-\sqrt{3}\right)i\right.$ c) 97-i d) 10+6i

a) $x=\frac{2}{13},\ y=\frac{23}{13}$

b) $x=\frac{47}{65},\ y=\frac{14}{65}$

c) $x=\frac{47}{65},\ y=\frac{1}{65}$

d) $x=\frac{31}{26},\ y=\frac{-20}{13}$

5. a) $\overline{Z_1+Z_2}=5, \overline{Z_1}+\overline{Z_2}=5, \overline{Z_1Z_2}=7+i,\ \overline{Z_1}\cdot\overline{Z_2}=7+i$

b) $\overline{Z_1+Z_2}=-1+i, \overline{Z_1}+\overline{Z_2}=-1+i, \overline{Z_1Z_2}=2+i,\ \overline{Z_1}\cdot\overline{Z_2}=2+i$

c) $\overline{Z_1+Z_2}=2-\sqrt{3}i, \overline{Z_1}+\overline{Z_2}=2-\sqrt{3}i, \overline{Z_1Z_2}=-2\sqrt{3}i=\overline{Z_1}\cdot\overline{Z_2}=-2\sqrt{3}i$

d) $\overline{Z_1+Z_2}=3+\left(2-\sqrt{5}\right)i, \overline{Z_1}+\overline{Z_2}=3+\left(2-\sqrt{5}\right)i, \overline{Z_1Z_2}=-2\sqrt{5}-3\sqrt{5}i,$
$\overline{Z_1}\cdot\overline{Z_2}=2\sqrt{5}-3\sqrt{5}i$

6) a) $\frac{23}{65}+\frac{184i}{65}$ b) $\frac{11}{10}+\frac{4i}{5}$ c) –i 7) a) 10 b) 1

8) a) $\cos300^0+i\sin300^0$ b) $\sqrt{17}\left[\cos\theta+i\sin\theta\right]$ *so that* $\tan\theta=$

d) $\frac{1}{\sqrt{2}}\left(\cos\frac{3\pi}{4}+i\sin\frac{3\pi}{4}\right)$ 9) a) -32i b) 1024-1024 I c) -i

10) a) $2(\cos60^0+i\sin60^0), 2(\cos180^0+i\sin180^0), 2\left[\cos300^0+i\sin300^0\right]$

b) $\sqrt[3]{2}\left[\cos80^0+i\sin80^0\right]$
$\sqrt[3]{2}\left[\cos200^0+i\sin200^0\right]$
$\sqrt[3]{2}\left[\cos320^0+i\sin320^0\right]$

11) $2\sqrt{2}+2\sqrt{6}i, -2\sqrt{2}-2\sqrt{6}i$

12) a) $x\frac{5}{13}, y=\frac{14}{13}$ b) x=-4,y=6 c) x=3,y=-1

13) a) center (-1,1), radius r= $\sqrt{2}$ b) center (2,-3), radius r=2

14) y=$\cos\theta\pm isn\theta$

Solution for Unit Nine

1. a) $\frac{\pi}{2}$ b) $\frac{5\pi}{4}, \frac{7\pi}{4}$ c) $\frac{\pi}{3}, \pi, \frac{5\pi}{3}$

2. a) 2, $\frac{\pi}{2}$, 2π, b) 1, $\frac{\pi}{8}$, π c) 4, $\frac{-\pi}{4}$, 3π d) $\frac{5}{4}, \frac{2\pi}{3}, \frac{2\pi}{3}$

3. a) $\frac{\pi}{8}, \frac{\pi}{2}$ b) -3π, 6π 5. a) $y=3\cos\left(\frac{3}{2}x+\frac{3\pi}{8}\right)$

b) $y=\tan\left(\frac{x}{2}-\frac{\pi}{4}\right)$ c) $y=\sec\left(\frac{x}{2}-\frac{3\pi}{8}\right)$ 6. a) $\frac{5\pi}{6}$ b) $\frac{-\pi}{4}$

c) $\frac{\pi}{3}$ d) ½ e) 0 f) $\frac{24}{25}$ g) $\frac{-\pi}{6}$ h) $\frac{1-\sqrt{1-x^2}}{x}$

a) $\frac{1}{\sqrt{1-x^2}}$ b) $2x\sqrt{1-x^2}$ c) $\sqrt{1-10x^2+9x^4}-3x^2$

a) $\frac{2+\sqrt{15}}{6}$ b) $\frac{1}{5}(3\sqrt{7}-4\sqrt{3})$ 9. Exercise left for reader

a) $y=\frac{1}{3}\tan 5x$ b) $y=\cos\left(x-\frac{\pi}{3}\right)+3$

a) $\frac{13}{5}$ b) $\frac{3}{2}$ c) $\frac{4}{5}$ (d, e) exercise left for reader.

a) $\frac{435}{533}$ b) $\frac{92}{525}$ c) $\frac{308}{533}$ d) quadrant I

Solution for unit ten

a. $A_1 = -4$ b. $A_{20} = A_1 + 19d = -4 + 57 = 53$

c. $A_n = 3_n - 7$ d. $A_{n+1} = 3n - 4$

a. $d=\frac{A_8-A_4}{8-4}=\frac{13-11}{4}=\frac{1}{2}$ b. $A_1=\frac{19}{2}$

c. $A_n=\frac{1}{2}n+9$

Since $d=\frac{A_k-A_m}{k-m}=\frac{2m+1-2k-1}{k-m}=-2\Rightarrow A_{k+m}=1$

$\{102, 108, \ldots 996\}\Rightarrow A_n = A_1 + (n-1)6\Rightarrow 996 = 102 + 6n - 6$
$\Rightarrow n = 150$

a. $d = 3$ b. $d=\frac{-7}{2}$

a. $d=\frac{a_{307}-a_{309}}{307-309}\Rightarrow a_{307}-a_{309}=6$

b. $d=\frac{a_{200}-a_{204}}{200-204}=-3\Rightarrow a_{200}-a_{204}=12$

a. $d=\frac{7-3}{n+1}=\frac{4}{n+1}$ b. $d=\frac{3-2}{n+1}=\frac{1}{n+1}$ c. $d=\frac{k-m}{n+1}$

a. $G_6=\frac{1}{4}$ b. $G_8=(8)\left(\frac{1}{27}\right)=\frac{1}{16}$

c. $G_n = 8\left(\frac{1}{2}\right)^{n-1} = 16\left(\frac{1}{2}\right)^n$

9. a. $r = 3$ b. $G_1 = \frac{1}{9}$ c. $G_n = G_1 r^n$

10. a. $r = 4$ b. $G_1 = \frac{1}{16}$ c. $G_n = 4^n$

3

11. $r = -\frac{1}{2}, G_1 = \sqrt{2}, G_5 = \frac{1}{\sqrt{128}}$ and $G_n = \sqrt{2}\left(-\frac{1}{2}\right)^{n-1}$

12. $\frac{9}{x} = \frac{y}{9} \Rightarrow xy = 81$ and $15 - = y - 15 \Rightarrow +y = 30 \Rightarrow x = 27, y = 3.$

13. $\frac{-5x+4}{x+1} = \frac{2x+8}{-5x+4} \Rightarrow (-5x+4)^2 = (x+1)(2x+8) \Rightarrow x = 2.$

14. a. $15 \times 6 = 90$ b. $(21-4+1)(9) = 162$ c. $(100-12+1)(7) = 623$
d. 110 e. 177

f. $S_n = 2 - \frac{2}{n+1} \Rightarrow S_{13} = 2 - \frac{2}{14} = \frac{26}{14} = \frac{13}{7}$

15. a. $S_{10} = \frac{10}{2}[A_1 + A_{10}] = 5\,(8 + 35) = 215$.

b. $S_{20} = \frac{20}{2}[A_1 + A_{20}] = 730$

c. $\sum_{n=1}^{24}(5+3n) = 12\,(8 + 77) = 1020$

d. $\sum_{n=1}^{12}(5+3n) = 6\,(8 + 4) = 294$

16. a. $A_8 = 19$, $A_4 = 11$ and $S_{10} = S\,[2A_1 + 9d] = S(28) = 140$
b. $A_1 = 5$, $A_{10} = 23$ and $A_n = 2n + 3$

17. a. $d = 2$ b. $A_n = 2n + 5$ c. $\sum_{n=1}^{n} 2n + 5 = n^2 + 6n$

d. $S_{20} = 520$

a. $S_n = \frac{n}{2}[4 + 3n + 1] = \frac{1}{2}(3n^2 + 5n)$ b. $S_n = n^2 + 4n$

c. $S_n = \frac{1}{2}(5n^2 + n)$

a. $A_n = S_n - S_{n-1} = 6n - 1$ b. $A_n = S_n - S_{n-1} = Sn -$

a. $S_{20} = 10\,[A_1 + A_{20}] = 10\,[2 + 59] = 610$

b. $S_{30} = \frac{30}{2}[A_1 + a_{30}] = 750$

$a_4 + a_5 + \ldots a_{29} \Rightarrow S_{26} = 13\,[2a_4 + 25(2)] = 13\,(18 + 52) = 910$

a. $S_{25} = \frac{25}{2}[104 + 296] = 5{,}000$ b. $S_{28} = \frac{28}{2}[105+294] = 551($

1830 24. $d = 4$, and $A_n = 4n - 5$

a. $S_{60} = 12{,}780$ b. $S_{15} = \frac{15}{2}[-395 + -33] = -3{,}12$

a. $S_{20} = 530$ b. $S_{19} = \frac{19}{2}[23 + 59] = 779$ c. $S_{15} = 6$

d. $S_{99} = \log 100 = 10$ e. $S_{100} = \frac{25}{51}$ f.

$S_{70} = 70x^3$

a. $S_{79} = 79 \times 979$ b. $S_{69} = \frac{69}{2}[104 + 988]$ c. S

20,424

a. 25 b. 15 c. $\frac{5n}{2}$ d. $\frac{n}{2}(m + k)$

$S_{16} = 8\,(2(8) + 30) = 368$

a. $S_{39} = 4017$ b. $S_{40} = 7980$

a. $S_6 = 195$ b. $S_8 = 381$ c. $S_{10} = 3069$

d. $3[2^n - 1]$

a. $S_7 = 381$ b. $S_7 = \frac{16}{3}[1 - 3^{-7}]$ c. $3\left[\left(\frac{3}{2}\right)^{10} - 1\right]$

a. $r = 2$ b. $S_6 = \frac{65}{2}$ c. $S_n = \frac{1}{2}(2^n - 1)$

34. a. $\frac{2}{3}$ b. 3 c. $\frac{1}{7}$ d. $\frac{3}{2}$

35. a. $\frac{10}{3}$ b. $\frac{49}{30}$ c. $\frac{5}{8}$ d. 200

e. $\frac{40}{3}$ f. $\frac{15}{4}$ g. $\frac{125}{4}$ h. $\frac{4}{3}$

i. 32 j. 108 k. $\frac{35}{44}$ l. 5

36. a. $3^{1+\frac{1}{3}+\frac{1}{9}+\ldots} = 3^{\frac{1}{1-\frac{1}{3}}} = 3\sqrt{3}$ b. $4^{\frac{1}{1-\frac{1}{5}}} = \sqrt[4]{1024}$

37. $k = \frac{2}{3}$

38. a. $k = -2$ b. $k = -3$ c. $k = -2$

Solution for unit eleven

1. a. i) $\left[\frac{5}{4}, \infty\right]$ ii. $\text{lub} = \frac{5}{4}$ iii. $\left(-\infty, \frac{-2}{3}\right)$ iv. $\text{glb} = \frac{-2}{3}$

b. i) $(1, \infty)$ ii. $\text{lub} = 1$ iii. $\left(-\infty, -\frac{1}{4}\right)$ iv. $\text{glb} = -\frac{1}{4}$

c. i) $\left[\frac{5}{3}, \infty\right)$ ii. $\text{lub} = \frac{5}{3}$ iii. $(-\infty, -4)$ iv. $\text{glb} = -4$

d. i) $[e^{-2}, \infty)$ ii. $\text{lub} = e^{-2}$ iii. $(-\infty, 0)$ iv. $\text{glb} = 0$

e. i) $\left[\frac{32}{3}, \infty\right]$ ii. $\text{lub} = \frac{32}{3}$ iii. $(-\infty, 0)$ iv. $\text{glb} = 0$

f. i) $\left[\frac{2}{3}, \infty\right)$ ii. $\text{lub} = \frac{2}{3}$ iii. $(-\infty, 0]$ iv. $\text{glb} = 0$

2. a. increasing b. st. decreasing c. st. increasing
d. st. decreasing e. st. decreasing f. increasing
g. decreasing h. neither i. st. increasing
j. decreasing k. increasing

3. a. convergent, limit value = 1
b. divergent, limit value doesn't exist (2 or –2)

c. convergent, limit value $\frac{1}{2}$

d. convergent, limit value e^2

e. divergent, doesn't exist

Limit value: a. $\frac{25}{9}$ b. $\frac{-11}{9}$

a. $\frac{2}{3}$ b. $-\frac{1}{3}$ c. $\frac{-\sqrt{3}}{2}$ d. $-\frac{1}{3}$

a. $\frac{1}{2}$ b. $\frac{-3}{10}$

a. $\frac{10}{7}$ b. $\frac{4}{3}$ c. $\frac{1}{25}$ d. $\frac{1}{3\sqrt[3]{a^2}}$

e. nt^{n-1} f. $-\frac{1}{9}$ g. $\frac{1}{6}$ h. 3

i. $\frac{1}{3\sqrt[3]{25}}$ j. 6 k. 1 l. $\frac{1}{\sqrt{6}}$

a. e^{-1} b. $e^{\frac{20}{3}}$ c. e^5 d. e^{-4}

a. $k = \frac{-\ln 5}{2}$ b. $k = \frac{-2}{3}\ln 4$

0. a. $\frac{1}{4}$ b. $\frac{1}{5}$ c. -1 d. $\frac{1}{\pi}$

e. -2 f. $\frac{1}{1+\pi}$ g. $-\frac{7}{2}$ h. 8

1. a. 4 b. $-\frac{1}{\sqrt{3}}$ c. 8

2. a. $-\infty$ b. $-\infty$ c. $-\infty$ d. $\pm\infty$
e. ∞ f. ∞ g. $-\infty$ h. $-\infty$ i. $-\infty$

3. a. $(-\infty,-2]$ and $[2,\infty)$ b. $\{x: -3 \le x \le 3\}$ c. $\{x: x \ne 3, -3\}$
d. $\{x \in \mathbb{R}: x \ne 3\}$ e. $\{x \in \mathbb{R}: x > 2\}$
f. for all integer k, $[k, k+1]$ is continuous
g. $(-\infty, 0) \cup [2, \infty)$ h. $(-\infty, \ln 2) \cup [3, \infty)$ i. $(0, \infty)$
j. $(-\infty, 2]$ and $(2, \infty)$ k. at each $x \in (-\infty, \infty)$

14. a. $c=\frac{-3}{2}$ b. $a=\frac{5}{4}$ c. $a=e^4$ d. $c=\pm 2$

Solution for unit – 12

1. a. $y=2x-2\pi+1$ b. $y=-6x+3$ c. $y=3x-3$

d. $y=10x-13$ e. $\frac{1}{2}x-\frac{1}{2}$ f. $y=\frac{3}{2}x$

g. $y=e^{-\sqrt{2}}x$ h. $y-\frac{1}{2}=\frac{\sqrt{3}}{2}\left(x-\frac{\pi}{6}\right)$

i. $y-1=\frac{x-8}{10\ln 10}$ j. $y=4x-2$ k. $y=3x+2$

l. $y=3x+1$ m. $y=e(4x+3)$ n. $y=-x$

2. a. $f'(x)=0\Rightarrow 6x^2-6=0\Rightarrow x=\pm 1$

a. $f'(x)>0\Rightarrow 6x^2-6>0\Rightarrow x\in(-\infty,-1)\cup(1,\infty)$

b. $f'(x)<0\Rightarrow 6x^2-6<0\Rightarrow x\in(-1,1)$

c. $f'(x)=2\Rightarrow 6x^2-6=2\Rightarrow x=\pm\frac{2\sqrt{3}}{3}$

d. $f'(x)=-2\Rightarrow 6x^2-6=-2\Rightarrow x=\pm\sqrt{\frac{2}{3}}$

3. a. $(0,2)\Rightarrow y-f(0)=f'(0)(x-0)\Rightarrow y-2=-15x\Rightarrow y=-15x+2$

$(3,2)\Rightarrow y-f(3)=f'(3)(x-3)\Rightarrow y-2=24(x-3)$

b. $(0,2)\Rightarrow y-2=-15x$

4. a. $(-\infty,\infty)$ b. $x\in[-5,\infty)$ c. $x\in\left[-\frac{5}{3},\frac{5}{3}\right]$ d. $(-\infty,3]$ and $[3,\infty)$

5. a. $\frac{-\ln 3}{x(\ln x)^2}$ b. $\frac{5}{2x\ln 4}$ c. $\frac{2x}{(\ln 2)(x^2+5)}$

d. $\frac{x}{(x^2+1)\ln 10}$ e. $\frac{6}{6x+5}-\frac{6x+1}{3x^2+x+1}$

f. $\frac{-\sin x}{\sqrt{3x-1}}-\frac{3}{2\sqrt{(3x-1)^2}}$ g. $2x\cos(x^2+1)$

h. $\frac{6x^2}{\sqrt{1+x^4}}-\frac{8x^6}{2\left(\sqrt{1+x^4}^3\right)}$ i. $\frac{2x+5}{2\sqrt{x^2+5x+6}}$

j. $\dfrac{x+2}{\sqrt{x^2+4x}}e^{\sqrt{x^2+4x}}$

k. $2x\left(\log_2^x\right)+\dfrac{x^2+3}{x\ln 2}$

l. $2\sec^2(x+3)\tan(x+3)$

m. $\dfrac{-5^x\ln 5}{(5^x+6)^2}$

n. $3^{6-x^2}\ln 3(-2x)$

o. $\dfrac{\cos\sqrt{x+4}\left(e^{\sin\sqrt{x+4}}\right)}{2\sqrt{x+4}}$

p. $\dfrac{9(4x-1)^2}{(2-5x)^4}$

6. a. 7 b. $\dfrac{47}{16}$ c. $\dfrac{-4\sqrt{5}}{5}$

7. $F'(-1)=-3$ 8. $f'(3)=\dfrac{1}{4}$

9. $F'(x)=3x^2f(x)+x^3f'(x)+f'(g(x))\,g'(x)\Rightarrow F'(1)=7e^3+4e^{15}$

10. $f''(x)=\dfrac{-2x^2+8x-16}{(x^2-4x)^2}\Rightarrow f''(-1)=\dfrac{-2-8-16}{25}=\dfrac{-26}{25}$

11. $G''(-1)=24$ 12. $G'(1)=\dfrac{-8}{9}$

13. a. $f^n(x)=(-4)^ne^{-4x}$ b. $f^n(0)=(-4)^n$

14. $f^n(x)=e^x(x+n)$ 15. $g'(1)=\dfrac{1}{6}(1-\ln 6)$

16. a. $y=-2x-4$ b. $5y-x=-42$

17. $f''(1)=-e^{-1}$ 18. $f'''(0)=(\pi\ln 3)^3$

19. $g^n(x)=f^n(2x+4)(2^n)$

20. a. $f^n(x)=\dfrac{n!}{(3-x)^{n+1}}$ b. $f^n(x)=\dfrac{(-1)^n n!}{(2+x)^{n+1}}$

Solution for unit – 13

1. a. i) 2, –2 ii) max. value 28 iii) min. value – 16 iv) [–16, 28]

b. i) 0, $\dfrac{1}{8}$ ii) max. value 3 iii) min. value 0 iv) [0, 5]

c. i) 1 ii) max. value 5 iii) min. value 0 iv) [0, 3]

d. i) $-\frac{1}{2}$ ii) max. value $\frac{7}{2}$ iii) min. value 0 iv) $\left[0, \frac{7}{2}\right]$

c. i) $\left[\frac{\pi}{6}, \frac{5\pi}{6}\right]$ ii) max. value $\left(\frac{\pi}{6} + \sqrt{3}\right)$ iii) min. value $\left(\frac{\pi}{6} - \sqrt{3}\right)$

f. i) –2, 0, 2 ii) max. value 27 iii) min. value –48

g. i) 1, 5 ii) max. value 143 iii) min. value 10 iv) 10, 143

h. i) 2 ii) max. value 3 iii) min. value 1 iv) [1, 3]

2. a) i) $(-\infty, -2]$ and $[2, \infty)$ ii) $-2 \le x \le 2$ iii) at $x = -2$ and $x = 2$

b) i) $[-4, 4]$ ii) $(-\infty, -4]$ and $[4, \infty)$ iii) at $x = -4$ and at $x = 4$

c) i) $(-\infty, \infty)$ ii) no interval iii) no extreme value

d) i) no interval ii) decreasing for all x $(-\infty, \infty)$ iii) no local extreme value

e) i) $(-\infty, -1]$ and $[1, \infty)$ ii) $[-1, 1]$ iii) at $x = -1$, max. at $x = 1$, min

f) i) $(-\infty, -1]$ and $[1, \infty)$ ii) $[-1, 1]$ iii) at $x = -1$ and at $x = 1$

g) i) $(-\infty, 3]$ ii) $[3, \infty)$ iii) at $x = 3$

h) i) no interval ii) $\mathbb{R}\backslash\{7\}$ iii) no extreme value

i) i) $[-1, \infty)$ ii) $(-\infty, 1]$ iii) at $x = -1$

3. a) $b = 1, a = 1$ b) $k = -4$ c) $k = \frac{1}{3}$ d) $k = 9$

4. a) i) $(0, \infty)$ ii) $(-\infty, 0)$ iii) at $(0, f(0))$

b) i) $(-\infty, \infty)$ ii) No iii) no inflection

c) i) $\left(-\infty, -\frac{1}{\sqrt{3}}\right)$ and $\left(\frac{1}{\sqrt{3}}, \infty\right)$ ii) $\left(-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$

iii) at $x = -\frac{1}{\sqrt{3}}$ and $x = \frac{1}{\sqrt{3}}$

d) i) $\left(\frac{1}{5}, \infty\right)$ ii) $\left(-\infty, \frac{1}{5}\right)$ iii) at $x = \frac{1}{5}$

e) i) $(-\infty, -3)$ ii) $(-3, -1)$ iii) at $x = -1$ and $x = -3$

5. a) 6 and 6 b) $\frac{15}{2}$ and $\frac{15}{2}$ c) $2\sqrt{6}$ and $2\sqrt{6}$ d) $\frac{4}{3}, 0, \frac{2}{3}$

a) $\frac{4}{5}$m and $\frac{4}{5}$m b) 30 m and 30 m c) $4\sqrt{2}$ and $4\sqrt{2}$ d) 9m and 9m

a) $\left(-\sqrt{3},3\right)$ and $\left(\sqrt{3},3\right)$ b. $\left(-\sqrt{\frac{5}{2}},\frac{3}{2}\right)$ and $\left(\frac{5}{2},\frac{3}{2}\right)$ c) $\left(\frac{3}{5},\sqrt{\frac{3}{5}}\right)$

$h=2$ cm, $r=10\sqrt{\frac{15}{2\pi}}$

$2\pi r+4x=24$ m$\Rightarrow x=6-\frac{1}{2}\pi r$ and $A=x^2+\pi r^2$

$\Rightarrow A=\left(6-\frac{1}{2}\pi r\right)^2+\pi r^2 \Rightarrow A'(r)=2\pi r+\pi^2 r-6\pi=A'(r)=0$

$\Rightarrow r=\frac{6}{\pi+2}$ and $x=\frac{3\pi+12}{\pi+2}$

10. a) 4π cm/sec b) 20π cm/sec 11. a) $\frac{1}{64\pi}$cm/sec b) $\frac{1}{2}$cm/sec

12. $6\sqrt{2}$ cm by $3\sqrt{2}$ 13. $\frac{9}{4}$ ft/sec

14. $v=\frac{4}{3}\pi r^3 \Rightarrow \frac{dv}{dt}=4\pi r^2\frac{dr}{dt}\Rightarrow \frac{1}{4\pi r^2}\left(\frac{dv}{dt}\right)$, then a) $r=2$ cm

$\Rightarrow \frac{dr}{dt}=\frac{1}{4\pi(4)}\left(\frac{3\text{cm}^3}{\text{min}}\right)=\frac{3}{16\pi}$cm/min b) $\frac{dr}{dt}=\frac{1}{48\pi}$cm min

15. $v=\frac{1}{3}\pi r^2 h$ and $\frac{r}{h}=\frac{2}{6}\Rightarrow r=\frac{1}{3}h\Rightarrow v=\frac{1}{3}\pi\frac{h^3}{9}$

$v=\frac{\pi h^3}{27}\Rightarrow \frac{dv}{dt}=\frac{3\pi h^2}{27}\frac{dh}{dt}\Rightarrow \frac{dh}{dt}=\frac{9}{\pi h^2}\left(\frac{dv}{dt}\right)=\frac{9}{\pi(16)}(2)=\frac{9}{8}$ cm min

Solution for unit – 14

1. a. $\sqrt{3}x+c$ b. $\frac{x^{\sqrt{2}+1}}{\sqrt{2}+1}+c$ c. $\frac{-\sqrt{3-4x}}{6}+c$

d. $\frac{1}{3}\sqrt{2x-1}+c$ e. $\frac{(3x-1)^5}{15}+c$ f. $\frac{1}{7}\left(\sqrt[3]{3x+1}\right)^7+c$

2. a. $\frac{e^{4x}}{4}+c$ b. $\frac{e^{3x-4}}{3}+c$ c. $\frac{3^{4x}}{4\ln 3}+c$

d. $\frac{-e^{-2x}}{2}+e^{-x}+c$ e. $\frac{-5}{2}e^{-2x}+e^{-x}+c$ f. $\frac{-2}{3}\cdot\frac{24^{-x}}{\ln 24}+c$

3. a. $\ln|x|+c$ b. $\frac{1}{4}\ln|4x-1|+c$ c. $-\frac{1}{5}\ln|2-5x|+c$

d. $\frac{-\sqrt{2}}{7}\ln|1-7x|+c$ e. $e^{-x}-e^{-x}+2x+c$ f. $2\sqrt{x-1}-\frac{1}{2}e^{2}$

g. $\int 2\sqrt{e^{x}}-e^{-x}+c$ h. $-\frac{1}{x}+c$ i. $-\frac{1}{x+3}+c$

j. $-\frac{1}{2}(x-4)^{-2}+c$ k. $\frac{1}{6-9x}+c$

4. a. $2e^{\sqrt{x+1}}+c$ b. $e^{x^4}+c$ c. $\frac{2}{3}\left(\sqrt{x+3}\right)^3-4\sqrt{x+3}+c$

d. $6\sqrt{x^2+4}+c$ e. $\frac{(4x^2+2x+7)}{14}+c$ f. $\frac{1}{2}\ln|1-2x+4x^2|+c$

g. $\frac{(2x+3)^9}{2}+c$ h. $\frac{(4x-3)^{11}}{44}+c$ i. $\frac{2}{3}\sqrt{x^3-2}^{3}+c$

5. a. $\frac{(\ln x)^3}{3}+c$ b. $\frac{-1}{2(\ln x)^2}+c$ c. $\ln|\ln x|+c$

6. a. $\frac{1}{12}\ln\left|\frac{2x-3}{2x+3}\right|+c$ b. $\frac{1}{2}\ln|x^2+4x+3|+c$

c. $\frac{3}{2}\ln|2x-1|-2\ln|x+1|$ d. $5\ln|x-2|-\ln|x+2|-3\ln|x|+c$

e. $\ln\left|\frac{x^2+1}{x}\right|+c$ f. $\frac{1}{16}\ln\left|\frac{4x^2-1}{4x^2+1}\right|+c$

7. a. $x^2e^x-2xe^x-2xe^x+2e^x+c$ b. $-e^{-x}(x^3+3x^2+6x+6)+c$

c. $\frac{1}{2}e^{2x}\left(x^2-x+\frac{1}{2}\right)+c$

d. $\frac{1}{2}x^2 \ln|x+1| - \frac{1}{4}x^2 + \frac{1}{2}x - \frac{1}{2}\ln|x+1| + c$

e. $x \ln|3x| - 3x + c$ f. $x(\ln x)^2 - 2x \ln x - 2x + c$

g. $x^2 \sin x + 2x \cos x - 2 \sin x + c$

h. $\frac{1}{2}x^3 \sin 2x + \frac{3}{4}x^2 \cos 2x + \frac{3}{4}x \sin 2x + \frac{3}{8}\cos 2x$

8. a. $\frac{e^x}{5}(\sin 2x - 2\cos 2x) + c$ b. $\frac{e^{2x}}{5}(2\cos x + \sin x) + c$

c. $\frac{e^{-x}}{2}(-\sin x - \cos x) + c$

9. a. $\frac{9}{2}$ b. $\frac{13}{2}$ c. $\frac{23}{3}$ d. 0

e. $\frac{54 - 4\sqrt{2}}{9}$ f. 2 g. $\frac{4}{15}$ h. $\frac{3\sqrt{3}}{4}$ i. $\frac{1}{3}$

10. a. $\ln 2$ b. $\frac{-22}{15}$ c. $\ln\sqrt{2}$ d. $e - 1$

e. $\frac{-\pi}{2}$ f. $\frac{\pi}{2} - 1$ g. $\frac{e}{2}[\sin(1) - \cos(1)] + \frac{1}{2}$

h. $\frac{e^2(\sin 2 - \cos 2) + 1}{2}$ i. $\frac{24 \ln 2 - 7}{9}$

11. a. π b. $\ln 2$ c. $\frac{\pi}{4}$ d. $2 \log 2 - 1$ e. $\sqrt{3} - \sqrt{2}$

f. $\frac{4}{3}$ g. $2 \log 2$ h. $2e^2 - e$ i. $\ln 4 - \frac{3}{4}$

j. $\frac{74}{9}$ k. $\frac{7}{18}$ l. $\frac{2}{3}$

12. a. $\frac{125}{6}$ b. $2\sqrt{2}$ c. 24 d. 8

e. $\frac{7}{6}$ f. $\frac{64}{3}$ g. $\frac{8}{\ln 9}$

13. a. $\frac{125}{6}$ b. $\frac{10}{3}$ c. $\frac{32}{3}$

14. a. $\frac{20}{3}\pi$ b. $\frac{196}{5}\pi$ c. $\frac{625}{6}\pi$

d. $v = 2\pi\int_0^2 (16x^2 - x^6)dx = 2\pi\left[\frac{16x^3}{3} - \frac{x^7}{7}\right]_0^2 = \frac{1024\pi}{21}$

e. $v = 2\pi\int_0^1 (x^2 - x^4)dx + 2\pi\int_1^3 (x^4 - x^2)dx$

$= \frac{4\pi}{15} + \frac{1188\pi}{15} = \frac{1192\pi}{15}$

15. a. $\text{vol} = \pi\int_2^6 \left(\sqrt{y-2}\right)^2 dy = \pi\left[\frac{y^2}{2} - 2y\right]_2^6 = 8\pi$

b. $\text{vol} = \pi\int_{-1}^2 \sqrt{y+1}^2 dy = \pi\left[\frac{y^2}{2} + y\right]_{-1}^2 = \frac{9\pi}{2}$

c. $\text{vol} = (e-2)\pi$

Solution for unit – 15

1. a. 3 b. 1 c. $\sqrt{13}$ d. $\sqrt{5}$ e. $\sqrt{10}$ f. 2
2. (0, 0, 0) and (0, 2, 0) 3. a. $(x-1)^2 + y^2 + (z+2)^2 = 21$
   b. $(x-3)^2 + (y+1)^2 + (z-2)^2 = 14$
4. $5 - 3 = 2$
5. a. outside sphere b. on the sphere c. inside the sphere
6. a. no intersection b. $(0, 3 \pm \sqrt{3}, 0)$ c. no intersection
7. a. (0, -4, 0) and (0, 6, 0) b. $(2 \pm 2\sqrt{7}, 0, 0)$
8. $r = 5$ 9. a. $k = \pm 6$
10. $(x-1)^2 + (y+2)^2 + (z+3)^2 = r^2 = -k + 14 \Rightarrow 0 - k + 14 = 5 \Rightarrow k = 9$

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